

Service Requests Portal — Developer Handoff

One document. It replaces the two that used to overlap (the root `README.md` and `docs/handoff/README.md`) and carries both halves: **how the thing works**, and **why it works that way**. The code is disposable; the decisions are not.

Audience: the engineering team who will rebuild this application on the organisation's own stack and standards. Also useful to anyone opening the repo.

Status: working prototype. It runs, it holds test data, people have used it, and it is the reference for a rebuild — not the thing that ships long-term. See `NEXT_STEPS.md` for the programme decision (it is a **Rebuild**).

How to read it. Every section answers *what it does* and *why*. Where a decision was forced by a constraint you may not share — a SQLite quirk, a hosting proxy, an unfinished vendor contract — it says so explicitly, so you can **delete the workaround instead of faithfully reproducing it**. [Part IX](#) is the chronological record of decisions made while building, kept in full because several of them were corrected once and the reasoning is the only thing that stops them being un-corrected.

What is authoritative.

Question	Authority
What the product does	this document + <code>USER_MANUAL.md</code> + the screenshots
Exact field and API shapes	the prototype source, cited by path throughout
What EasyVista accepts	nobody yet — see Known gaps
Current data	the live database, <i>not</i> this document

Screenshots in `handoff/screenshots/` were captured on **2026-08-08** by `client/scripts/capture-screenshots.mjs` against the hosted database, which holds a purpose-built demonstration set. It is entirely test data, and the owner confirmed it is fine to photograph. `handoff/screenshot-manifest.json` is written **by** that script, so it cannot describe a shot the script did not take. See [Part VII](#).

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Part I — Orientation

1. The problem

The **Customer Interactions / Product Owners** team fields a constant stream of work from **field representatives**: Billing Center and Policy Center defect reports, enhancement requests, and — added later — **requests for reports and dashboards**. They triage it, prioritise it, and decide what is escalated to **Tier 2 GTS**, who work tickets in **EasyVista** (surfaced throughout the app as "Service Desk", see [§14](#)).

Before this portal there was no system of record:

- Defect reports lived in email threads, chat and spreadsheets — no audit trail, and duplicates were near-impossible to spot.
- Enhancement requests had no structured intake, so details arrived incomplete.
- Product Owners had no single queue to triage from.
- Historical records sat in Excel files that could not be searched.
- Escalating to EasyVista meant manual copy-paste.
- **Reps had no visibility**. They could not see whether an issue was already known, or what happened to something they filed — so they filed it again. Duplicate intake was the largest single source of wasted triage effort.
- Nobody knew when new work arrived without checking.
- Report and dashboard requests arrived through the same informal channels and had the same problems, plus one of their own: nobody could say how much analyst time went into them.

Almost every non-obvious decision in this codebase traces back to two goals: reduce duplicate intake, and make state legible to the person who reported it. When a design choice looks fussy, that is usually why. A third joined them when report requests landed: **make analyst effort visible**, which is why hours are a child table and not a column.

2. Who uses it

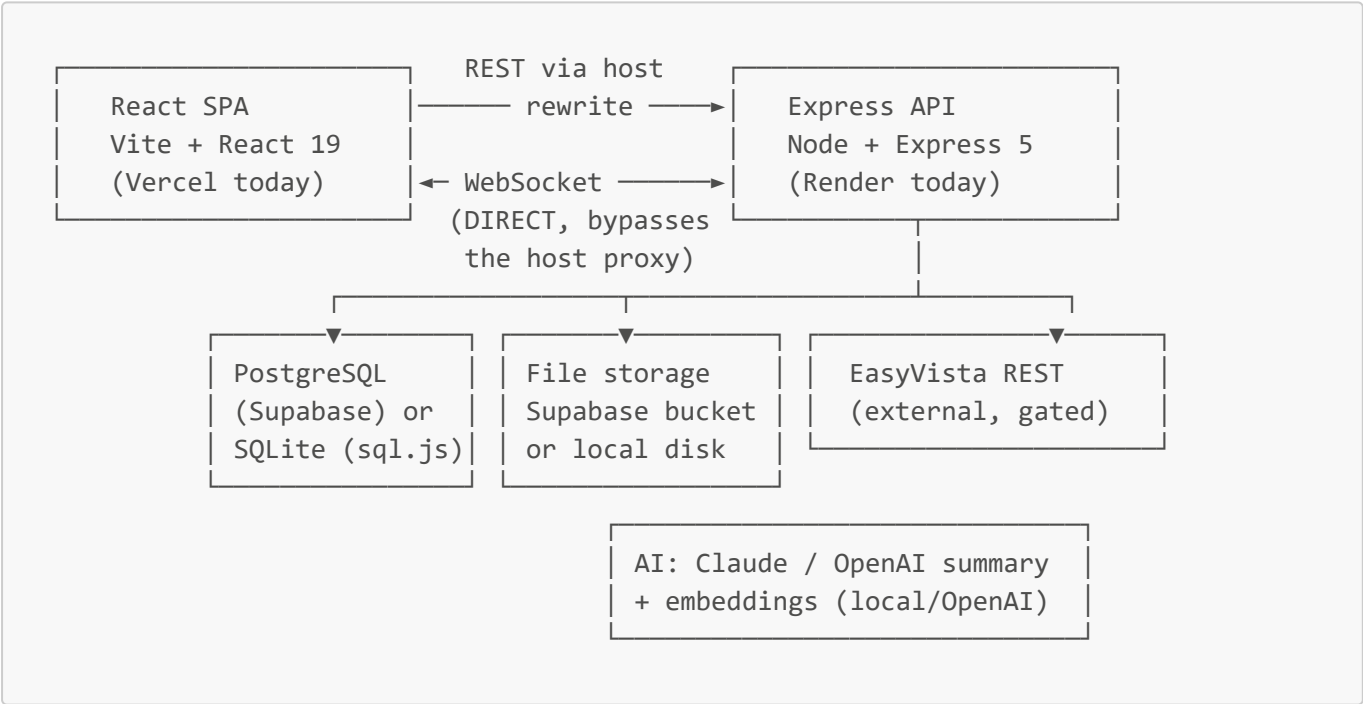
Role	Signs in?	Can do
Field representative (<code>users.role = 'rep'</code>)	Yes — required.	File any request; check for duplicates before filing; follow the status board and the requests they filed. Holds no grants , so <code>ensureAdmin</code> refuses them like a stranger and the header shows them no Admin link.
Product Owner / admin (<code>role = 'admin'</code>)	Yes	Everything in the queues they are granted : triage, edit, status, attachments, redirect, Service Desk send, public visibility, retire, bulk actions, Excel round-trip, and the read half of metadata.

Role	Signs in?	Can do
Reporting analyst	Yes	An admin whose grant is narrowed to report . There is no fourth role — see §10.
Manager	Yes	A rank above admin, per application. Gates exactly one thing: seeing other people's throughput numbers.
Viewer	Yes	Read one application's queue and export it. Changes nothing.
Super user (<code>is_super_user</code>)	Yes	Everything above across every application, plus the Access page and the metadata writes . The single privilege bypass in the system.

Roles are **per application**. "Admin" is not a global role — someone can administer Billing Center and have no access at all to Policy Center.

`users.role` is the door; the grants are the rooms. Two values may sign in (`ACCOUNT_ROLES_THAT_MAY_SIGN_IN` in `server/src/constants.js`): `admin` and `rep`. Anything else is refused at the login endpoint regardless of grants.

3. Architecture



The host names are the **prototype's** hosting. The internal deployment will be company servers and a company database. The application structure does not change. **Four things in this codebase exist only because of that hosting and should be deleted** — see §21.

Why the WebSocket bypasses the host proxy. Vercel rewrites cannot carry WebSocket upgrades, so a same-origin socket degrades to perpetual HTTP long-polling — billing a flood of requests. Connecting directly to the API host gives a real WebSocket: one persistent connection, ~zero ongoing requests.

That creates an auth problem, because the session cookie is not sent cross-origin. The client therefore fetches a **short-lived HMAC-signed token** from a same-origin, session-authenticated endpoint and passes it in the

socket handshake. **This whole mechanism exists for one hosting constraint.** If your platform carries WebSocket upgrades end to end, delete it and use the session.

4. Tech stack and layout

Layer	Technology	Notes for the rebuild
Frontend	React 19.2, React Router 7.13, Vite 5.4	Vite is the build tool , not a host. It stays.
UI system	Custom — client/src/components/bite-size/BitsizeUI.jsx + vanilla CSS, bs-prefix	No MUI, Tailwind or Bootstrap. Replace with your design system; the behaviours documented here are what must survive, not the CSS. Custom dialogs and notices throughout — no native alert()/confirm() .
Backend	Express 5.2, Sequelize 6.37	Thin routes → services.
Database	PostgreSQL (pg 8.16) or SQLite (sql.js 1.13)	Dual-provider was a prototype convenience. See portability traps .
Real-time	Socket.IO 4.8	
Sessions	express-session + connect-pg-simple	Postgres-backed store; falls back to MemoryStore on sql.js. See §12.
Auth	bcrypt	Placeholder. Target is SSO / Active Directory.
Uploads	multer 2.0	Image-only for attachments, deliberately.
Excel	xlsx (SheetJS) 0.18	
AI summary	@anthropic-ai/sdk 0.112 or OpenAI via fetch	One env var switches the whole stack.
AI embeddings	@huggingface/transformers 4.2 in-process / OpenAI / Voyage	Local option: no vendor, no key, no per-call cost, text never leaves the server.

client/src/	
pages/	7 route components
components/	
admin/	queue, command bar, modals, detail/ (13 sub-components)
public/	status board, submit form pieces
bite-size/	BitsizeUI design system + Layout/AppShell
common/	AiSearchPanel, PaginationControls
hooks/	one per feature area
lib/	api.js (the single request helper), socket.js
utils/	filter/sort/format helpers shared admin+public
constants/	column, filter, sort and status registries
client/scripts/	the verification + screenshot harness (see Part VII)
server/src/	
routes/	13 thin route modules; validation at the boundary

services/ helpers/	9 services; all business logic mappers (public allow-list), easyVistaPayload, lookups, storage, reportVisibility, submissionInsert, timeline
middleware/ constants.js	cors, session, csrf, upload, rateLimit, errorHandler role ladder, status registries, allow-lists, sentinels
server/db/ models/index.js	the whole schema + migration logic + portability traps
sequelize.js	provider selection
server/scripts/ docs/	migrations, backfills, seeds, purge – all dry-run by default this document, USER_MANUAL.md, NEXT_STEPS.md, screenshots

Part II — The product

5. Domain vocabulary — read this first

Getting these words wrong produces a subtly wrong system. They are not interchangeable.

Term	Means
Submission	One request. A row in <code>submissions</code> . Also "ticket".
Type	<code>defect</code> , <code>enhancement</code> or <code>report</code> . A lookup value , not an enum — the Metadata page manages the list. Drives required fields, the status vocabulary, and what the Service Desk is told.
Application	A product queue — <code>Billing Center</code> , <code>Policy Center</code> , <code>Other</code> . Owns triage rights.
Other	The catch-all working list , covering two cases: a system with no configured application to submit to the Service Desk directly, and a system nobody has identified yet. Either way the work is still tracked — reports get built from here, and a defect or enhancement is raised on the Service Desk by hand with its incident number typed back in. Not a waiting room; the owner's phrase is " <i>almost like a task list</i> ".
Reports-only application	An application a reporting analyst created by typing its name in (<code>applications.reports_only = 1</code>). It takes report requests and nothing else , and is granted to everybody who works report requests.
Soft association	<code>submissions.working_application_id</code> — a queue an <code>Other</code> ticket also appears in , without leaving <code>Other</code> . The only place two columns answer "whose queue is this", and it never decides who may edit.
Cleanup task	Internal work item (<code>is_cleanup</code>). Tagged <code>defect</code> , <code>enhancement</code> or <code>cleanup_only</code> . A flag on a defect or an enhancement, never a type of its own.
Retire	Soft archive. Nothing in this app hard-deletes a submission.
Public	Appears on the status board and in public AI search.
Workaround request	A rep is blocked on a live case <i>now</i> . Two columns: <code>needs_workaround</code> (the ask), <code>workaround_provided</code> (the team closing it).

Term	Means
Redirect	Hand a ticket to another application's queue. The ticket moves .
Resubmission	Re-send to the Service Desk after changes. Creates a new submission and a new incident; the original is untouched.
Viewer envelope	The single server answer to "who is this caller and what may they see". <code>GET /api/viewer</code> .
Home application	The application a person most likely wants preselected. A prefill , never a lock.
Pinned application	The queue an admin <i>decided</i> to land on. Distinct from "the last one I looked at".
Level of effort	An analyst's size estimate for a report request. A lookup (S/M/L/XL).
Hours	Analyst time, one row per sitting in <code>request_time_entries</code> . Never a column.

Five distinctions that are easy to collapse and must not be

1. Redirect moves; resubmission forks. A redirect changes `submissions.application_id` on the existing row and writes a ledger entry. A resubmission inserts a *new* row. Conflating them yields either two tickets for one problem, or one ticket two teams each think the other owns.

"The ticket MOVES. It is not copied and not mirrored: a copy would give the reporter two tickets for one problem and leave two teams each assuming the other owned it." —

`server/src/services/redirectService.js`

2. Read scope is wider than write scope. A team that hands a ticket on **keeps reading it** and **stops writing it**, the instant it moves. Read comes from the routing ledger; write asks only about the ticket's *current* application. The sending team needs to see where their ticket went.

3. `needs_workaround` and `workaround_provided` are two columns, not one flag. Handling the request must not erase the fact that it was made. "Open request" means the first without the second. A single tri-state column would lose the timing, and the pair is what lets history show how long the rep waited.

4. A cleanup task is a flag, not a type. `is_cleanup` plus `cleanup_tag_type` on a defect or an enhancement. This is why an "application admin for defects and enhancements" needs no cleanup grant — cleanups ride along with the type they are tagged as. `cleanup_only` means it was never anybody's bug report.

5. `submissions.assigned_to` is the current holder; `request_assignments` is the trail. The first is cheap to query and index for "my queue". The second cannot be reconstructed after the fact, which is why it shipped **with** the feature — without it, reassignment silently erases everyone who held a request before the last person.

6. `application_id` owns the ticket; `working_application_id` only shows it. The soft association is the one place two columns answer "whose queue is this", and it is safe only because the second one answers strictly less:

	<code>application_id</code>	<code>working_application_id</code>
Who may edit	decides it (<code>canMutateApplication</code>)	never consulted
Who may read	decides it	widens, but only into a queue the caller already holds
Which queue lists it	yes	yes (OR on the scope filter)
Set on	every ticket	only a ticket in Other
Changed by	a redirect	the analyst working it

Collapse the two and you get one of two failures: either an admin gains write access to a ticket nobody granted them, or an analyst's own list stops showing work they are doing. The rule that keeps it honest is that the soft column is **read-only in every sense that matters** — see `server/src/helpers/softAssignment.js`.

6. The four kinds of work

Kind	Stored as	Asked for	Ends at
Defect	<code>type = defect</code>	Screen, date/time, what you saw, steps	Service Desk (Deployed)
Enhancement	<code>type = enhancement</code>	The request and what it would save	Service Desk (Deployed)
Cleanup task	<code>is_cleanup</code> flag on either of the above	An internal description	Cleanup Status: Completed
Report request	<code>type = report</code>	Which application's DATA, what it is for, measures and sources or which report and what should change	Delivered — built in the portal's own team, never handed to the Service Desk

Report requests are the second half of the product

They arrived after the defect/enhancement portal was working, and they are not a fourth flavour of the same thing. What is different:

- **Two branches.** `is_new_dashboard` decides which half of the form is asked and which half is stored. A **new** dashboard states its measures and their sources; a **change** names the report and what should change about it. Only the chosen branch's answers are stored, so a hand-rolled request that sends both cannot leave the other branch's answer contradicting the one that was asked.
- **The application is the DATA's, not the requester's.** Asked on the form and refused at the endpoint if blank, because it decides which analysts ever see the request. A defect still derives it — a bug happened where the person was.
- **Other is a real application row**, for when the honest answer to "whose data is this?" is "both" or "I do not know".

- **They are private.** Only the person who filed one may see it on the board.
- **They have their own status vocabulary** — nine words, three of them exclusive.
- **They carry an analyst's half:** level of effort, hours, an approval, an assignment trail, and delivery notes.
- **Nothing ships.** No release number, no release notes, no Service Desk send — the Service Desk send **refuses a report request outright**, because it would set **Submitted**, a word outside the nine.

7. Statuses and the lifecycle

One table, scoped per type

`submissions.status_id` points at `defect_enhancement_statuses`. **Sixteen rows**, seeded but editable at runtime, so **nothing may hardcode the full list**.

New · Approved · Redirected · Backlog - Monitoring Impact · Future Consideration · Deferred - Not in Current Scope · Rejected · Duplicate · Submitted · Deployed · Retired · Requires Additional Review · Pending Management Approval · In progress · Delivered · On hold

The last three belong to **report requests alone**. `statusesForRequestType(type, statuses)` in `server/src/constants.js` decides which words each type may hold: a report request gets its nine in *registry* order, every other type gets the table minus those three. Mirrored in `client/src/constants/statusConstants.js` — **change both**.

Why one table and not two. Two options were on the table and neither was taken:

- *A second status column* is two columns for one fact — the same defect the source field list has with Complete / Completed / Complete Date — and something every read and every write would have to branch on.
- *A resolver that picks the table by type* makes `status_id` an id whose MEANING depends on another column. It cannot be joined or foreign-keyed, and two tables would both hold a row called **Retired**.

Six of a report request's nine statuses are already rows in that table and mean the same thing on both types — "Approved" is "Approved". So the table keeps one vocabulary, three rows were added, and a code registry scopes them. **What a requester sees is identical to a separate table. What a maintainer sees is one column, one FK, one join.** A status an admin adds on the Metadata page still reaches defects exactly as it does today.

Enforced on **create, update and backdated history**, offered per type in the queue's inline select, the detail modal's status select, and the Add-a-ticket dialog. The Metadata page marks the three with **Report requests only**, because one list otherwise gives an admin no way to know which dropdown a value appears in.

The public board turns status into position

Two tracks, chosen per type, agreeing on what each POSITION means — so one `STATUS_STAGE` map and one set of pip colours cover both:

defect / enhancement	Reported — Approved — With Service Desk — Deployed
report request	Reported — Approved — In progress ————— Delivered

Rules (`client/src/components/public/StatusBoardRow.jsx`):

- **Position derives from the CURRENT status, never the furthest timestamp.** A redirect resets a ticket to **New** for the receiving team while the sending team's **Approved** timestamp stays in history. A "furthest-timestamp-wins" reading credited the previous team's progress to the new team. **This was a real bug; do not reintroduce it.**
- A later stop being reached implies the earlier ones were. **Approved** is the only stop a ticket can skip.
- **Closed outcomes** (**Rejected**, **Duplicate**, **Retired**, ...) get an outcome pill and a one-line explanation **instead of a track** — drawing a track would be a lie, because nothing further is coming.
- **Parked** statuses (**Backlog**, **Future Consideration**, **Deferred**, **On hold**) say so rather than drawing a stalled track.
- An unrecognised status renders as a neutral "holding" badge, never unstyled.

The stage tiles name **both** vocabularies at the two positions where the words differ — "With Service Desk / In progress" and "Deployed / Delivered" — because a tile counts both types while a row track names only the one that ticket travels.

A retired status must not hide a live ticket

When every selectable status is chosen — the default and reset state — the status whitelist is **dropped entirely** rather than applied. Both the admin queue and the public board do this, for the same reason.

Creation paths

Six, recorded in `submission_sources` and visible as "Created via":

Source	Who	Public by default?
<code>rep_form</code>	A signed-in requester, via /	Yes
<code>admin_manual</code>	Admin, in-queue create	Yes
<code>admin_backdated</code>	Admin, for something reported before the portal existed	Yes
<code>admin_cleanup</code>	Admin, internal cleanup task	No if <code>cleanup_only</code>
<code>admin_excel_import</code>	Bulk historical load	Yes unless cleanup-only
<code>admin_easyvista_resubmission</code>	The fork of a resubmit	inherits

Public by default is the rule; internal cleanup-only tasks stay private. An explicit choice from the caller always wins.

An admin-created report request has no reporter. `reporter_user_id` is written only by the public submit route, so an admin recording somebody else's request cannot claim it for them. Coherent — but it means the Add-a-ticket dialog cannot put a report request in front of the person who asked for it, and such a request is on the admin queue and on nobody's board.

8. Screen by screen

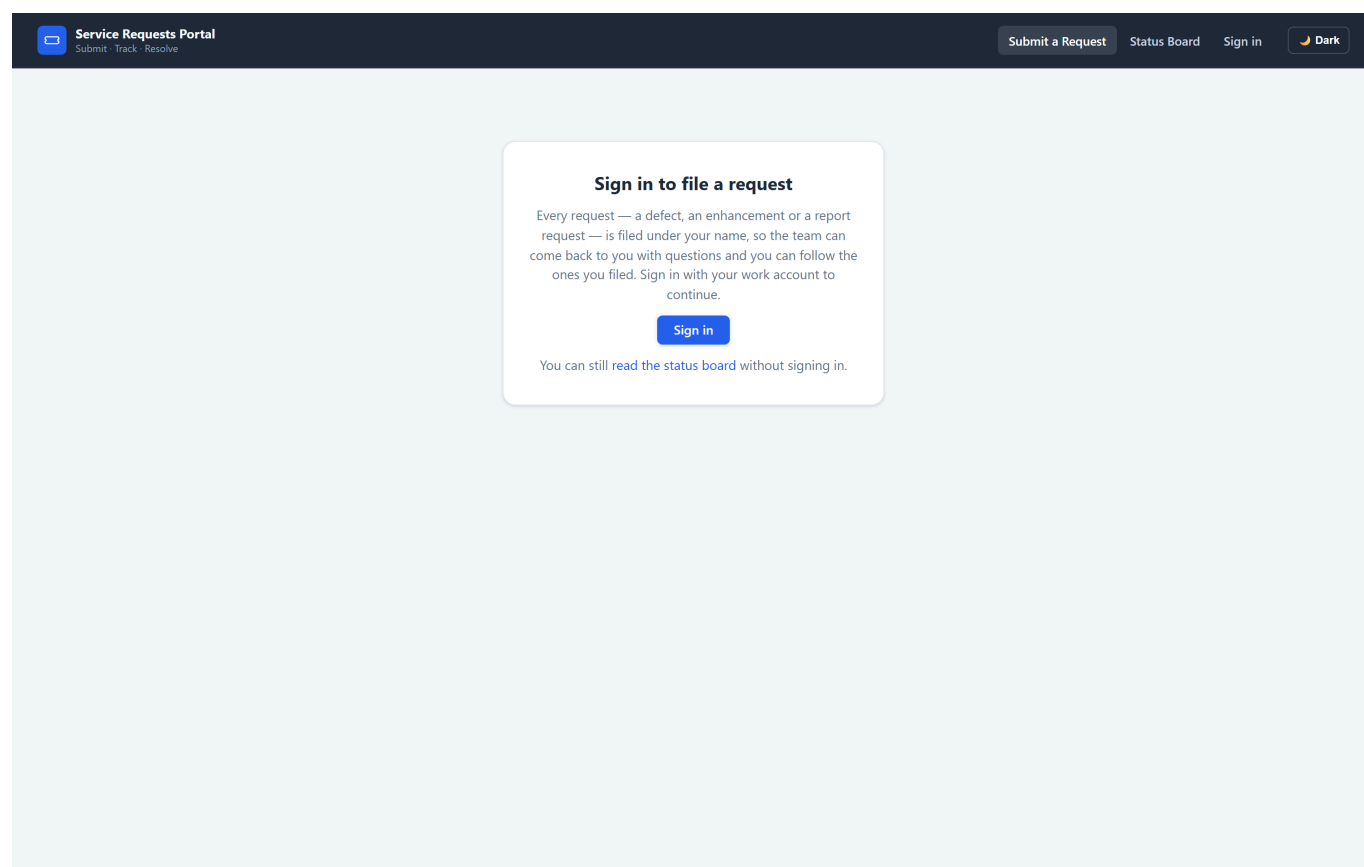
Seven routes (`client/src/App.jsx`):

Route	Page	Gate
<code>/</code>	Submit a service request	signed in
<code>/public</code>	Status Board	none
<code>/admin/login</code>	Sign In	none
<code>/admin</code>	Admin Queue	admin session
<code>/admin/metadata</code>	Manage metadata	admin session (writes: super user)
<code>/admin/throughput</code>	Reporting throughput	admin session
<code>/admin/access</code>	Access Management	admin session + super user

Client-side gates are **signposting only**. Every endpoint re-checks server-side. `RequireSuperUser` waits for the real `GET /api/viewer` answer rather than guessing from the session — it exists so a non-super-user does not land on a page that could only show them errors.

Detailed, screenshot-by-screenshot walkthroughs of each screen are in `USER_MANUAL.md`. What follows is the **design reasoning** a rebuild needs, not a tour.

8.1 Submit a service request (`/`)



Filing requires a signed-in person, for every request type. Not a disabled form — the form is not rendered at all, and the endpoint refuses the POST regardless. See §9 for the history of that switch and why it defaults on in code rather than in an environment variable.

Service Requests Portal
Submit · Track · Resolve

Submit a Request | Status Board | Sign out | Dark

Submit a service request

Defects, enhancements and report requests all go to the same triage queue. You'll get a reference number and can follow it on the Status Board.

YOUR REQUEST

☒ Defect ☐ Enhancement ☐ Report request

Filing as **Bailey Rep** bc.rep@example.invalid

Summarize it in one line * 0 / 140

e.g. Renewal invoice shows the prior term's installment amount

Which application is this about? *

Where you saw it. If it is not in the list, or you are not sure, choose Other — the triage team will route it to the right queue.

Billing Center

Already reported? Write your one-line summary above and we will check it against existing defects and enhancements. Check for duplicates

WHERE IT HAPPENED

Screen title * e.g. Invoice Details

Date it happened * mm/dd/yyyy

Time OPTIONAL --:--

Reference numbers OPTIONAL

Policy number e.g. 40-123456

Account number e.g. 8004521

Transaction number e.g. 90211884

WHAT HAPPENED

Exactly what you saw * 0 characters

What you expected, what appeared instead, and any error message word-for-word.

BEFORE YOU SUBMIT

☐ One-line summary

☐ Screen title

☐ Date it happened

☐ What you saw

☒ Which application

No screenshot yet
Defects with a screenshot are far more likely to be reproduced and fixed.

Submit request

You can press Submit with fields empty — we will point them out first.

WHAT HAPPENS NEXT

- Reported**
Your ticket appears on the Status Board straight away with a reference number.
- Triaged**
The triage team reviews it and either approves it, links it to an existing ticket, or explains why not.
- Submitted to Service Desk**
Approved items go to the Service Desk and get a ticket number you can track.

One column of form, one rail of guidance. Sections: **what are you reporting** → **your request** → (per type) → **do you need a workaround** → **screenshots**.

- **Required fields differ by type.** The client mirrors the server's per-type checks — and says so in a comment naming the exact server lines to keep in step, because anything the server rejects that the client does not catch reaches the requester as a bare 400 instead of an inline prompt.
- **The heading names the service, not an application.** It was `Submit a {application} request`, derived from the viewer, so it read "Submit a Billing Center request" to somebody filing against Policy Center. It is `Submit a service request`. The ticket still records the application and the confirmation still says which one it went to.
- **Who, then what, one field per row, in both branches.** The summary carries the whole request; it used to share a ~814px row with a 250px name box whose 140-character counter was squeezed against the label.
- **Validation appears only after a submit attempt**, and focus moves to the first field that needs attention. The copy under the button is a promise the form keeps: *"You can press Submit with fields empty — we will point them out first."*

The **pre-submit duplicate check** is the highest-value feature on the page and the direct answer to the duplicate-intake problem (`client/src/components/public/DuplicateCheck.jsx`):

- **Deliberately not the `AiSearchPanel` component.** That one is a search *tool* — it owns a query box, a scope and a time window. Here there is nothing to configure: the query *is* the summary already typed, and the window is **all time**, because a defect reported two years ago and since deployed is exactly the answer this requester needs. Both go through the same `api.aiSearch`.
- Disabled until the summary is long enough to be searchable.
- It remembers the exact text that produced the result, so editing the summary afterwards offers a **re-check** rather than silently showing matches for the old wording.

- **It searches only the kind of request being filed.** A report request is only ever a duplicate of another report request — "the unapplied cash dashboard needs a write-off column" has nothing to do with a broken invoice screen. Both directions are **hard filters in the query** (`loadCandidates`), not a post-filter, so a wrong-kind ticket never takes a top-K slot from a right-kind one. A defect and an enhancement stay eligible for each other, because which of the two a sentence describes is a triage decision; the same kind gets a **0.05 preference on the display blend only** (`AI_SEARCH_SAME_TYPE_WEIGHT`, smaller than the 0.15 recency weight), never on `match` — so it settles ties and cannot lift a ticket past the similarity floor.
- The panel **says what it searched**, and the response carries `meta.searchedOnlyType` / `meta.excludedType` so it cannot claim more than it looked at. *A narrowed search reporting "nothing like this" without naming its scope is a bigger claim than it can support.*
- `Op.ne` alone would have dropped every row with a NULL `type_id` — SQL says NULL is neither equal nor unequal to anything, and historical rows exist with no type. **Excluding one kind of ticket must not also exclude the untyped ones.**

Screenshot attachment has three ways in, because reps get screenshots three ways: drag, Browse, and — the common one — **PrintScreen then Ctrl+V**. The paste path has real subtleties (`ScreenshotDropZone.jsx`):

- The paste listener is **window-level**, so a paste lands wherever the requester happens to be — they have just come back from another window.
- Clicking the zone **focuses** it rather than opening the file picker, because Chrome does not dispatch paste events while an `input[type=file]` is focused. The input is a **sibling** of the zone, not a child, so its synthetic click cannot bubble back in.
- A clipboard screenshot usually arrives **with no filename**, which would fail the extension check — so one is derived from its MIME type.
- The client mirrors the server's allow-list, so a file the API would reject is refused here with a reason.
- **The native file input is kept and only its button redrawn.** The native control carries the accept filter, multi-select and drag-and-drop; replacing it means rebuilding all three.

Cap: 3 files, PNG/JPG/GIF/WEBP/BMP/HEIC, 10 MB each.

Who the ticket is from: the server decides, not the form. A signed-in reporter's own name is used and the submitted one **discarded**, so nobody can file under someone else's name. The form stops asking — a name box whose value is thrown away on arrival is worse than no box. See

`server/src/services/reporterService.js`.

Service Requests Portal

Submit · Track · Resolve

Submit a Request

Status Board

Sign out

Dark

Submit a service request

Defects, enhancements and report requests all go to the same triage queue. You'll get a reference number and can follow it on the Status Board.

YOUR REQUEST

Defect

Enhancement

Report request

Filing as **Bailey Rep** bc.rep@example.invalid

Summarize it in one line *

0 / 140

e.g. Weekly unapplied cash by billing centre

Which application is the data from? *

The system the numbers come out of, not the team asking for them. If it spans more than one, or you are not sure, choose Other — it goes to every reporting analyst and they will route it.

Select one

Already reported?

Write your one-line summary above and we will check it against existing report requests.

Check for duplicates

WHAT YOU NEED

Something new

A change to one you already use

Describe what you need *

0 characters

What it should show, who will read it, and what decision it helps them make.

What data does it need?

OPTIONAL

Fields, systems, date ranges — anything you know it has to pull from.

BEFORE YOU SUBMIT

One-line summary

Description

Which application

Measures and sources

Submit request

You can press Submit with fields empty — we will point them out first.

WHAT HAPPENS NEXT

1 Reported

Your request appears on the Status Board straight away with a reference number.

2 Picked up

A reporting analyst on the team takes it and sizes the work.

3 Delivered

You'll see it marked complete on the board, with the date it was finished.

8.2 Status Board (/public)

Service Requests Portal

Submit · Track · Resolve

Submit a Request

Status Board

Sign in

Dark

Status Board

Every issue the team has been told about, and where each one stands. Find yours by number, or check whether something has already been reported.

Ticket number, incident number, policy, account, reporter, or a keyword...

Filters

All applications

Active

Retired

All

AI Ticket Search — describe the problem in plain language and let AI find near matches

WHOLE BOARD

22 tickets

Your filters never change these numbers. Pick one to narrow the list.

22 Everything

2 Reported

5 Approved

4 With Service Desk / In progress

3 Deployed / Delivered

8 9 other outcomes

21 of 22 tickets

All applications · no filters applied

Sort

Last status update

Newest first

Showing 1–21 of 21

Per page: 50

Page 1 of 1

TICKET	TYPE	SUMMARY	STAGE	REPORTED BY	APPLICATION	UPDATED
#275	Defect	Second payment taken in the same minute does not show in unapplied funds	2 Duplicate	Alina Brzezinski	Billing Center	1 day ago
#300	Enhancement	Pre-fill the garaging address from the mailing address	Reported	Rebecca Lindgren	Policy Center	2 days ago
#274	Defect	Payment allocation loses the suspense entry when two payments post in the same minute	Reported	Tomas Ferreira	Billing Center	3 days ago
#370	Defect	Commission statement PDF drops the agency code on multi-page runs	Approved	Bailey Rep	Other	6 days ago
#296	Defect	Policy change wizard loses the additional interest when you step back	Approved	Nathan Okonjo	Policy Center	6 days ago
#272	Defect	Write-off screen rejects amounts over \$1,000 without saying why	Approved	Marcus Delaney	Billing Center	7 days ago
#299	Enhancement	Add a producer code filter to the submission worklist	Approved	Nathan Okonjo	Policy Center	10 days ago
#283	Enhancement	Add a bulk Apply Payment action to the unapplied funds screen	Approved	Dana Whitfield	Billing Center	12 days ago
#301	Enhancement	Decode the VIN to fill vehicle make, model and year automatically	Approved	Sofia Marchetti	Policy Center	14 days ago

"Every issue the team has been told about, and where each one stands." Updates over WebSocket as the team works, with no refresh.

Two count scopes, kept deliberately apart — a pattern repeated on the admin queue and worth preserving:

13 / 92

- **WHOLE BOARD** — the tiles. Badged, and captioned *"Your filters never change these numbers."*
- **The list band** — *10 of 10 tickets*, carrying its own denominator and the filters currently applied.

Two identical-looking rows of numbers meaning different things is a readability trap. Three things keep them apart: a badge naming each scope, a line stating whether filters affect it, and separate visual treatment with the filtered band joined to the table it describes. Tiles are also **toggles** — clicking the one you are on clears back to everything, so a tile is never a one-way trip.

Rows expand in place. A row is one scannable line — ref, type, summary, four track pips, reporter, application, updated. Expanding reveals the description, the dated track, and (when it happened) **"Moved between teams"**. The reference shown is the **incident number when there is one, otherwise #id** — kept in step with the reference the AI search cites, so a reporter can match an AI result to a row without expanding it.

Filters are drawn closed, with a count badge and individually removable chips. Two controls are promoted into the command row: **search** (the most-used) and **retiredFilter** (Active / Retired / All) — because it is a **scope, not a filter**: it changes the meaning of every count on the page.

The chips, the **Filters** badge, the band's summary line and the no-matches state are all derived from **one** `getActivePublicFilters` call — deriving it in more than one place is exactly how those four drift apart. **Stage is not state of its own**: the Stage select and the tiles are two ways of writing `filters.statuses`, the same way column headers and the sort control are two ways of writing `filters.sort`.

All four data-surface states are implemented on both list surfaces (skeleton / empty / error / data). The skeleton is **shaped like the real rows at the real row height**, so nothing jumps when data lands, and it is hidden from assistive tech.

The admin queue's loading state has a history worth repeating: it used to be a **Loading...** line above the **stale** rows, leaving out-of-date tickets on screen presenting themselves as current — *"the worst of the loading failure modes"*.

"My reports" appears only when there is a "mine" to show. A toggle that can only ever return nothing is worse than no toggle. Ownership has two sources and one decider (`useViewer.js`): the server's `is_mine` for a signed-in reporter **always wins**; otherwise, ticket ids this browser remembers filing (`localStorage`). **The browser list is the one acknowledged throwaway in the identity design — delete it once SSO lands.**

Service Requests Portal

Submit · Track · Resolve

Submit a Request

Status Board

Sign out

Dark

Live · updates as the team makes changes

BA Bailey Rep

Report an issue →

Ticket number, incident number, policy, account, reporter, or a keyword...

Filters

All applications

Active

Retired

All

All reports

My reports 10

AI Ticket Search — describe the problem in plain language and let AI find near matches

Open

WHOLE BOARD

31 tickets

Your filters never change these numbers. Pick one to narrow the list.

31 Everything

4 Reported

6 Approved

5 With Service Desk / In progress

6 Deployed / Delivered

10 9 other outcomes

30 of 31 tickets

All applications · no filters applied

Sort

Last status update

Newest first

Showing 1–30 of 30

Per page: 50

Page 1 of 1

TICKET	TYPE	SUMMARY	STAGE	REPORTED BY	APPLICATION	UPDATED
#275	Defect	Second payment taken in the same minute does not show in unapplied funds	2 Duplicate	Alina Brzezinski	Billing Center	1 day ago
#300	Enhancement	Pre-fill the garaging address from the mailing address	Reported	Rebecca Lindgren	Policy Center	2 days ago
#274	Defect	Payment allocation loses the suspense entry when two payments post in the same minute	Reported	Tomas Ferreira	Billing Center	3 days ago
#309	Report	Accounts with a policy in one system and no billing account in the other	Delivered	you	Billing Center	4 days ago

WHAT WAS REPORTED

A reconciliation list of accounts that exist on one side and not the other, so the gaps can be worked through.

Reported by you · Jun 23, 2026 · Billing Center

MOVED BETWEEN TEAMS

Other → Billing Center

WHERE IT STANDS

Reported Jun 23

Approved Jun 28

In progress Jul 20

Delivered Aug 4

REFERENCE NUMBERS

Ticket #309

Only New wears the queue's left stripe. Every status had its own colour, so every row carried one and the stripe distinguished nothing — a wall of colour reads as decoration. The status is already stated in words, in its own column, with its own badge; the stripe now marks the one thing that column cannot say at a glance: *nobody has looked at this yet*. The `row-status--*` class is still written for every row — it is the row's state in the DOM, and what the browser check reads — but only the `new` rule paints.

8.3 Ticket search (both surfaces)

Service Requests Portal

Submit · Track · Resolve

Submit a Request

Status Board

Sign in

Dark

Status Board

Live · updates as the team makes changes

Report an issue →

Ticket number, incident number, policy, account, reporter, or a keyword...

Filters

All applications

Active

Retired

All

AI Ticket Search

Hide AI search

Search

System

Time frame

e.g. customer was double-charged on a renewal invoice — or paste a ticket, incident, or policy number

All systems

Reported: last 24 months

Search

WHOLE BOARD

22 tickets

Your filters never change these numbers. Pick one to narrow the list.

22 Everything

2 Reported

5 Approved

4 With Service Desk / In progress

3 Deployed / Delivered

8 9 other outcomes

21 of 22 tickets

All applications · no filters applied

Sort

Last status update

Newest first

Showing 1–21 of 21

Per page: 50

Page 1 of 1

TICKET	TYPE	SUMMARY	STAGE	REPORTED BY	APPLICATION	UPDATED
#275	Defect	Second payment taken in the same minute does not show in unapplied funds	Duplicate	Alina Brzezinski	Billing Center	1 day ago
#388	Enhancement	Pre-fill the garaging address from the mailing address	Reported	Rebecca Lindgren	Policy Center	2 days ago
#274	Defect	Payment allocation loses the suspense entry when two payments post in the same minute	Reported	Tomas Ferreira	Billing Center	3 days ago
#378	Defect	Commission statement PDF drops the agency code on multi-page runs	Approved	Bailey Rep	Other	6 days ago
#296	Defect	Policy change wizard loses the additional interest when you step back	Approved	Nathan Okonjo	Policy Center	6 days ago
#272	Defect	Write-off screen rejects amounts over \$1,000 without saying why	Approved	Marcus Delaney	Billing Center	7 days ago

One reusable component (`client/src/components/common/AiSearchPanel.jsx`) on both the public board and the admin queue. The time-frame control encodes **both the dimension and the window** — "reported in the last 90 days" vs "resolved in the last 30 days" — because those are different questions.

Results come back in two labelled sections, and this is the important part:

- **Closest matches** — tickets the model endorsed, by relevance tier.
- **Keyword matches** — "Tickets whose ID, incident or Jira number, policy, account, reporter, or text literally contains what you typed — not ranked by the AI."

They answer different questions. Semantic search answers *"has anyone reported this problem before"*. A pasted incident number is a **lookup**, which cosine similarity is structurally bad at. Merging them destroys the distinction between *"the AI thinks this is relevant"* and *"this literally contains the number you pasted"*. A ticket appears in one section, never both.

The panel describes; it does not rule. The prompt already banned "Yes, this has been reported" and the model still produced rulings, because *"do not say X" leaves every other way of saying X available*. The **shape** is prescribed now — sentence one must be the closest match, its ref, what it is about, its status — and the ban is stated as a class. The panel is titled "Closest matches", not "AI summary".

On the admin queue the panel starts **collapsed**, so the ticket table stays above the fold; on the public and requester surfaces — where searching *is* the task — it stays open.

8.4 Admin Queue (`/admin`)

The main working surface. Top to bottom: header and three menus → alert banners → **WHOLE QUEUE** scope strip → command bar → applied filter chips → collapsed ticket search → **FILTERED VIEW** band with impact totals → the table.

Actions live behind three menus, not a flat row: **Add a ticket...** (primary), **Data** (import/export), and the account menu (Manage metadata · Reporting throughput · Manage access · Sign out). This replaced a flat six-button row. Every action is still reachable, one level down. *Manage access* is **hidden** rather than shown-and-refused for non-super-users — and the route and every endpoint behind it check again server-side. Menus close on outside click and Escape and **return focus to the trigger**.

Two count scopes again. **WHOLE QUEUE** is filter-independent and says so; **FILTERED VIEW** describes the rows below and carries the impact totals. One subtlety: cleanup-only items display under a **Cleanup Only** pseudo-status rather than their underlying status, and are counted separately — without that, **total** would not equal the sum of the cards. The "other statuses" card is deliberately **not** a quick filter: no existing filter expresses "cleanup-only" exactly, so it reports the count rather than pretending to filter by it.

Two queues, two column sets. The report queue draws ID · Reported/Updated · Summary · Status · Assigned To · Public, keeps its own saved layout, and calls the status column plain "Status". Both layouts live in the one `columns_json` as `{ default, report }`; a row still holding a bare array reads as the default set with no report set, which is right for anyone who saved a view before this.

A one-click switch between the two kinds of work in the command row: All kinds · Defects & enhancements · Report requests. It writes `filters.types` — the same value the filter panel's multi-select writes — so the segments, the chips and the table cannot disagree, and a hand-picked combination in the panel simply presses no segment. Labels come from the **live type list**, not hardcoded, so renaming the lookup value cannot silently stop matching.

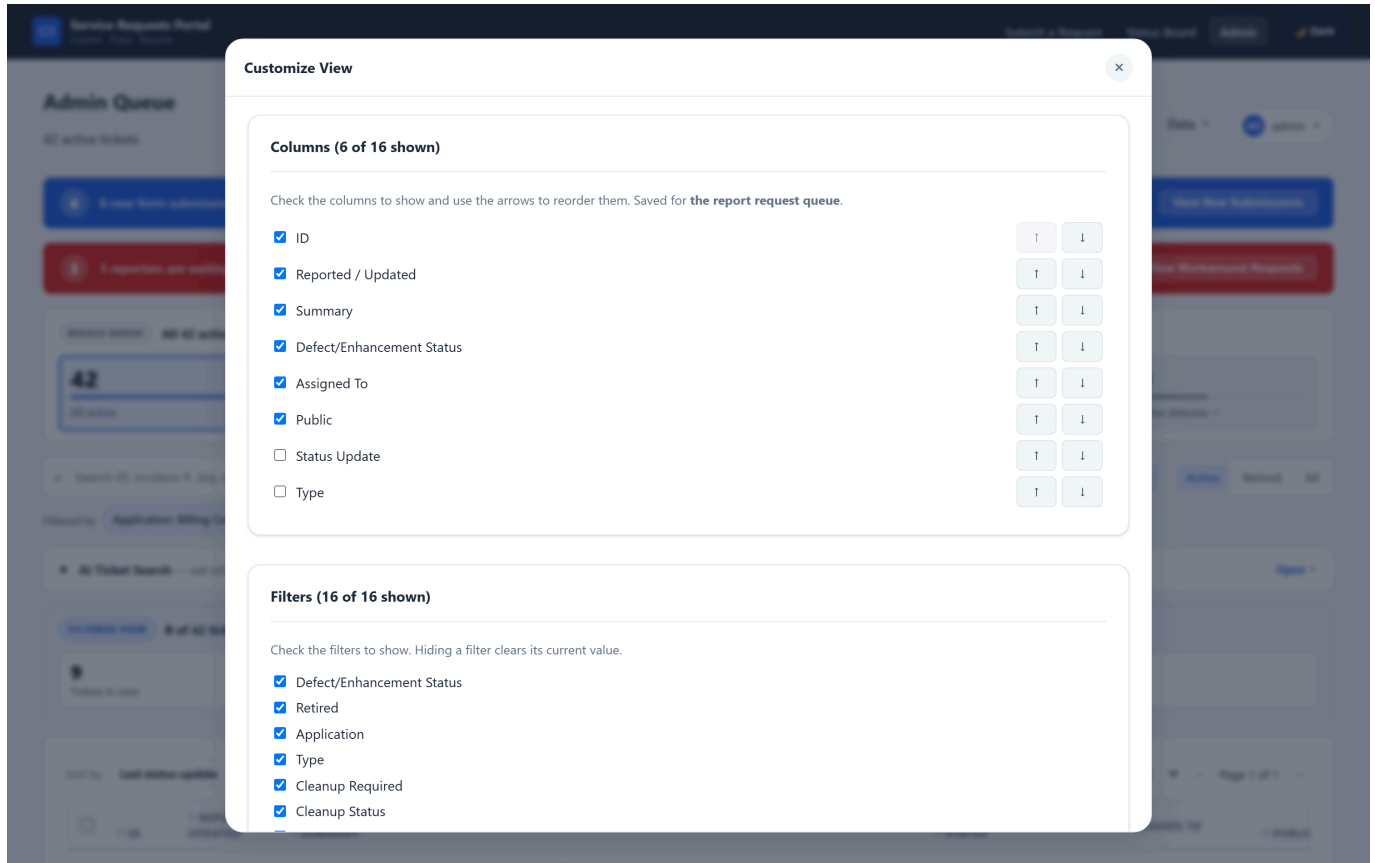
A way back from the new-submissions view. The banner replaces the whole filter set, and for a while the only way back was Clear all, which also threw away the application an admin was scoped to: answering "what's new?" cost them their place. `filtersBeforeJump` remembers what was on screen and offers it back as one chip ("Back to Billing Center"), withdrawn once taken or once Clear all says start from nothing. **Held in state, not localStorage** — "where I just came from" stops meaning anything after a reload.

Filters: a grouped panel, two promoted. Fourteen flat labels forced an admin to read all of them to find one; named groups let them jump. `search` and `retiredFilter` are promoted for the same reasons as on the public board; `application` joins them when the caller can see more than one — for that person it is a **scope**, not a filter. Both promoted controls still honour the admin's visible-filter set, because the page **resets the value of any hidden filter**: rendering a hidden control would show a value being cleared behind the scenes, and leaving a hidden filter's value in place would let it silently constrain the table. The workaround filter has **three** states, not two — a ticket nobody flagged is neither open nor handled.

Application scope and the pin. The scope select is a look; **pinning is a decision**. Separate clicks on purpose. Which queue an admin lands on resolves once per session: the application they **pinned** (an explicit decision, so it always wins) → their **home application** (AD group, else most-filed) → **every** application. Two notes that cost real debugging time: seeding waits for **both** the saved preferences *and* the viewer envelope (waiting only on preferences ran the seed while the envelope was still the anonymous placeholder, spending the one chance to seed on landing at `All`); and a pin on an application since renamed resolves to nothing and falls

through rather than to an empty queue. The select is hidden entirely for someone who administers exactly one application.

Sorting is independent of visible columns. Two paths write the **same** `filters.sort` value. They are decoupled because hiding a column used to hide its sort. Direction wording follows the field's **type** — "Newest first" is meaningless for Summary — and mirrors the comparator the server uses.



Customize View is saved **server-side** per admin so it follows them across devices; `localStorage` is only a cache to avoid a flash. Two rules that matter: the server allow-lists column and filter keys and **drops unknown ones**, so client/server drift fails safe; and sanitisation runs against the **full** registries, not the default visible sets — narrowing to defaults would silently strip an admin's kept columns on every load.

Inline editing. Four fields are editable in the table. Editable cells **remount when the row value changes**, so live updates from other admins actually appear — `defaultValue` only applies on mount.

Service Requests Portal
Submit · Track · Resolve

Submit a Request Status Board Admin Dark

Filtered by: Application: Billing Center × Type: Report × Clear all

AI Ticket Search — ask whether an issue has been reported before, and what happened to it [Open](#)

FILTERED VIEW 9 of 42 tickets — the rows in this table Changes with every filter · Application: Billing Center · Type: Report

9 Tickets in view \$0.00 Policy Premium Impact \$0.00 Direct Dollar Impact 0 Policies Impacted

Sort by: Last status update Newest first Showing 1–9 of 9 Per page: 50 Page 1 of 1

ID	REPORTED / UPDATED	SUMMARY	STATUS	ASSIGNED TO	PUBLIC
☒ #309	6/23/2026 upd 8/4/2026	Accounts with a policy in one system and no billing account in the other • report • Billing Center	Delivered	Billing Center Report Analyst	Yes
☒ #292	8/4/2026 upd 8/4/2026	Payment plan take-up by product • report • Billing Center	New	Unassigned	Yes
☐ #290	7/28/2026 upd 7/31/2026	Daily suspense clearing worklist • report • Billing Center	Approved	Billing Center Report Analyst	Yes
☐ #288	7/7/2026 upd 7/30/2026	Add a write-off column to the Monthly Billing Summary • report • Billing Center	In progress	Billing Center Owner & Analyst	Yes
☐ #294	7/19/2026 upd 7/25/2026	Refunds issued after cancellation, by reason • report • Billing Center	New	Unassigned	Yes
☐ #287	5/25/2026 upd 7/22/2026	Unapplied cash by agency, aged • report • Billing Center	Delivered	Billing Center Report Analyst	Yes

2 tickets selected — across every page of this filtered view

Make public Make private **Retire** Unretire Clear

Bulk actions. The bar is pinned to the viewport bottom so it stays reachable while scrolling a long selection. The dangerous part is the **scope**, and it is stated in words: the master checkbox acts on the **entire filtered set across every page**. Three guards:

1. Changing filters/search **clears** the selection — a selection must never straddle two different filtered sets. Keyed on **filters**, *not* on **rows**, so benign live refreshes leave it intact.
2. Opening the confirm modal takes a **snapshot** of the ids, so a background reload cannot empty the selection between opening and confirming.
3. On apply, the snapshot is **re-intersected with the current rows** — the hard guarantee that a bulk change never touches a ticket outside the current view.

Server side: capped ids per request, and the loop reuses the **per-row** update path so socket emits, status-history logging and embedding refresh match the single-ticket action exactly. A single failing id lands in **failed** and never aborts the batch. The viewer rides along, so a batch cannot reach tickets the same admin would be refused one at a time.

Add a ticket has two modes (New / Historical) across four type segments (Defect · Enhancement · Cleanup · Report request), the report segment carrying its own two sub-branches. The type is one segmented control plus a computed **data-branch**, which is why a fourth type was a segment and a branch rather than a rewrite.

8.5 The detail modal

Where triage happens. **Six tabs**, and the structural rule is: identity, alerts and the action bar live **outside** the tab strip, so nothing needing attention can hide behind an inactive tab — and a tab that *is* hiding something required says so on its label. Tab order follows the ticket's life: *what came in* → *its evidence* → *what has happened to it* → *the internal call* → *the outbound hand-off* last.

The sixth slot is **per type**: **Delivery** on a report request, **Service Desk Submission** on everything else.

Identity band. **Defect** · **New** · **Public**, the summary, then **Billing Center** · **Reported 8/6/2026 by ...** · **Updated 2 days ago**. Collapsible to one line. Badges read from the **edit draft**, not the saved record, so they track the dropdowns live. A **Cleanup Only** task's **type badge is withheld** — showing "Defect" next to a "choose a type" prompt reads as a contradiction. Its collapsed state is deliberately **not** keyed to the open ticket: an admin who wants the compact header wants it on every ticket.

Tab 1 — Report. The form **as it came in**, read-only, because *this is a record, not a working copy*. It reads from **detail**, not the **edit** draft. Values render as text under a rule, never in an input box, so they cannot be mistaken for something typeable. (The rule sits *underneath* the value — a rule between label and value reads as separating the value from the label above, making every value look like a heading.)

A **report request has its own Report layout**: what it is for, who to ask or which report, and what they need. Before that it asked a defect's questions — policy number, screen, time it happened — and showed the summary alone, so **an analyst could open a report request and not read what had been asked for**. The eight requester fields were stored, exported and imported and drawn **nowhere**.

Honest caveat, shown in the UI: *"Not yet sent to the Service Desk. Showing the current saved values."*

True snapshots do not exist. See [Known gaps](#).

Tab 2 — Files. The per-file remove action is a **quiet text button**, not a red danger button — five attachments used to mean five red buttons competing with the one real destructive action in the footer. The grid caps its own height.

Tab 3 — History. The status trail, newest first, in one list with its own scroll boundary, plus provenance, external identifiers and release metadata. Previously the latest event was always expanded with older ones behind a second nested disclosure, so a long-lived ticket buried everything below it.

Tab 4 — Triage. The tab you land on: status, priority, duplicate reference, decision notes, Jira number, public visibility, retire. **Reviewer fills itself in on SAVE**, not on open — prefilling would mark every ticket edited the moment somebody looked at one.

Tab 5 — Impact. One judgement, one section: policy premium impact, direct dollar impact, policies affected, frequency, impact notes. Previously split across four boundaries for a single decision.

A report request's Impact tab is impact notes and nothing else. Dollar impact, policies affected and an occurrence rate are defect measures — a dashboard that does not exist yet affects no policies and recurs no number of times a month. Its SIZE lives on Delivery.

Tab 6a — Service Desk Submission (defects, enhancements, cleanups). The most consequential tab, and it exists because **two things were invisible before it**: re-submitting does not update the existing incident (it forks), and **most of the modal's fields never reach the Service Desk at all**. It shows exactly what will be sent, editable in place, what sending will do, and what is stopping it. §14 has the mechanics.

Release # and Release Notes and the Triage tab's Workaround are **gone from a report request** — nothing ships, and nothing is broken. The **Service Desk number is editable behind an unlock**: an application with no catalog has its tickets raised by hand and the number had nowhere to go, but for every ticket the portal DID send, that number is the server's own record of the hand-off.

Tab 6b — Delivery (report requests). The analyst's half:

The screenshot shows a modal window titled "Submission #309" with a close button (X) in the top right corner. The modal has a header bar with three tabs: "Report", "Delivered", and "Public", with "Public" currently selected. A "Hide details" link is on the right. Below the header, the title "Accounts with a policy in one system and no billing account in the other" is displayed, followed by the text "Billing Center · Reported 6/23/2026 by Bailey Rep · Updated 4 days ago".

The modal is divided into several sections:

- Report Files History Triage Impact Delivery**: A horizontal navigation bar with "Delivery" selected.
- ASSIGNMENT**: Contains a dropdown menu for "Assigned to" (currently "Billing Center Report Analyst") and a "Take it from Billing" button.
- SIZING**: Contains a dropdown menu for "Level of effort" (currently "M — up to a week") and a dropdown menu for "Priority" (currently "3 - Medium").
- DELIVERY NOTES**: Contains a text area for "What was delivered" with the text "Delivered as a monthly reconciliation list rather than a dashboard — it is a worklist and it is meant to shrink to nothing. Matchin is on account number.".
- GO-AHEAD**: Contains a green box with the text "Approved by Yusuf Adeyemi (Controller) on 6/28/2026" and a "DERIVED" status. Below this is a "Clear it" button.
- Who approved it**: A dropdown menu for "Who approved it" (currently "Yusuf Adeyemi (Controller)") and a date field for "On" (currently "6/28/2026").

At the bottom of the modal, there is a "No unsaved changes." message and two buttons: "Save Changes" and "More".

- **Go-ahead** — an approver's typed **name** and a date, plus `approval_recorded_by`, which is an id the server fills in. Both the name and the date are needed for `is_approved`: *a name with no date is half-typed, not an approval*. The approver is a name rather than an id because they are often not a portal user — a manager who replied to an email — so there is no id to hold.
- **What it is approving**: a manager or supervisor okaying the **resources** to build the report. So it is a gate **before** work starts, which is how it is built. It is genuinely a **second gate**, not a restatement of the `Approved` status: triage accepting a request as valid and a manager authorising the spend are different people saying yes to different questions. Keep them separate, and keep the group labelled "Go-ahead".
- **Level of effort** — a lookup, so the Metadata page can rename or retire values.
- **Hours** — one row per sitting, per person, per day worked.
- **The assignment trail** — who has held it and who moved it.

- **Delivery notes** — what was actually delivered, in the analyst's words. The report-request counterpart to `release_notes`, which is deploy language a report request never uses.

Action bar. Pinned. **One primary** (`Save Changes`), the outbound action beside it, and the rarely-used actions behind a `...` `More` overflow. These four used to sit inside the scroll region in one flat row — primary Save eight pixels from a red Retire, with the irreversible hand-off styled more quietly than either. Expanding any section pushed all of them out of view, which is why Save had to be duplicated into the header. Retiring asks first. The overflow menu is a **DOM descendant of the modal, never a portal** — a portal's clicks would land on the backdrop, whose close handler discards staged attachments. `Redirect` is hidden when there is nowhere to send it.

Alerts. Every warning in **one region ordered by severity**, capped and scrollable past two. Previously twelve independent slots stacked ahead of the first field, so a retired + resubmitted + remotely-changed ticket opened on nothing but banners.

Two locks, deliberately not merged:

Lock	Overridable?	Why
Presence — another admin has this ticket open	Yes — "edit anyway"	Advisory. They may have walked away.
Foreign application — the ticket lives in a queue you do not administer	No	The server refuses the write. An override would only produce a 403.

The workaround alert is keyed off the **saved** record, not the draft, so ticking "Handled" does not make the alert vanish before the change is saved — it changes tone instead. `Mark handled writes immediately` rather than staging: *"Mark handled" reads as a verb*, so requiring a second trip to Save Changes meant admins ticked it, closed the ticket, and the request stayed open. It is built from the **saved** record so this one-click action never quietly commits unrelated staged edits.

8.6 Manage metadata (`/admin/metadata`)

Service Requests Portal
Submit · Track · Resolve

Submit a Request
Status Board
Admin
Dark

Metadata

Every dropdown in the portal reads from a list here. Renaming a value renames it everywhere it already appears; switching one off keeps the tickets that use it and only stops it being offered on new ones.

Back to Admin Queue

10

Lists

55

Values across them

8

Values no ticket uses

LISTS

Defect/Enhancement Statuses

16 values

Submission Types

3 values

Cleanup Statuses

4 values

Cleanup Tag Types

3 values

Applications

4 values

Enhancement Request Types

5 values

Priority Levels

4 values

Occurrence Timeframes

5 values

Levels of Effort

1 value

Defect/Enhancement Statuses

16 values

The Status column and Triage tab in the admin queue, the Status filter, and the four-stop track on the public board.

Add a value to defect/enhancement statuses

Add

It goes to the bottom of the list and is offered straight away.

VALUE	IN USE	OFFERED ON NEW TICKETS	ORDER
New	11 tickets	☒ Offered	↑ ↓
Approved	9 tickets	☒ Offered	↑ ↓
Redirected	0 tickets	☒ Offered	↑ ↓
Backlog - Monitoring Impact	1 ticket	☒ Offered	↑ ↓
Future Consideration	1 ticket	☒ Offered	↑ ↓
Deferred - Not in Current Scope	1 ticket	☒ Offered	↑ ↓
Rejected	2 tickets	☒ Offered	↑ ↓

Every dropdown in the app is data: submission types, statuses, cleanup statuses, cleanup tag types, applications, enhancement request types, priority levels, submission sources, occurrence timeframes, **levels of effort**.

- **Deactivate, never delete.** Existing tickets keep pointing at retired lookups.
- `sort_order` drives display order everywhere.
- The `Retired` status is **protected** from deletion and reorder.
- A **retired status must not hide a live ticket** (see §7).
- The page states the **consequence** of an edit before it happens, and locks the name of a value that is in use.

Managing metadata is super-user only — the three WRITES, not the read. Editing a lookup renames or withdraws a value on every ticket that holds it, on the public board and in every export — and it is **not scoped by the per-application grants** the rest of the admin side runs on, so an admin for one application was editing every application's vocabulary.

The READ stays open to every admin, and must. `GET /api/admin/meta/options` is where the queue's filters and the detail modal's dropdowns come from. Narrowing it too would take every other admin's dropdowns away to protect values they can already read off any ticket.

That asymmetry reads like an inconsistency, which is what a later tidy-up would "fix", so `test/metaRouteGuards.test.js` pins it **from the router's own stack** — including a sweep over every mutating `/api/admin/meta` route, so one added later without the guard fails too. All four tests were confirmed to fail against a deliberately weakened route before being kept.

8.7 Access Management (`/admin/access`) — super users only

Service Requests Portal
Submit · Track · Resolve

Submit a Request Status Board Admin Dark

Access

Triage rights are granted here, one application at a time. An account with nothing granted sees an empty queue — never everyone's.

[Back to Admin Queue](#)

4
Applications

11
Accounts

2
See no tickets at all

! bc_rep, pc_rep hold nothing, so their queue is empty. Give them a role below to put tickets in front of them.

Who can triage what

Changes save as you make them and take effect on that person's next request.

	ACCOUNT	SUPER USER	BILLING CENTER 27 tickets	POLICY CENTER 12 tickets	OTHER 4 tickets	MARKETING ANALYTICS 2 tickets	SEES
☐ admin No display name set	☒	Admin of every application, including tickets with none set					• All 45
☐ bc_app_admin Billing Center App Admin	☒	Admin Defects & enhan	No access	Admin Defects & enhan	No access	• Billing Center · Admin · defects & enhancements • Other · Admin · defects & enhancements	
☐ bc_owner_analyst Billing Center Owner & Analyst	☒	Admin Every type	Admin Every type	Admin Every type	Admin Report requests	• Billing Center · Admin • Policy Center · Admin • Other · Admin • Marketing Analytics · Admin · reports only	
☐ bc_rep Bailey Rep	☒	No access	No access	No access	No access	• Nothing	
☐ bc_report_analyst Billing Center	☒	Admin	No access	Admin	Admin	• Billing Center · Admin · reports only • Other · Admin · reports only	

A grid of people × applications, each cell a **role and a scope**. Plus per-application ticket counts, so revoking someone is a decision made **with the size of the queue in view** rather than blind.

Grants carry their type scope end to end. Each cell offers Every type · Defects & enhancements · Report requests only — the three shapes that exist in practice. A combination those cannot express reads as **Mixed** rather than being rounded to the nearest one, which would rewrite it on the next save. Bulk grants clear only the types they name, except an every-type grant, which supersedes the narrower ones. The page also offers **Manager**, which the server always accepted and the page never listed — so the only way to hand it out was a database write.

This page could not make an analyst, and unmade the ones that existed. It read and wrote only (user, application, role), and setUserGrants deleted every row for a user then re-inserted without request_type, which defaults to '' — **every type**. So saving anybody's row promoted a report-only analyst to a full admin on that application, from a dropdown, with no screen that would show it. Two accounts were one save away from it. 12 tests now pin the scope.

Other design notes:

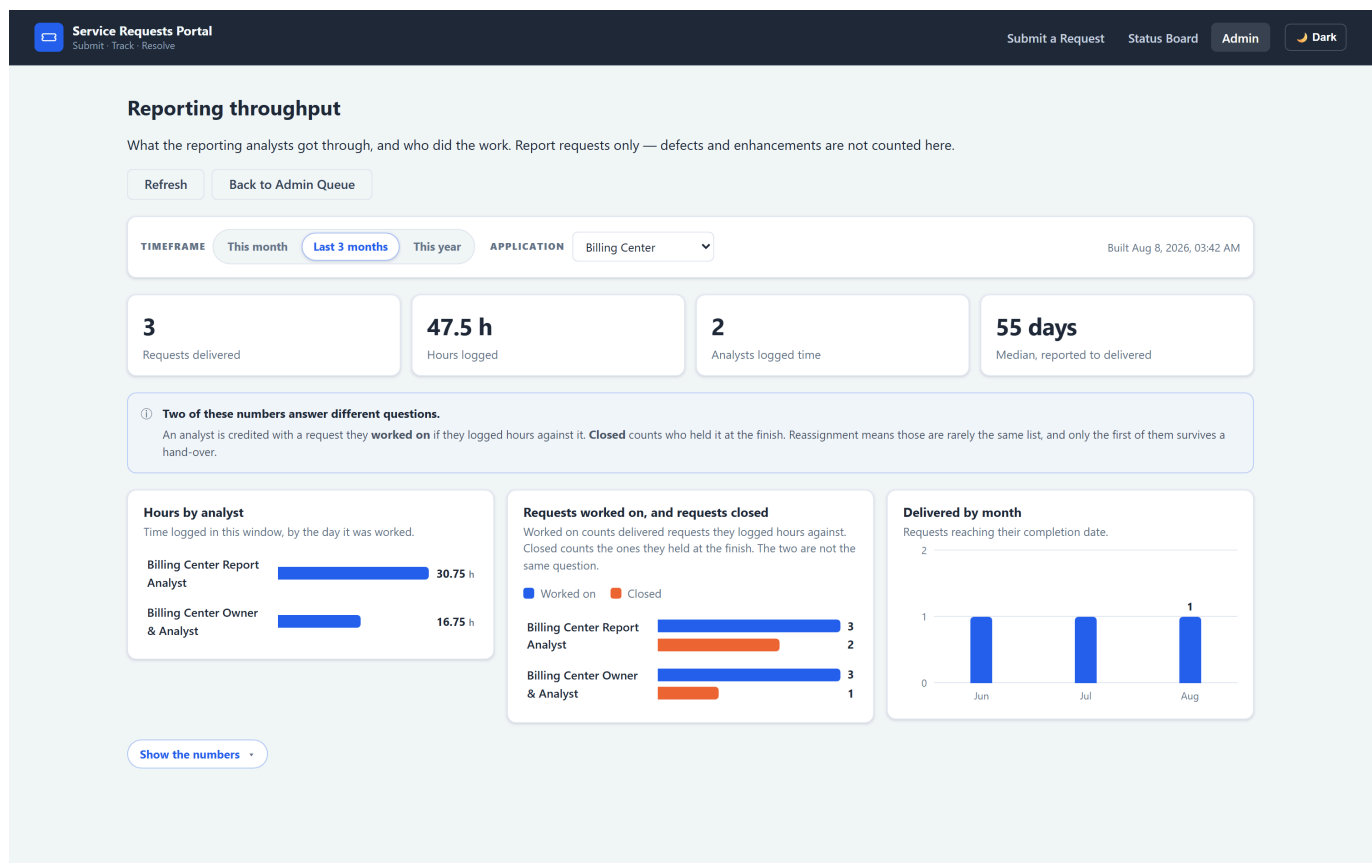
- Cells are **tinted by state** — a wall of untinted dropdowns all reads alike, and finding who is missing access would mean reading each one.
- no access is an **option**, not the absence of one, so every cell answers the same question the same way.
- Rows are sorted deterministically, so a diff of two screenshots means something.
- Editing one person sends their **whole grant set as a replacement**, because the page edits a set — sending the whole set means two super users editing the same person cannot interleave into a state neither chose.

- Bulk grant/revoke is **all-or-nothing, validated in full before anything is written**: a batch naming one bad application changes nobody, *because a partially applied access change is the hardest kind to notice*.
- Each mutation **applies the server response** rather than guessing.
- **The portal refuses to remove its last super user**. Without one, nobody can reach this page to grant anything, and fail-closed scoping means every queue would be empty for everyone — a state no one could undo from inside the app.

It also lists **AD group** → **application** mappings. These set a **default application**, not an entitlement.

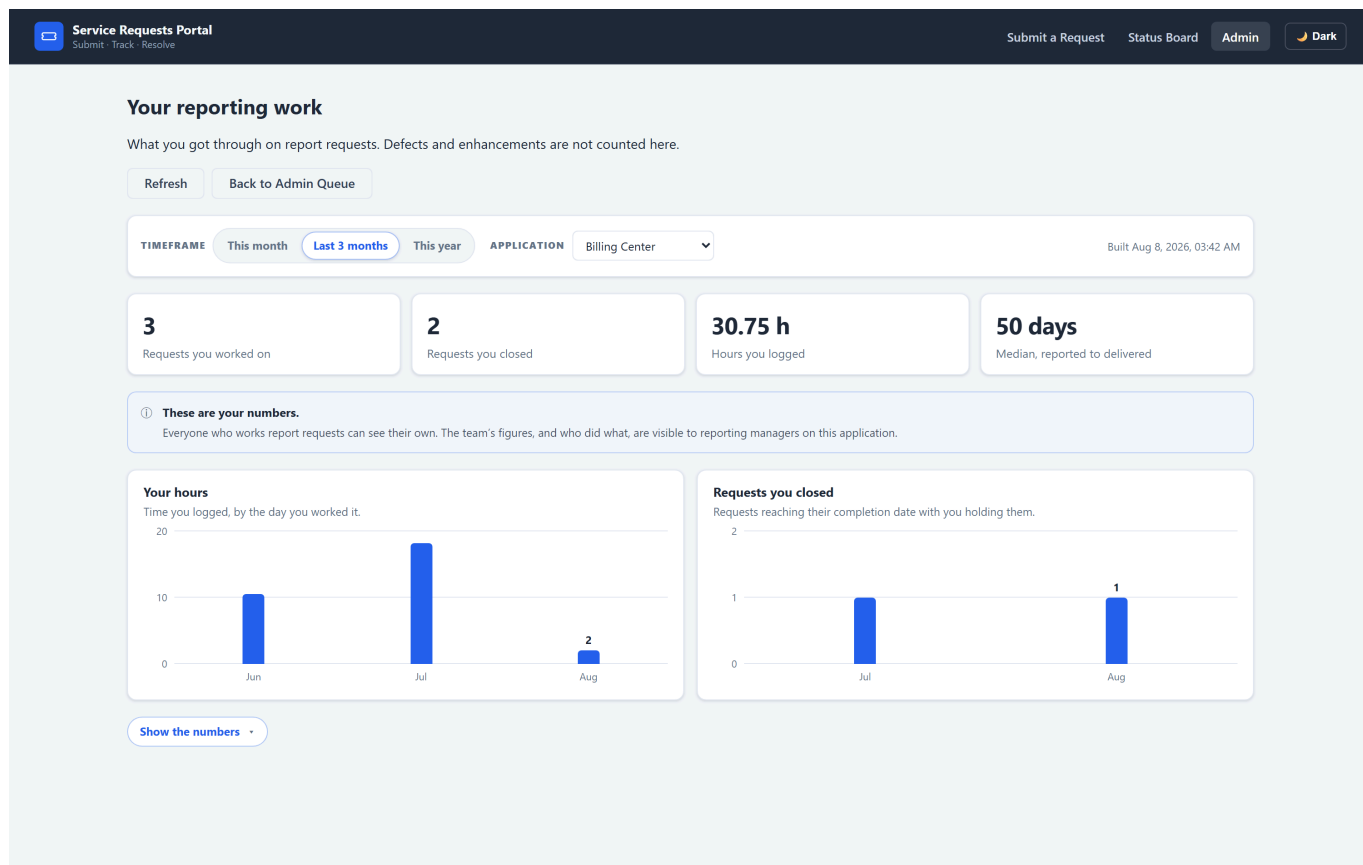
The EasyVista catalogs card was removed from this page. A catalog GUID is an identifier *inside* EasyVista, so the team that runs EasyVista owns the value and nobody using this portal has it. The card asked super users for an answer they were never going to have, and showed three applications as misconfigured when nothing was wrong. **Removing it was not enough on its own** — it was the only way to set a catalog, so deleting it alone would have left two of three applications permanently unable to get one, and the refusal message pointed at the page being deleted. The configuration moved to the environment; see §14.

8.8 Reporting throughput (/admin/throughput)



How much report-request work is being delivered, and by whom.

The server picks the view. `scope` comes back on the response (`'team'` or `'self'`) and the page draws the composition it names. The analyst's view is a **different composition**, not the manager's with rows hidden: four tiles about them, two column charts, and **no .tp-bars at all**, so there is no per-colleague mark anywhere on it. The narrowing is in the **query** (`onlyUserId`), not in the browser — *filtering in the browser would ship the whole team to it and call that privacy*.



"All applications" is offered only to someone who **manages every application they can read** — the same rule the endpoint applies when it chooses between the two answers. Otherwise the page would have to be two shapes at once.

Everything is **computed**; there is no stored total anywhere, so the page can never disagree with the tickets it describes. (Created Month in the source field list was exactly this mistake: a stored month that can drift from its own date.)

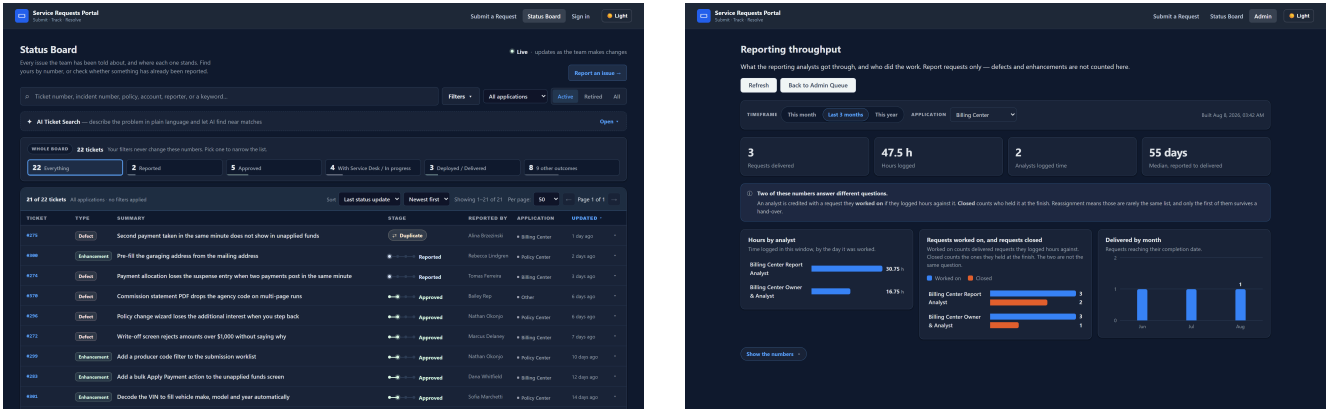
Two subtleties in the numbers:

- **Hours are windowed by the day WORKED**, not the day typed in — an analyst catching up on Friday for Tuesday's work belongs in Tuesday.
- **"Who worked a delivered request" is deliberately NOT windowed**. A request delivered in August was often worked in July, and crediting only the hours inside the window would erase that.

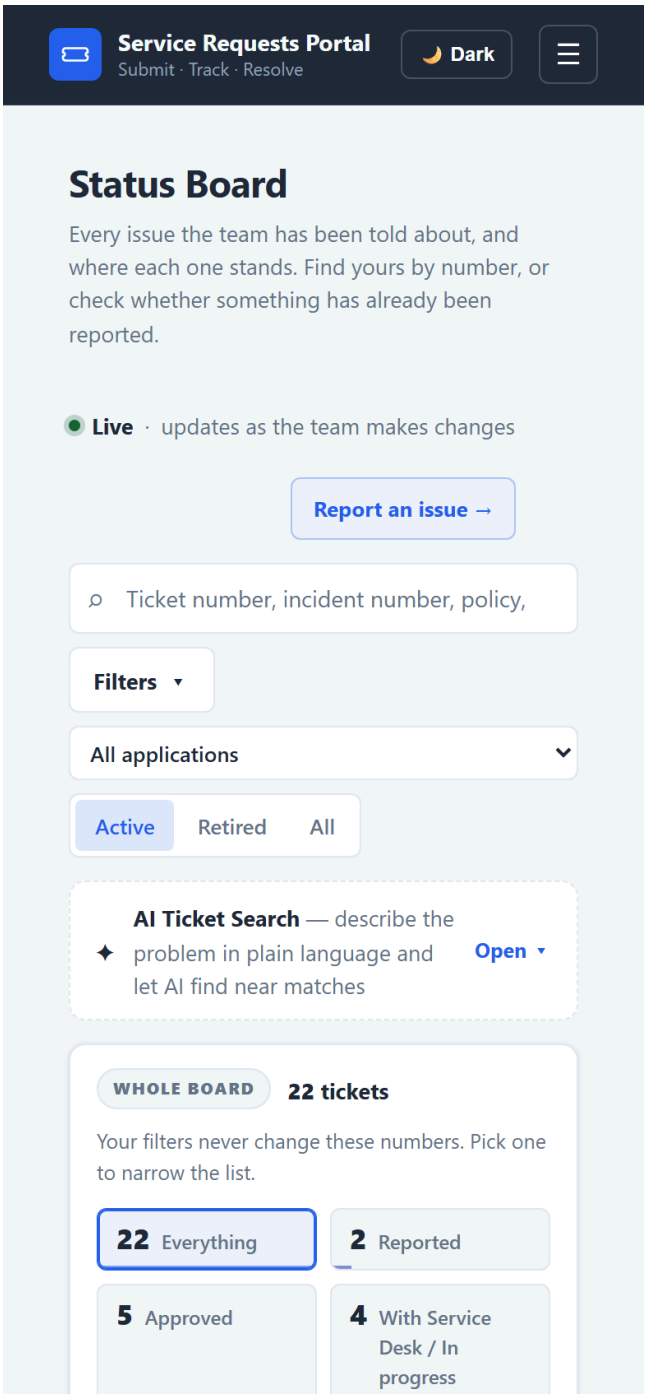
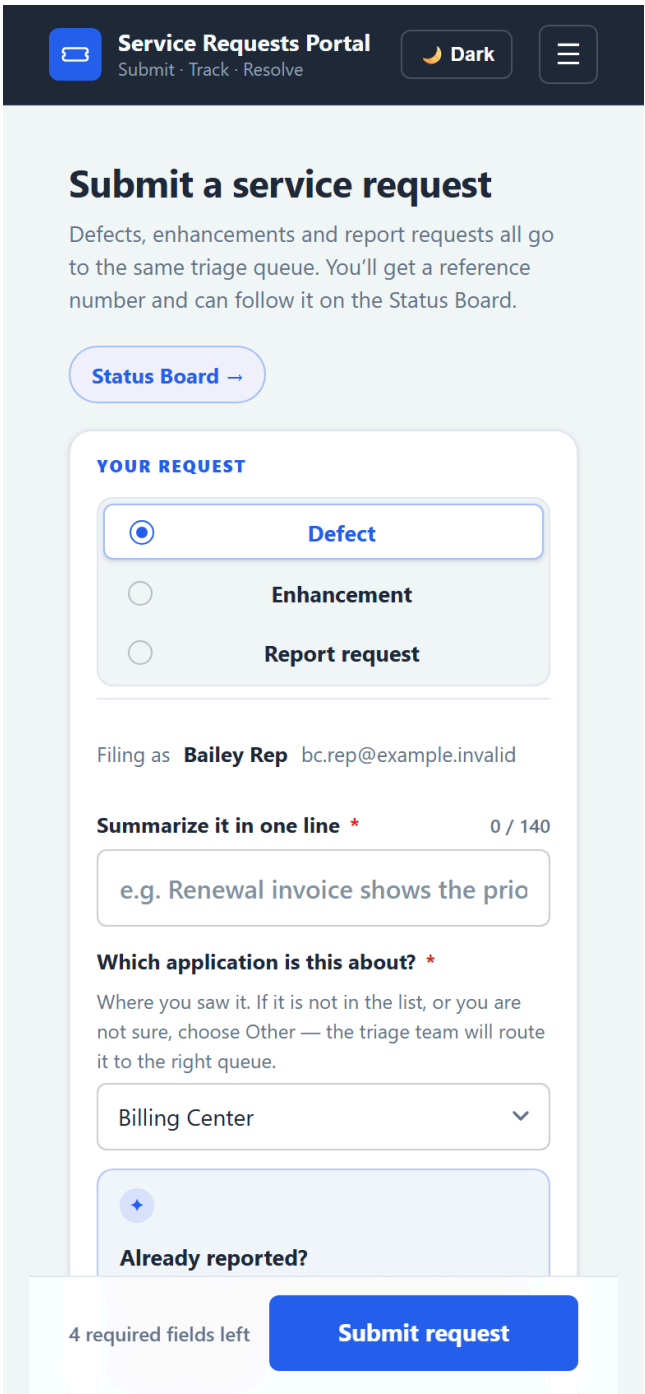
Deliberately **not live-updating**, and it says when it was built.

8.9 Theme and responsiveness

Light and dark, toggled in the header, stored in `localStorage['bc-theme']`, applied as `data-theme` on `<html>`. Initial value follows `prefers-color-scheme`.



Every surface works at 390px.



On narrow screens the submit form hides the rail's copy of the primary button and shows a sticky bar instead. The detail modal's tab strip swaps for a labelled `<select>` carrying the same badges as text — **CSS decides which is visible, so both are always in the DOM and always in step.**

The admin queue at 390px is the weakest surface in the prototype — the table degrades to stacked rows and is usable but cramped. Treat it as a known shortfall to design properly rather than a pattern to copy.

Part III — Mechanisms

9. Identity and the viewer envelope

The single most important architectural idea in this app.

There is **one** server answer to *"who is this caller and what may they see"*, and every surface reads it:

```
GET /api/viewer → {
  authenticated, user, isSuperUser,
  applicationRoles: { [appId]: 'viewer'|'admin'|'manager' },
  applicationGrants: [{ applicationId, role, requestType }],
  memberApplicationIds, readableApplicationIds, homeApplicationId,
  submitRequiresAuth, reportRequiresAuth, impersonating, ...
}
```

No page reads the session, a cookie, or localStorage to decide what a person may see. They all read this, through one hook, `useViewer`.

Deliberately **not** behind `ensureAdmin`: the status board must work for a caller with no session, and this is how the client learns there isn't one. The anonymous envelope carries no user and no rights, so being unauthenticated is a **shape**, not an error. **It always returns 200.**

The SSO seam

`viewerService.resolveSessionIdentity` is the *one function* that reads the session. Today it reads what the local login wrote. Under SSO it reads what the provider asserted — including `groups`, which is **already honoured everywhere downstream**. Group-driven home applications start working the moment the assertion carries them, with no other change. `reporterService` mirrors the same seam for submissions.

This is the pattern to keep. Whatever your identity provider is, funnel it through one resolver and one envelope endpoint.

Who may file

Filing requires a signed-in person, for every request type. `config.SUBMIT_REQUIRES_AUTH` defaults **on**, in code rather than in an environment variable, because a deploy takes the code and not one machine's `.env`. `SUBMIT_REQUIRES_AUTH=false` re-opens the anonymous path for an environment that has to have it.

This flipped on 2026-08-07, and the history matters because the old comment asserted a constraint that no longer holds:

- It used to default to `AUTH_MODE === 'sso'`. The reason was real at the time: SSO was going to be the only way a **rep** could sign in, the local login was admin-only, and arming this while `AUTH_MODE=local` would have left the submit form reachable by nobody and taken the portal's whole purpose offline.
- **That constraint is gone.** `users.role` now has a second value that may sign in — **rep** — so a requester signs in through the local login today, with no identity provider involved.
- The **report** branch had required a session since the eleventh pass for a different and stronger reason (below). There was never an argument for the other two types being different; only the missing login.

`filingRequiresSignIn(requestType, submitRequiresAuth)` in `server/src/constants.js` holds both clauses as one pure function, and the second is the one that is easy to lose:

```
if (submitRequiresAuth) return true;           // covers every type
return requestType === 'report';              // survives the switch being OFF
```

A report request requires a session even if the global switch is turned off, because it follows from the visibility rule rather than from a preference: a report request is visible only to the person who filed it, so an anonymous one belongs to nobody — unfindable by its own requester and hidden from everyone else. Filing it would be writing a row nobody can ever read. `test/submitRequiresAuth.test.js` pins the whole matrix.

The client's copy of `submitRequiresAuth` defaults to `false` so a failed `/api/viewer` never locks the form: the server refuses an unsigned submission on its own, and guessing "locked" would take the form offline over a transient fetch error. When filing is locked and there is no session, the page shows **an honest wall instead of the form** — a form that cannot be submitted is worse than a clear message — with a sign-in link and a note that the status board is still readable.

The gate is on FILING, not on reading. `verify-submit-form.mjs` asserts that the board still renders with no session, because "everything needs a login" is easy to over-implement into locking the board, which carries no private data for a stranger (a report request is already withheld from one).

Two honest failures, told apart

`express-session` mints a fresh empty session for an id it does not know, so `req.session` alone cannot distinguish "never signed in" from "signed in, and it is gone". **The cookie the browser sent can.**

Case	Answer
A dead <code>bc_sid</code> and no typed name	401 <code>sessionExpired: true</code> — "Your session has expired"
No cookie and no name	400 "Requester Name is required" (the right words for a form that IS asking)
Sign-in required for this type	401 <code>authRequired: true</code> — a different way in, not a failure to recover from

Five cases are pinned in `test/reporter.test.js`, including that a CSRF cookie alone is not a session and that a dead cookie **with** a typed name still files anonymously (when the anonymous path is open at all).

This distinction was written for a real report: the owner, signed in as `admin`, saw "Filing as admin" on the form and got "*That did not send — Requester Name is required*" for a field the form deliberately stops showing. The cause was sessions living in MemoryStore and being dropped by a deploy; the **form now recovers** instead of scolding — it says what happened, keeps every answer, and re-reads the viewer, so the name field comes back and the request can be sent without signing in again if that is quicker.

Dev impersonation

Per-application roles cannot be tested before SSO exists without a way to become another user. `POST /api/dev/impersonate` is that way, and it is a **password-free login by design** — so it is gated on **three independent conditions**, and the route is **not even registered** unless all three hold: `AUTH_MODE=local`, `NODE_ENV != production`, `DEV_IMPERSONATION=true`.

Any one being false makes the path **404 rather than exist-and-refuse**, so a single mis-set variable cannot open it. Every handler re-checks the flag anyway, so requiring the module directly from a script cannot bypass the gate. It can only assume an **existing users** row, so it cannot invent rights.

Do not carry this into production. Delete it once SSO lands, or reproduce the three-gate pattern exactly.

10. Access control

The model

Mechanism	Grants	Where
<code>users.role</code>	Whether you can sign in at all, and as what kind of account	<code>admin</code> and <code>rep</code> may sign in; anything else is refused
<code>user_application_roles</code>	Queue rights, per application, per request type	the only source of triage rights
<code>users.is_super_user</code>	Every application's queue, the Access page, the metadata writes	one column, one bypass
<code>application_ad_groups</code>	A default application. No rights whatsoever.	prefill only

The role ladder

Ordered weakest first, and **the order is load-bearing** — a role confers everything the roles before it do, so "at least viewer" is an index comparison rather than a list of exceptions.

- `viewer` — read the application's queue and its tickets; export. Changes nothing.
- `admin` — everything in that application.
- `manager` — everything an admin can do, plus **seeing other people's throughput numbers**. That is the only thing it gates.

Deliberately a **code-level catalog, not a lookup table** — unlike statuses, a role means nothing without the code paths that honour it, so a row someone added by hand could only ever be a role that does nothing. Unknown roles **fail closed**.

Nobody was seeded with manager. It is a privacy decision, and handing it out because an account sounds senior is how one gets made by accident. Add it deliberately if a product owner should see their team's figures rather than only their own.

The type scope

`user_application_roles` is `(user_id, application_id, role, request_type)`.

- An **application admin for defects and enhancements** is **two rows**, one per type — not one row with two values. Cleanups ride along with them.
- An **analyst** is **one row narrowed to report**. That is the whole of what "analysts are admins configured to certain types of requests" means. **There is no fourth role.**
- Somebody who does both is **three rows**. Grants add up; they do not conflict.

`request_type` stores `' '`, not `NULL`, for **"covers every type"**. Both SQLite and Postgres treat `NULL`s in a unique index as distinct from each other, so a nullable column would let the same person hold two conflicting all-types grants on one application and silently lose the guarantee the index exists for. The unique index is `(user_id, application_id, request_type)` and the old two-column one is dropped (`replaces:` in `RAW_UNIQUE_INDEXES` — leaving it in place would keep rejecting the second row, and `CREATE INDEX IF NOT EXISTS` cannot notice that). **Probed with real inserts, not reasoned about.**

That table is synced **without alter** (SQLite would corrupt its composite uniqueness), so a new column on it never arrives via the boot sync. `ensureColumn` in `db/models/index.js` adds it explicitly. **Any future column on that table needs the same treatment.**

`canMutateApplication(viewer, applicationId, requestType)`

The third argument is load-bearing. Omitting it asks the weaker question "may they work in this queue at all", which is right for a queue-level check and **wrong for a write**. All six call sites pass it: `can_edit`, both attachment routes, redirect, create, update. **A new write path MUST pass it too.**

`test/typeScopedAccess.test.js` is the regression net — 13 tests.

The two rules that must survive the rebuild

1. No row is no access. An admin with no grants sees **no tickets**, never all of them — enforced by *construction*, not by a conditional:

```
// viewerService.buildApplicationScopeWhere
// no grants → { application_id: [] } → SQL "IN (NULL)" → matches nothing
// super user → {} → the only bypass
```

"There is no code path where 'no roles' silently becomes 'no filter'."

The service layer reinforces it: `scope` is a **required** parameter, and omitting it returns nothing rather than everything, so a new caller that forgets to resolve one fails closed instead of leaking another team's queue.

2. Access scoping runs first and unconditionally. Everything after it is presentation filtering driven by the query string, and **no query parameter may widen what a caller can see**. Choosing an application you do not hold simply returns nothing.

Read scope vs write scope

	Source	Includes
Read (<code>resolveAdminReadScope</code>)	grants + the routing ledger, both scoped by request type	applications you hold <i>for that type</i> , the soft association, plus tickets your queue handed on
Write (<code>canMutateApplication</code>)	grants only, against the ticket's current application, for its type	<code>admin</code> and above

A ticket outside the caller's read scope reads as **absent (404)**, **not forbidden (403)**, so the queue cannot be walked by id to learn what other teams handle. Legacy tickets with **no** application stay visible to super users only.

Read ignored the request type until 2026-08-08, and that was a leak. `readableApplicationIds` collapses a caller's grants down to the applications they touch, losing the type each grant was narrowed to — and read scope was built from that list alone. A reporting analyst granted `report` on Policy Center could read **every Policy Center defect and enhancement**: in the queue, through the detail endpoint, and in the Excel export.

The owner found it: *"I signed in as pc_report_analyst and should only have access to view report requests on the admin side, and yet I can see all the defects and enhancements."* This document claimed the opposite — §8.4 said "No defect is on this screen ... it is what the server sends" — and that was true only because the queue's kind switch happens to **open on a filter that hides them**. A filter is not a permission.

`resolveAdminReadScope` now carries `typeIdsByApplication` (application → allowed type ids, `null` for an all-types grant) and `canReadSubmissionRow` checks it on all three paths: the application, the soft association, and the hand-off ledger. **By type ID, not name** — the list path scopes RAW rows before hydration, where `row.type` is undefined and only `type_id` exists.

Two deliberate asymmetries: an envelope with **no** per-type detail keeps the old all-types behaviour, so a stale envelope cannot blank somebody's queue; and a row with **no** type is admitted only by an all-types grant, so it fails closed.

Being told is the same permission as being able to see

`emitAdminNotification` broadcast to **every admin**, unscoped — so the same analyst had a banner announcing every new defect and every workaround request. The owner found that one in the same sitting: *"since I don't work those, I shouldn't see anything related to defects, enhancements or cleanups if I don't have that role."*

`resolveAdminAudienceForRow` answers the same question as `canReadSubmissionRow` from the other end — *who may see this row* rather than *may this caller see it* — and reads the same three things, so the two

cannot drift. Each admin socket joins a room of its own (`admin-user:<id>`) and the event goes to the entitled ones.

Two rules worth keeping:

- **null means "cannot tell", and goes to everybody.** An import summary carries no submission; silencing it would break a working notification to close a gap it does not have.
- **An exception fails closed and emits nothing.** `null` is a known shape; a throw means the scoping did not run, and broadcasting then is the leak this exists to close.

Middleware division of labour

- `ensureAdmin` — *whether* the caller is an admin at all.
- `resolveViewer` → `req.viewer` — *what* they administer, resolved **once per request** so every handler in a chain scopes off the same answer. Resolution failure is an error, never an empty-but-successful viewer.
- `ensureSuperUser` — re-reads `is_super_user` **from the users row, not the session**, so a demotion takes effect on the demoted person's very next request, including one already in flight against the Access page.

AD groups grant nothing

Worth stating loudly because it is the natural mistake:

*"An AD group says which applications a person **works in**, not what they may triage. Triage is granted deliberately, by a super user, one application at a time — so nobody acquires the ability to change other teams' tickets by being added to a distribution group."*

`resolveApplicationRoles` deliberately does **not** read AD groups. `resolveMemberApplicationIds` and `resolveHomeApplicationId` do.

Super user is granted out-of-band

By script (`npm run grant:super-user <user> --apply`) or from the Access page — **deliberately not by migration**. If a deploy-time migration promoted every `role='admin'` user, demoting someone from the Access page would be silently undone by the next deploy: the schema would keep overruling an administrator's decision.

11. Public data boundary

Public API responses are **field-allow-listed** through one function: `mapPublicSubmission` (`server/src/helpers/mappers.js`).

Never leaves the server on a public surface:

`created_by_email` · `reviewer` · `decision_notes` · `impact_details` · `impact_notes` · `policy_premium_impact` · `direct_dollar_impact` · `policies_affected_count` · `fingerprint` · `reporter_user_id` · `easyvista_submitted_by` · `delivery_notes` · `approved_by_name` · the report-request requester fields · routing `note`

The allow-list is enforced in **five** places, and all five must hold:

1. Public REST list and detail endpoints.
2. **Socket broadcasts** to `public:update` — watchers include unauthenticated sockets.
3. **Public AI search results** — mapped through the same function.
4. **The text embedded for public semantic search** — a separate public-scope vector built from public-safe text only.
5. **The literal keyword/identifier lookup doc** for public scope.

Points 3–5 are the ones a rebuild will most easily miss. It is not enough to filter the response — if the *embedding* or the *keyword doc* is built from internal text, a public search can surface a ticket **because of** a decision note, which leaks the fact of its content even when the note itself is not returned.

Two fields are attached **after** mapping rather than inside it, because they are facts about the *viewer*, not the row:

- `is_mine` — compared server-side against `reporter_user_id`, returned as a bare boolean. The socket broadcast reaches every watcher at once and so cannot carry it.
- `routings` — the hand-off trail, stripped by `mapPublicRouting`. The reporter sees **that** their ticket moved, when, and between which teams — never the note, which *"is triage talk between admins and can name colleagues or judge their work."*

A ticket that never moved carries **no `routings` key at all**, rather than an empty array.

A report request is private

Signed out, the board listed everybody's. **It names an internal dataset, a department, and often what somebody is trying to measure.** Now: only the person who filed it.

The rule is **ONE function** (`server/src/helpers/reportVisibility.js`) because **four surfaces enforce it** — the board list, the board's by-id route, the public semantic search, and the live socket broadcast. Three agreeing and one not is the shape of a leak, **and that is exactly what happened while building it**: the by-id route went on answering 200 because `getSubmissionByIdWithLookups` hydrates the type as `model_type_name` while the list path writes `type`, so the check read `undefined` and let the row through. **Found by a browser check, not by reading the code.** The rule knows both spellings now.

The socket needed a **THIRD audience**, not a second: `{}` is the whole board and `{ nobody: true }` is nobody. Collapsing them would broadcast a private row to every watcher.

Reading it by its own number must fail the same way a number that does not exist fails — **404, not 403** — or guessing ids confirms which ones are out there.

Consequence worth knowing: the duplicate check on the report branch now only finds the requester's **own** report requests. Two people asking for the same dashboard will not be warned about each other. That follows directly from the rule. A count without content ("2 similar requests exist") would close the gap without breaking it; nobody has asked for it.

12. Real-time, presence and concurrency

Events

Event	Direction	Audience
<code>admin:notification</code>	server → client	admins room
<code>public:update</code>	server → client	everyone — allow-listed fields only , and report requests to their reporter alone
<code>ticket:presence</code>	server → client	admins viewing a ticket
<code>submission:new</code> / <code>submission:updated</code> / <code>submission:redirected</code>	internal emit	fan out to the above
<code>attachment:added</code> / <code>attachment:removed</code>	internal emit	"
<code>ticket:enter</code> / <code>ticket:leave</code> / <code>ticket:activity</code>	client → server	presence

A redirect emits to **both** queues: the ticket leaves one board and appears on the other, and neither admin should have to refresh. The reporter follows their own ticket across the hand-off — **and the note is not part of that payload and must never become part of it.**

New-submission alerts fire only for tickets from the **public form**, not admin-created entries.

The client calls `resetSocket()` after login and logout, because the server assigns rooms and presence handlers **at connect time only** — without a fresh handshake an anonymous socket keeps missing admin events after login, and a signed-out admin keeps receiving them.

Presence — an advisory soft lock

`submissionId` → `Map<socketId, { username, openedAt, lastActivityAt }>`. The **holder** is the earliest opener still connected (`Map` preserves insertion order). State is **in-memory and ephemeral**: it auto-clears on disconnect, which is exactly right for "they closed the tab / walked away".

The client re-announces on **every (re)connect**, because presence is tracked per socket connection and wiped on disconnect — otherwise a network blip silently drops the soft lock while the modal is still open. A throttled activity ping keeps "last active" fresh.

It is **advisory**. The real protection is optimistic concurrency.

Optimistic concurrency

Two independent checks, and you need both:

1. **At save time**, the caller sends the row version (`updated_at`) it loaded. If the row has changed since → **409**.
2. **Inside the UPDATE's WHERE clause**, repeated. Two admins who both passed the read-time check cannot both write: whichever update lands second matches **0 rows** and gets the same 409.

Check 1 alone is a race. Check 2 is what actually makes it safe. The authorisation check runs **before** the conflict check, so an unauthorised caller learns nothing about the row's edit history.

Conflict resolution UI

On a 409 the modal does not just complain. `ConflictReviewPanel` performs a **three-way diff**: the base snapshot taken when the modal opened, the user's draft, and the now-current server version. It lists only fields where the draft differs from current, classifies each as *your change* / *their change* / *both*, and lets the user take the current value or keep theirs.

Live-update handling is nuanced and worth copying:

- A `submission:updated` for the open ticket by **another** admin raises a banner **only if this viewer has unsaved edits to lose**.
- A **pure viewer** (no unsaved edits) gets a silent live refresh and the fresh version adopted as the new edit base. Keeping the stale snapshot would make the form show outdated values, and a follow-up Save would silently revert the other admin's change.
- In-progress drafts are persisted to localStorage (debounced), and persistence is **paused** while a recovered draft is being offered, so it is not wiped before the user decides.

Session persistence

`connect-pg-simple` **against the same Postgres, into a `user_sessions` table**. One dependency, one additive table, no change to any route. Sessions used to live in `express-session`'s default MemoryStore, so **every restart dropped all of them** — including every deploy, which is how a requester came to see "Requester Name is required" for a field the form was not showing.

Conditional, because it has to be. `SESSION_STORE=auto` (the default, and what deploys run) uses the store when the app is already on Postgres and MemoryStore otherwise — local development runs on sql.js, where a Postgres store cannot work. `pg` and `memory` force either one; `memory` is the escape hatch if the store ever misbehaves in production, flippable as an env var without a code change.

`pg` **with no `DATABASE_URL` throws rather than falling back**: falling back there would quietly reinstate the exact bug in the one environment whose operator explicitly asked for the opposite. Nine tests in `test/sessionStore.test.js` pin the fallback as hard as the happy path.

The store announces itself on boot — `[sessionStore] Postgres, table "user_sessions" (DB_PROVIDER=postgres)` or `[sessionStore] MemoryStore (...)`, with the reason. That line is the only honest way to confirm which branch an environment took, short of an endpoint that reports configuration.

The honest expired path is kept, not replaced. It should stop firing on deploys, and that is all: an 8-hour expiry, a pruned row, and a local sql.js box all still reach it.

Its own two-connection pool, sharing Sequelize's SSL treatment via a `normalizeDatabaseUrl` helper exported from `db/sequelize.js` — the Supabase URL's `sslmode=require` makes `pg` verify a chain the pooler's certificate fails. Expired rows are swept every 15 minutes, not the store's default 60 seconds.

The fallback is load-bearing, not a nicety. Anything that touches `middleware/session.js` has to keep the MemoryStore branch working, and `test/sessionStore.test.js` is what says so.

13. AI semantic search

Optional and self-disabling. With no summary key set, `/api/ai-search/status` reports `enabled: false` and every AI surface renders nothing. The app is fully functional without it. Docs: `server/docs/ai-search.md`.

Provider configuration

`AI_PROVIDER` is a **master switch** driving both the summary vendor and the embeddings vendor, so one line picks the whole stack and never a mix:

<code>AI_PROVIDER</code>	Summary	Embeddings	Third parties
<code>openai</code>	OpenAI Chat Completions	OpenAI embeddings	1
<code>anthropic</code>	Claude (<code>@anthropic-ai/sdk</code>)	local , in-process	0

Claude has no embeddings API, hence the split. The `local` provider runs a small open-source model in-process via `transformers.js` — no vendor, no key, no per-call cost, and **ticket text never leaves the server**. First use downloads ~90MB of weights, then runs on CPU. Granular `AI_SUMMARY_PROVIDER` / `EMBEDDINGS_PROVIDER` overrides still win if set.

The pipeline

1. Cheap DB pre-filter → candidates (application, time window, scope, TYPE)
2. Ensure candidate embeddings exist (bounded self-heal)
3. Embed query, rank by cosine
4. LLM ranks + writes a grounded summary over the candidates
5. Return the summary + the REAL hydrated DB rows, in TWO sections

Step 5 is a hard rule: ticket data always comes from the database row, never from the model's text. The model may only reference the tickets it was given — it never invents a ticket, status, or date.

Two grounding guards in `aiSummary.js`:

- OpenAI **strict structured output**, so the model cannot emit a shape outside the schema.
- A **self-consistency check**: an explicit `has_relevant_match === false` forces `matches = []` no matter what the model listed — *it must not affirm tickets it just called irrelevant*.

A provider or parse failure **never breaks the request**. The route still returns the similarity-ranked tickets; the summary is just empty.

Ranking, and a bug worth not repeating

- **Top-K selection is by RAW cosine similarity** (after a similarity floor).
- The **recency-blended score only tiebreaks display order**.

`final = match + weight * recency`, where recency halves every `AI_SEARCH_RECENCY_HALFLIFE_DAYS`.

They are separated because letting the blended score drive *selection* was **evicting the best semantic match from the top-K**. The floor is applied to the raw match too — recency must not rescue an irrelevant ticket past it. The same-type preference (`AI_SEARCH_SAME_TYPE_WEIGHT`, 0.05) rides on the display blend only, for the same reason.

The literal-match safety net

Identifiers are matched **literally, and are deliberately never embedded**. Two reasons:

1. **Identifier strings are semantic noise**. Embedding `I250101_0001` adds no meaning and dilutes the topical signal cosine ranking depends on.
2. **Any change to an embedded doc changes its `content_hash`**, which would re-embed the entire corpus — a CPU pass locally, a billable re-index hosted — and leave search degraded until the backfill finishes.

So there is a third, non-embedded, never-hashed **keyword doc** per ticket. Matching is careful about false positives:

- Query terms: lowercased, punctuation stripped, stopwords and -char terms dropped, plus a trailing-s-trimmed variant so "invoices" hits "invoice".
- Identifier-shaped tokens (anything containing a digit) **skip** the stopwords and length rules — a ticket really can be `#42` — because they match against identifier **fields**, not free text.
- A term may match *inside* an identifier only when distinctive enough not to collide: **5+ characters, or 3+ mixing letters and digits**. So a bare year like `2026` can only match a field that *is* `2026`, never a policy number that happens to contain it.
- The numeric ticket id is **equality-only**, so `42` finds `#42` and not `#1420`.

Literal matching runs over **every** window-surviving row, not just vectorised ones, so a ticket created minutes ago is findable by its incident number before the backfill reaches it.

Storage and scoping

`submission_embeddings`: **one row per (submission, scope)**.

- `admin` — always; embedded from full internal text.
- `public` — only when `is_public = 1`; embedded from **public-safe text only**.

Vectors are stored as a **JSON float array in a TEXT column**, so this works identically on SQLite and Postgres with **no pgvector dependency**.

For a Postgres-only rebuild, use `pgvector`. The JSON-in-TEXT choice exists solely to keep the local SQLite path working, and it does the cosine ranking in application memory with a safety cap on candidate rows.

`content_hash` skips re-embedding when source text is unchanged, so steady-state searches do zero embedding work. Unpublishing a ticket **deletes its public vector** — no stale public embedding may survive.

Public scope fails closed at every step: `is_public = 1` is hard-forced, retrieval uses the public vectors, the LLM cards are built from public-safe fields only, keyword and identifier matching run against public-safe fields

only, the report-visibility rule applies, and every result is mapped through `mapPublicSubmission`. The public endpoint is **rate-limited per IP**.

Backfill: `npm run backfill:embeddings` — idempotent, batched.

14. Service Desk / EasyVista

Where the portal hands off to Tier 2 GTS. **The least finished part of the system**, and the part most needing attention.

One display label. `TRACKER_LABEL = 'Service Desk'` in **two** places that must change together: `client/src/constants/tracker.js` and `server/src/constants.js`. Deliberately a **display name only** — `easyvista_ticket_id`, the `easyvista-preview` route, `EASYVISTA_*` env vars and `server/src/easyvista.js` keep the vendor name, because renaming them is a migration with no user-facing gain.

Two independent switches

Variable	Default	Meaning
<code>EASYVISTA_ENABLED</code>	off	Whether a send actually leaves the app
<code>EASYVISTA_DEMO_MODE</code>	on	Whether an un-wired send is presented as though real

Credentials alone are not enough:

"The payload shape, the endpoint path and the response parsing are all still unconfirmed, so an environment that happens to have a base URL and a key configured must not start transmitting on its own. Turning this on is the conscious act of saying the integration is ready."

Demo mode is only ever consulted when the integration is **not** live, so it can never quiet a warning about a real transmission. It exists so stakeholders can be walked through the flow end to end before EasyVista is switched on.

The repurposed-fields problem

Read this before designing the real integration. EasyVista's Billing Center catalog **does not have fields named after the things we send**. Existing fields are **repurposed** — `E_KCL_CHECK_VOID_REASON` carries the summary, `E_KCL_MKT_AUDIENCE` carries what happened, and so on.

This mapping lives in **exactly one place** (`server/src/helpers/easyVistaPayload.js`), and the admin modal shows both names side by side *"so the repurposing is visible rather than folklore."* Because the repurposed fields are not surfaced anywhere readable in the EasyVista UI, **everything is also rendered into Description as an HTML table**.

⚠ Known issue, EasyVista side: EV **overwrites Description** with its own form-question results, which come through empty. Sending the table is what can be done from here; making it stick is an EV-side fix. **Raise this with the EasyVista team before rebuilding.**

`Urgency_ID` is derived from our priority level's leading digit ("**1 - Urgent**" → **1**), falling back to Medium (3) when a ticket has no priority — the normal case for defects. The requestor EasyVista sees is **the admin who**

pressed send, mapped through `EASYVISTA_ADMIN_MAILS` until `users.email` is populated (and it prefers a real `user.email` the moment one exists). **Field order in the map is the wire format** for the HTML table; reordering it changes every future ticket.

The catalog is per application, and lives in the environment

Which EasyVista catalog a ticket is raised in is **per application, not global**, because the outbound payload's repurposed field names belong to one specific catalog. With a single global catalog, adding an application through Manage Metadata gave it a queue, access and a board lane while its tickets would have posted silently into the first application's catalog. **Absent means NOT CONFIGURED, and the send is refused rather than misrouted.**

The value comes from `EASYVISTA_CATALOG_GUIDS / _CODES` — a catalog per named application in the same `Name:value,Name:value` shape as `EASYVISTA_ADMIN_MAILS`, beside the API key where GTS already works. Resolution order: the application column (nothing writes it now, but a direct fix is still honoured) → the map → the single-catalog form for the one application `EASYVISTA_DEFAULT_APPLICATION` names. **Nothing inherits another application's catalog at any step** — that was the original bug.

An application with no catalog: the button says so, and says what to do

The catalog check used to run **only on the live path**: with `EASYVISTA_ENABLED` off nothing is transmitted, so there was no catalog to land in and an unconfigured application demonstrated a send exactly like a configured one. That made the portal unable to show the case it actually has — an application the Service Desk is not wired up to, where the admin raises the ticket by hand — and it meant **Submit** cheerfully invented an incident number for a hand-off that never happened.

So the refusal now applies on **both** paths, and it surfaces three ways from one call to `easyVistaCatalogStatus`:

Where	What it does
GET <code>/api/admin/submissions/:id</code>	returns <code>easyvista_catalog: { configured, demoOnly, reason }</code> — <code>null</code> for a report request, which has no button
The detail modal footer	disables Submit , and prints the reason on its own line under the actions (<code>.dm-foot-blocked</code>)
POST <code>.../easyvista</code>	refuses with the same <code>reason</code> as a 400

One call, so a greyed-out button and a refused POST can never disagree.

The reason is the whole procedure, not a diagnosis — every step of it already exists, so naming them turns a dead end into an instruction:

"Other is not wired up to Service Desk, so this cannot be sent from the portal. Raise it in Service Desk by hand, then come back, unlock the Service Desk ticket number on this tab and enter the number it gave you, and set the status to Submitted."

The number is editable behind the unlock on that same tab (§8.5), and `Submitted` is in the status dropdown.

DEMO- catalogs, and why the placeholder announces itself

The demo site has to show **both halves**: Billing Center and Policy Center pretend-sending end to end, and **Other** refusing. That needs the two to count as configured — and writing a plausible-looking GUID into their rows would mean that on the day the integration is switched on, a real send posts into a catalog that does not exist.

So the placeholder says what it is, and `easyVistaCatalogStatus` **stops honouring it the moment** `easyVistaIsLive()` **is true**:

Catalog value	Demo path	Live path
a real GUID/code	configured	configured
DEMO-...	configured (<code>demoOnly: true</code>)	refused — "set up with a demonstration catalog rather than a real one"
nothing	refused — raise it by hand	refused

Fail closed, out loud. `npm run set:demo-catalogs` writes them (dry-run by default, `--clear` to undo) and **never overwrites a real value**. Before go-live, clear them and put the real IDs in `EASYVISTA_CATALOG_GUIDS` — an application's own column wins over the environment, so a leftover placeholder would keep winning.

`easyVistaIsLive` and `easyVistaDemoMode` moved to `src/helpers/easyVistaMode.js` so the payload helper could ask the question. `src/easyvista.js` requires the payload helper at its top, so requiring back the other way returned a half-built module and `easyVistaIsLive` came out **undefined** — which reads as `false`, and would have made a demonstration catalog look real on a live server. Failing open, in the one place the guard exists to fail closed. `src/easyvista.js` re-exports both, so every existing importer is untouched.

One payload builder, shared by preview and send

The preview and the real request are built by the **same function**, and the dry-run endpoint **runs the real submit path and returns just before the API call**:

"So the preview cannot disagree with the request."

A preview built from a second, hand-maintained copy of the format drifts silently and the admin trusts it anyway. **This is the single most reusable idea in the integration — keep it.**

POST rather than **GET** for the preview, because it carries the admin's unsaved draft. It writes nothing. A real first-time send ignores the draft (the client saves the row first, then submits); a dry run happens *before* that save, so it merges the draft itself — otherwise the preview would show stale values for exactly the case the admin is checking. The preview is debounced and only runs while the tab is mounted.

Send-as type, and what it refuses

EasyVista accepts a defect **or** an enhancement and nothing else. For an ordinary ticket the choice is pre-filled with its own type. A **Cleanup Only task has no sensible default**, so it must be chosen — which is also how a cleanup task reaches EasyVista at all, without being reclassified first.

Which fields **block** a send follows the **chosen** type, not the ticket's type. A blocked send **pulls the admin to the Service Desk tab**, where the offending fields are editable. The action bar can send outright only when the send is unambiguous; three cases route to the tab first, because each needs a decision made there: a **resubmit** (it forks the record), a **missing required field**, and a **Cleanup Only task** with no type chosen.

A report request is refused outright. It would have set **Submitted**, a word outside its nine — *and a vocabulary enforced on one path and not another is not enforced.*

Attachments

At most **four** per submission — a genuine choice, not a list. Files added on the tab go through the normal attachment upload, so they land on the ticket too; there is no second EasyVista-only pile. **null** selection means "all of them, up to the cap", so an older client keeps working. Selection is filtered against **the ticket's own rows**, which is what stops an id from another submission being attached.

 **This is the one piece still waiting on EasyVista.** Everything deciding *which* files go — the picker, the cap, validation, the confirm dialog — is built and tested. What remains, once the contract is known: the endpoint (same call, or a follow-up against the new ticket id?), multipart vs base64-in-JSON, the field name, whether several go per request, and the per-file size cap.

The attachment send **deliberately never throws**: the ticket already exists by the time it runs, so failing must not turn a successful submission into an error. It warns and reports what it did.

Resubmission

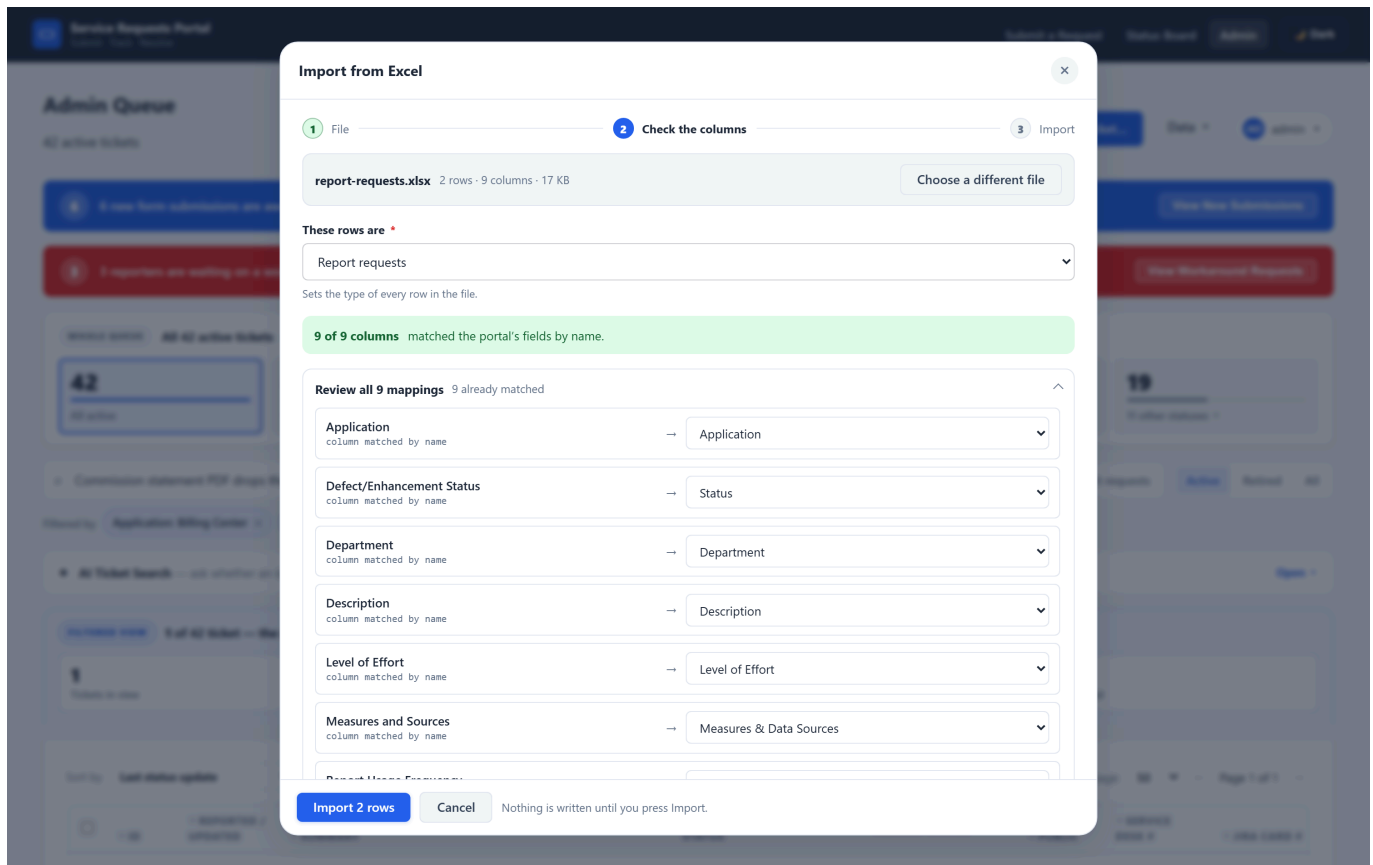
Creates a **new** submission, already set to **Submitted**, and a new EasyVista ticket, copying attachments across. The original is **otherwise untouched**. Bookkeeping columns link them both ways.

Every lookup is resolved and validated **before** inserting, so a missing lookup can never leave an orphaned resubmission row with a null status. Likewise on first send, the type lookup is resolved **before** the outbound call — a missing lookup must not leave an EasyVista ticket created against a record we then failed to tag.

easyvista_application_id snapshots what the incident was raised under at send time. Deliberately **not** derived from **application_id**, because a redirect after the send would then silently rewrite what was transmitted.

15. Excel round-trip and file storage

Import



Three steps and two phases. `POST .../import-xlsx/analyze` parses the workbook, proposes column mappings and reports valid/invalid counts **without writing anything**; the admin confirms; then `POST .../import-xlsx` commits. Every run is recorded in `excel_import_runs` — bulk historical loads are exactly the operation you want an audit trail for.

Note for a rebuild: the client's step 3 sends **no dryRun** flag, so pressing Import writes. The endpoint supports `dryRun=true`; the UI does not use it, because step 2 already reports the counts without writing.

The mode is the type, and it forces every row. `importMode` accepts `defect` | `enhancement` | `cleanup` | `report`. A report sheet could not be imported at all before the fourth entry existed: it came in as defects and its report columns were never read. With it:

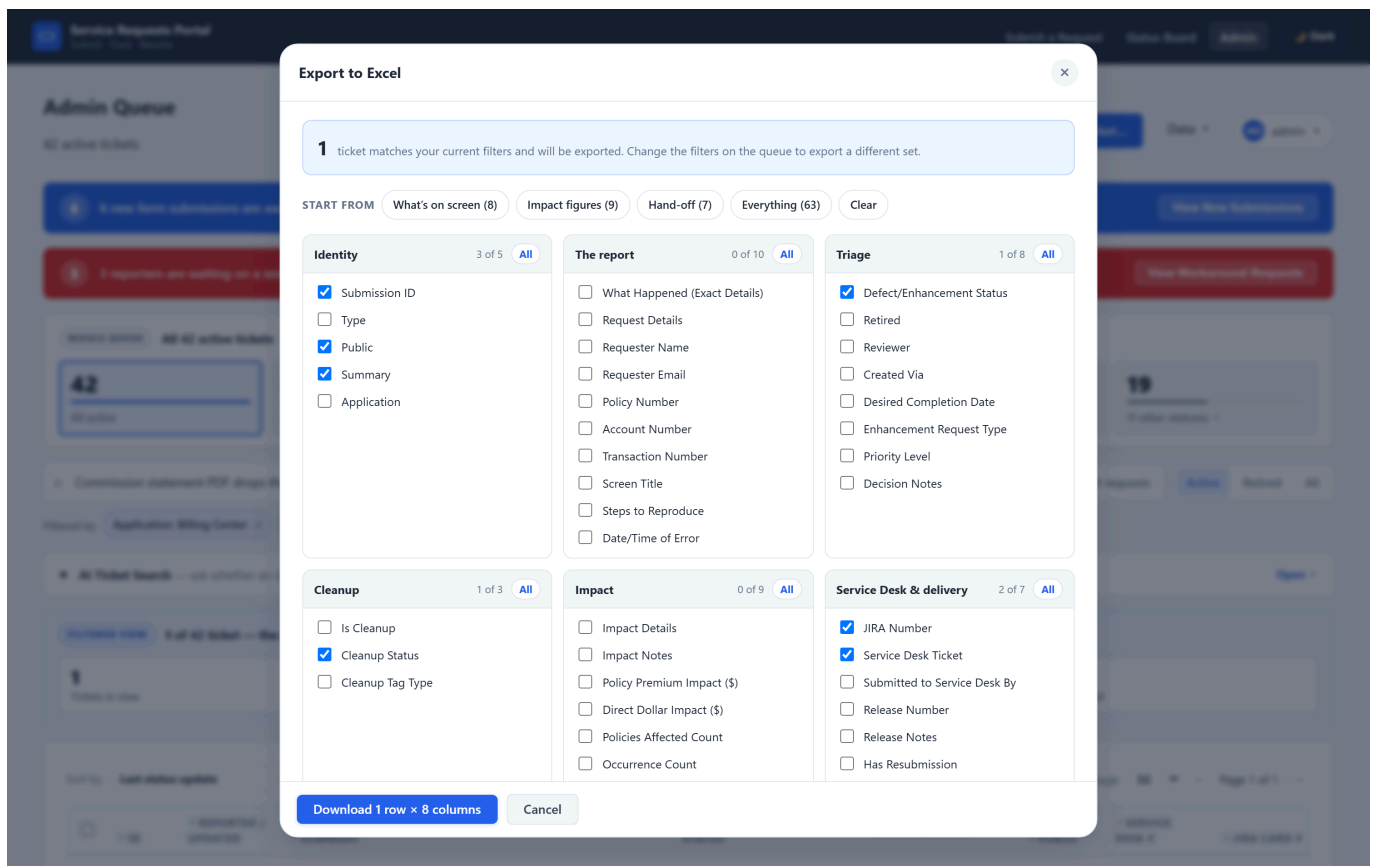
- **A report request is never a cleanup task** whatever a stray Cleanup column says.
- **Statuses are scoped by mode**, on the analyze step as well as the import: a report sheet carrying `Deployed` comes back as an unknown value needing a decision, and the "map it to" list offers the nine rather than the fourteen.
- **The Policy/Account requirement is lifted for a report sheet.** It refused every file with no policy column — which is every report-request file, because nothing about a dashboard involves a policy number.
- **Assigned To arrives as a name and is stored as a user id, or not at all.** `resolveImportedAssignee` matches a display name, username or email, and refuses three ways: unknown, **AMBIGUOUS** (two people, one spelling), or somebody with no grant on the row's application. Each refusal imports the row unassigned and says why. **Never stored as text, never guessed** — `test/importAssignee.test.js` is the net.
- **Duration becomes ONE time entry**, credited to that assignee on the day the request completed, because hours have to belong to a person and a day. With nobody to credit, the number stays out of

the ledger and the row says so: *throughput reporting that invents an owner is worse than throughput reporting with a gap*. One number in one cell cannot be split across people or days; a request two analysts shared has to be corrected on the Delivery pane afterwards.

- **Level of Effort resolves against the offered values**; anything else leaves the request unsized and is reported.
- **A warnings array on the response**, rendered under its own quieter banner. *Rows that landed minus a field are not rows that were skipped, and one list for both would make the difference invisible.*
- **approval_recorded_by is not importable, deliberately**. That column is the id of whoever entered an approval **in this portal**. Nobody did for an imported row, so it stays null rather than borrowing the importer's name; **approved_by_name** and **approved_at** carry what the sheet actually knows.

Visibility mirrors the create path: honour an explicitly mapped **is_public** column, but when unmapped or blank **default to public** — unless the row is a cleanup-only task. Imported tickets are indexed for AI search in the **background, batched and non-blocking** (lookup maps built once, not per row), and it is a no-op when AI search is not configured.

Export



A grouped field picker, then `GET /api/admin/submissions/export-xlsx`.

The export reads through the same access scope as the queue, so what an admin can download is exactly what they can see on screen. This is the kind of thing that quietly becomes a data-leak path in a rebuild if the export takes its own query path. The picker's groups come from the **server's own field list** (`ADMIN_EXPORT_FIELDS`), which is what makes the server-side grouping safe — and an export field with no **group** fails `test/exportFields.test.js`, which is the point.

Hours export as Hours Logged — `SUM(hours)` from one grouped query per list (`sumHoursBySubmission`), never a stored column and never a query per row. Blank rather than 0 when nobody has logged anything.

delivery_notes exports and imports. The first reasoning for leaving it out — "a delivery note is written after the work, and an import loads history from elsewhere" — was exactly backwards: **a backdated migration is the case where delivery notes ALREADY EXIST**, and dropping the column that says what was handed over loses the point of the record. Report-sheet only on import: a defect has no Delivery pane, so mapping it there would store a value nothing ever displays.

The round trip is verified end to end

`client/scripts/verify-report-import-export.mjs` exports a fully-filled report request, **imports the file the portal itself wrote**, and compares every column on the copy against the original. **That is the only test that can catch a header which does not round-trip** — the two that never did ("Reported Date", "Request Details") are why the pattern exists. The same run asserts all three assignee refusal paths and the `Deployed` rejection.

File storage

Attachments are **images only** — PNG, JPG, GIF, WEBP, BMP, HEIC, 10 MB each, 3 per submission from the form. **Extension and MIME type are both checked**, so arbitrary HTML/SVG cannot be stored and later served same-origin from `/uploads`.

Two backends, chosen at runtime (`server/src/helpers/storage.js`):

Condition	Backend	<code>attachments.file_path</code> holds
<code>SUPABASE_URL + SUPABASE_SERVICE_ROLE_KEY + bucket</code> all set	Supabase Storage	the public URL
otherwise	local filesystem, <code>server/uploads/<submissionId>/</code>	a path relative to <code>server/</code>

That relative base is `server/`, **not** the repo root — `helpers/storage.js` computes it from `server/src/helpers`. Getting it wrong makes every local file look already-missing, which is exactly what the purge script did on its first run.

Filenames are sanitised to `[a-zA-Z0-9._-]` and prefixed with a timestamp (plus a random segment on the Supabase path) so two uploads of `screenshot.png` cannot collide. The temp file is removed in a `finally` block whichever backend runs.

`attachments.purpose` is null for a screenshot and `'approval'` for the evidence behind a report request's go-ahead — **one nullable column on the table screenshots already use**, rather than a second attachments table: one upload path, one delete path, one storage helper. An `'approval'` file must **not** be readable from the unauthenticated `/uploads` path; it is served only by `GET /api/admin/attachments/:id/file`.

Two consequences the rebuild must address:

1. **On an ephemeral filesystem, the local backend loses every attachment on restart** — and leaves `attachments` rows pointing at files that no longer exist.
2. **The Supabase bucket is public.** An attachment URL is reachable by anyone who has it — unguessable, but **not behind authorisation**. Since these are screenshots that may contain customer policy and account data, this is the weakest point in the app's data boundary. **Serve attachments through an authorising endpoint, or use signed expiring URLs.** Everything else on the public surface is carefully allow-listed; this route around it was not intended as a design, it is a prototype shortcut.

Deleting an attachment is authorised **against the parent ticket**, not the attachment id — the file carries no application of its own, and a missing parent is refused rather than treated as unowned.

One attachment row can share another's stored object. A resubmission copies the row and keeps the same `file_path`, so anything that deletes files must dedupe by target or it will report a second success for a file already gone.

16. Reporting throughput

The numbers behind `/admin/throughput`, in `server/src/services/deliveryService.js`.

Hours are a child table, and that is the whole design. `request_time_entries` is one row per sitting: (`submission_id`, `user_id`, `hours`, `worked_on`, `note`). Not a column on `submissions`, because hours accumulate **across sittings AND across people** — a single number would be overwritten by whoever saved last, and "who actually did the work" would be unanswerable. `Duration` on a request is `SUM(hours)` computed on read; **it cannot drift from these rows because it IS these rows.**

No unique constraint: many rows per submission is the point, and one person can legitimately log twice on the same day.

`hours` is `DECIMAL(6,2)` for the same reason money is — this project has already had single-precision float silently corrupt stored values, and these numbers get summed.

Three counts that are deliberately different things:

Number	Means
<code>delivered</code>	Report requests whose <code>completed_at</code> falls in the window
<code>hours</code> per analyst	Hours whose <code>worked_on</code> falls in the window
<code>worked</code> per analyst	Delivered-in-window requests they logged any hours against, whenever they logged them

The third is not windowed on purpose: a request delivered in August was often worked in July, and crediting only the hours inside the window would erase that.

`median_days` is the median of `completed_at - created_at` over the delivered set.

`is_complete` and `is_approved` are DERIVED, never stored. The source field list had `Complete`, `Completed` and `Complete Date` — three fields for one fact, which is three chances for them to disagree. There is **one** timestamp (`completed_at`) and one pair (`approved_at` + `approved_by_name`), and the booleans read off

them in `mapSubmission`. Marking a report request `Delivered` fills `completed_at`, because the throughput page counts by that column and the board's own word for the end state must not leave it empty.

The two chart colours (`--chart-1` / `--chart-2`) have one pair per theme and both were run through a contrast validator: light `#2563eb`, `#eb6834` on `#ffffff` and dark `#3b82f6`, `#e2622f` on `#1b2638`, all six checks passing. `client/scripts/lib/chart-scale-probe.mjs` additionally asserts every bar sits where its own axis says — it exists because a chart drew 27 above the line marked 30 while passing every other check.

Part IV — Reference

17. Data model

23 tables (22 Sequelize models plus `user_sessions`, which the session store owns). `submissions` is the aggregate root; everything else is lookups, ledgers, children or access control.

`submissions` — 71 columns

Group	Columns
Identity	<code>id</code> , <code>created_at</code> , <code>updated_at</code> , <code>created_via_id</code>
Reporter	<code>created_by</code> , <code>created_by_email</code> , <code>reporter_user_id</code>
Classification	<code>type_id</code> , <code>application_id</code> , <code>working_application_id</code> , <code>status_id</code> , <code>priority_level_id</code> , <code>enhancement_request_type_id</code>
The report	<code>summary_of_issue</code> , <code>what_happened_exact_details</code> , <code>steps_to_reproduce</code> , <code>request</code> , <code>screen_title</code> , <code>date_time_of_error</code>
References	<code>policy_num</code> , <code>account_num</code> , <code>transaction_num</code> , <code>jira_number</code>
Triage	<code>reviewer</code> , <code>decision_notes</code> , <code>duplicate_reference</code> , <code>duplicate_of</code> , <code>fingerprint</code>
Impact	<code>impact_details</code> , <code>impact_notes</code> , <code>policy_premium_impact</code> , <code>direct_dollar_impact</code> , <code>policies_affected_count</code> , <code>occurrence_count</code> , <code>occurrence_timeframe_count</code> , <code>occurrence_timeframe_id</code> , <code>occurrence_rate</code>
Workaround	<code>needs_workaround</code> , <code>workaround_provided</code>
Cleanup	<code>is_cleanup</code> , <code>cleanup_status_id</code> , <code>cleanup_tag_type_id</code>
Service Desk	<code>easyvista_ticket_id</code> , <code>easyvista_submitted_by</code> , <code>easyvista_application_id</code>
Resubmission	<code>is_resubmission</code> , <code>resubmission_of_submission_id</code> , <code>resubmission_of_easyvista_ticket_id</code> , <code>has_resubmission</code> , <code>latest_resubmission_submission_id</code> , <code>latest_resubmission_easyvista_ticket_id</code>
Flags	<code>is_retired</code> , <code>is_public</code> , <code>logged_defect</code>
Release	<code>release_number</code> , <code>release_notes</code> , <code>desired_completion_date</code>

Group	Columns
Report request — requester	is_new_dashboard, needed_data, measures_and_sources, primary_contact, existing_report_link, changes_requested, report_usage_frequency, department
Report request — analyst	assigned_to, level_of_effort_id, completed_at, approved_at, approved_by_name, approval_recorded_by, delivery_notes

Lookups are FK-only. The eight legacy text columns were dropped (`scripts/dropLegacyTextColumns.js`); rows store only `*_id`, and text names are hydrated at read time by `helpers/lookups.js`. **No redundant text columns** — which means anything reading `row.status` directly gets `undefined`. This caused a real bug: `redirectService` recorded every hand-off as `New` until it resolved from `status_id` instead.

Column choices with reasons worth carrying:

- **reporter_user_id, not created_by, answers "is this mine"**. A rename or a typo would silently unlink someone's whole history, and two people share a name. Null on historic rows and anything filed without a session — *"those tickets belong to nobody, which is the truth rather than a guess."*
- **easyvista_application_id is a snapshot**, not derived, so a later redirect cannot rewrite what was transmitted.
- **working_application_id is the soft association**, and it is nullable, null on every row that predates it, and **only ever set on a ticket in Other**. It widens which queues LIST a ticket and nothing else — never who may edit it, and never read access beyond a queue the caller already holds. Cleared automatically if the ticket is redirected out of `Other`, because the queue that was watching it has its answer now. Indexed (`idx_submissions_working_application_id`): the scope filter reads it in the same `OR` as `application_id` on every admin queue query, so an unindexed column there would put a scan on the admin side's hot path. The rule lives in `server/src/helpers/softAssignment.js` and is unit-tested (`test/softAssignment.test.js`, 14 tests) — the resolver takes `isUnknownQueue` as a **value** rather than looking it up, precisely so a rule that decides whose list a ticket lands on can be pinned without a database.
- **applications.reports_only** marks an application a reporting analyst created. `NOT NULL DEFAULT 0`, so every application that predates it takes every type.
- **assigned_to is a user id, never a name** — a rename must not silently unlink someone's work.
- **approved_by_name IS a name**, because the approver is usually not a portal user. The accountability is `approval_recorded_by`, which is an id the server fills in: *a typed name with nobody behind it is a claim, not a record*.
- **is_new_dashboard stays TRI-STATE** in the mapper. `Boolean()` would turn "not a report request" into "a change to an existing report", which is a different answer, so null survives as null.
- The report-request columns are **plain columns, not a JSON blob and not an EAV table**: the confirmed field list is a `SAMPLE` and will move, so adding a field has to stay a one-line migration plus a form control. An EAV design buys flexibility nobody asked for and makes every read worse.
- **Three of the requester's fields are NOT here because they already have a column**: Title is `summary_of_issue`, Description is `what_happened_exact_details`, "what's not working" is `request`, and Requested Implementation Date is `desired_completion_date`. A second column for any of them would be the same defect the source list has.

`SUBMISSION_INSERT_COLUMNS` is a positional contract

`server/src/helpers/submissionInsert.js` holds the column list shared by the admin create path and the Excel import per-row insert, and **three parallel values arrays are zipped against it**.

The trap, paid in full. `delivery_notes` was first slotted in beside `release_notes`. Shifting the list by one broke the Excel import completely — 0 rows in, "Cannot read properties of undefined". The existing test pinned only the report-request **tail**, which a mid-list insert leaves untouched, so it passed while the import was broken and a **browser** check found it instead. The test now pins the whole list in order. **A new column goes on the END.**

Lookups

All share { `id`, `name`, `sort_order`, `is_active` }. Runtime-editable, deactivated rather than deleted.

`submission_types` · `defect_enhancement_statuses` (+`is_retired`) · `cleanup_statuses` · `cleanup_tag_types` · `applications` (+`easyvista_catalog_guid/_code`) · `enhancement_request_types` · `priority_levels` · `submission_sources` · `occurrence_timeframes` (+`days_equivalent`) · `levels_of_effort`

Ledgers and children

Table	Purpose
<code>submission_status_events</code>	Every status change: <code>submission_id</code> , <code>status</code> , <code>changed_at</code> , <code>changed_by</code> . Append-only. A status change is written as <code>Defect/Enhancement Status: <name></code> for every type; the public board strips that prefix back off (<code>normalizeEventStatus</code>).
<code>submission_routings</code>	The custody chain. One row per hand-off. <code>from_application_id</code> null marks the original filing rather than a hand-off, so the ledger reads as a complete chain instead of starting mid-story. <code>status_at_handoff</code> preserves what it was when it left, because the move resets the live status. <code>note</code> is immutable and internal . No unique constraint — many rows per submission is the point, and a ticket may legitimately come back (A → B → A).
<code>request_time_entries</code>	Analyst hours, one row per sitting. <code>DECIMAL(6,2)</code> . <code>worked_on</code> is the day the work happened, not the day it was typed in.
<code>request_assignments</code>	Who has held a request, and who moved it. <code>assigned_to</code> null records an UNASSIGNMENT — someone taking a request off a person without giving it to another, which is a real event and not the absence of one.
<code>attachments</code>	<code>submission_id</code> , <code>filename</code> , <code>mime_type</code> , <code>file_path</code> , <code>uploaded_by_role</code> , <code>purpose</code>
<code>excel_import_runs</code>	Import audit trail. Not a child of a submission — no FK — and it stays true after the rows it inserted are gone.
<code>submission_embeddings</code>	One row per (<code>submission_id</code> , <code>scope</code>); <code>model</code> , <code>content_hash</code> , <code>vector</code> (JSON in TEXT)

Access, preferences and sessions

Table	Purpose
<code>users</code>	<code>username</code> , <code>password_hash</code> , <code>role</code> , <code>is_super_user</code> , <code>external_id</code> (the IdP's stable key — objectGUID or UPN, so a person keeps their history through a name change), <code>display_name</code> , <code>email</code>
<code>user_application_roles</code>	A grant. (<code>user_id</code> , <code>application_id</code> , <code>role</code> , <code>request_type</code>), <code>granted_at</code> , <code>granted_by</code> — audited. No row is no access.
<code>application_ad_groups</code>	(<code>application_id</code> , <code>group_name</code> , <code>role</code>). Sets a default application. Grants nothing. The stored role is fixed rather than accepted from the caller, so a mapping cannot be created that looks like it grants something.
<code>admin_view_preferences</code>	Per admin: <code>columns_json</code> (now { <code>default</code> , <code>report</code> }), <code>filters_json</code> , <code>pinned_application</code>
<code>user_sessions</code>	Owned by <code>connect-pg-simple</code> . Named constraint and index rather than inheriting the bundled file's <code>session_pkey</code> on a table that is not called <code>session</code> .

A **hard delete** is used for AD-group mappings, deliberately: an unmapped group is *the absence of a default*, not a state worth keeping history of, and inactive rows would make the list read as though the mapping still meant something.

Indexes worth keeping

`submissions` carries indexes on `status_id`, `type_id`, `is_public`, `application_id`, `reporter_user_id`, `assigned_to` and `completed_at`. The last two exist for the throughput page, which groups by assignee and windows by completion date — neither is a column the queue ever filtered on.

18. API surface

All under `/api`. Admin routes require a session; `/api/admin/*` mutations also require the CSRF header.

Auth, identity, health

Method	Path	Notes
POST	<code>/api/auth/login</code>	10 attempts / 15 min / IP, in-memory
POST	<code>/api/auth/logout</code>	
GET	<code>/api/auth/me</code>	
GET	<code>/api/viewer</code>	The envelope. Always 200; anonymous shape when unauthenticated.
GET	<code>/api/realtime/token</code>	Short-lived socket token. 401 for non-admins.
GET	<code>/api/health</code> , <code>/health</code>	

Metadata

Method	Path	Guard
GET	/api/meta/options	public
GET	/api/admin/meta/options	ensureAdmin — stays open to every admin
POST	/api/admin/meta/:category	ensureSuperUser
PUT	/api/admin/meta/:category/:id	ensureSuperUser
POST	/api/admin/meta/:category/reorder	ensureSuperUser

POST /api/admin/applications is **deliberately NOT on this router**. A reporting analyst adds an application by typing its name in, and every write above is super-user-only because editing a lookup renames or withdraws a value on *every ticket that holds it*, across every application, unscoped by the per-application grants the rest of the admin side runs on — and `test/metaRouteGuards.test.js` sweeps this router's stack so a route added here later without the guard fails too.

Creating is not editing: it touches no existing ticket. So it gets its own door with its own narrower rule — CREATE only, and what it creates is always reports-only — rather than a hole in a guard a test polices on purpose. Renaming or retiring an application is still a super user's job on the Metadata page.

Method	Path	Guard
POST	/api/admin/applications	ensureAdmin + a report grant anywhere (<code>canCreateReportApplication</code>)

Body { `name` }; returns { `id`, `name`, `reportsOnly`, `grantedTo` }. It validates the name, refuses a duplicate **case-insensitively including against a switched-off application** (whose row nobody can currently see, so the message says a super user must switch it back on), and creates the row **and its grants in one transaction** — an application is a queue, and a new one with no grants is visible to nobody but a super user, which is the exact failure `Other` exists to avoid.

Submissions — public / requester

Method	Path	Notes
POST	/api/submissions	Filing. Reporter resolved server-side; requires a session .
GET	/api/public/submissions	Allow-listed, <code>is_public</code> gated, report-visibility filtered
GET	/api/public/submissions/:id	" — 404, not 403 , for one you may not see

Submissions — admin

Method	Path	Notes
GET	/api/admin/submissions	Scoped
GET	/api/admin/submissions/:id	Out-of-scope → 404, not 403 . Carries <code>easyvista_catalog: { configured,</code>

Method	Path	Notes
		<code>demoOnly</code> , <code>reason</code> } (null for a report request) so the hand-off button can be disabled before any click
POST	<code>/api/admin/submissions</code>	
PUT	<code>/api/admin/submissions/:id</code>	Optimistic concurrency → 409. Also owns <code>working_application_id</code> — the soft association (§5)
POST	<code>/api/admin/submissions/bulk-visibility</code>	
POST	<code>/api/admin/submissions/bulk-retire</code>	
POST	<code>/api/admin/submissions/:id/redirect</code>	Moves the ticket. Refuses a non-report type into a reports-only application — it is the fifth path that sets <code>application_id</code> , and the one <code>helpers/applicationScope.js</code> originally missed
POST	<code>/api/admin/submissions/:id/easyvista-preview</code>	Dry run; writes nothing
POST	<code>/api/admin/submissions/:id/submit-easyvista</code>	Refuses a report request
POST	<code>/api/admin/submissions/ai-search</code>	
GET	<code>/api/admin/submissions/export-fields</code>	
GET	<code>/api/admin/submissions/export-xlsx</code>	Same scope as the queue
POST	<code>/api/admin/submissions/import-xlsx/analyze</code>	
POST	<code>/api/admin/submissions/import-xlsx</code>	<code>dryRun=true</code> supported
GET	<code>/api/admin/submissions/import-xlsx/history</code>	

Static paths (`bulk-visibility`, `export-xlsx`, ...) are registered **before** the `PUT /:id` param route. Different methods, so they cannot be captured either way, but the ordering is intentional and commented.

Delivery, hours and throughput

Method	Path	Notes
GET	<code>/api/admin/submissions/:id/hours</code>	
POST	<code>/api/admin/submissions/:id/hours</code>	Quarter-hour granularity; a typo of 80 for 8 is refused

Method	Path	Notes
DELETE	/api/admin/hours/:id	
GET	/api/admin/submissions/:id/assignments	The hand-over trail
GET	/api/admin/throughput	from, to, optional application_id. The response's scope says which shape it is.

Attachments, access, preferences, AI

Method	Path	Notes
POST	/api/admin/submissions/:id/attachments	Image-only
GET	/api/admin/attachments/:id/file	The only way to read an approval file
DELETE	/api/admin/attachments/:id	Authorised against the parent ticket, not the attachment id
GET	/api/admin/access	Super user
PUT	/api/admin/access/users/:id/grants	" Whole set replaced, with scopes
POST	/api/admin/access/bulk	" All-or-nothing
PUT	/api/admin/access/users/:id/super-user	" Refuses to remove the last
POST / DELETE	/api/admin/access/ad-groups[:id]	"
GET / PUT / DELETE	/api/admin/view-preferences	PUT replaces the whole row
GET	/api/ai-search/status, /api/admin/ai-search/status	{ enabled, summaryEnabled }
POST	/api/ai-search	Public. Rate-limited per IP.

Dev only — never in production

GET /api/dev/impersonate/users · POST /api/dev/impersonate · POST /api/dev/impersonate/stop — triple-gated; the route is **not registered** unless all three conditions hold.

Error contract

4xx may surface their message; **5xx stay generic in production** so DB and internal details never reach a client.

Security headers and CSRF

- helmet — with contentSecurityPolicy: false (this is an API server; the SPA host owns its own CSP) and crossOriginResourcePolicy: 'cross-origin' so the separate frontend origin can load

`/uploads` images.

- **CSRF: double-submit cookie**, no external dependency. A non-httpOnly `bc_csrf` cookie is issued to every client; state-changing requests to `/api/admin/*` must echo it in `X-CSRF-Token`. The client does this centrally in `lib/api.js`'s shared `request()` helper — **keep it centralised**.
- `/uploads` is served with `X-Content-Type-Options: nosniff`.
- **Two upload configurations, deliberately:** a generic temp upload for trusted, separately-validated files (the admin Excel import), and an **image-only** upload for attachments, *"so that arbitrary (e.g. HTML/SVG) content cannot be stored and later served same-origin from `/uploads`."*
- CORS is an explicit allow-list from `CLIENT_ORIGIN` (comma-separated) with `credentials: true`.

19. Configuration

`server/.env`. **No secret values appear in this document** — names only. See `server/.env.example`.

Core

Variable	Default	Notes
<code>PORT</code>	<code>4000</code>	
<code>NODE_ENV</code>	<code>development</code>	<code>production</code> enables proxy trust, boot self-sync, secure cookies, generic 5xx
<code>CLIENT_ORIGIN</code>	<code>http://localhost:5173</code>	Comma-separated CORS allow-list
<code>SESSION_SECRET</code>	dev default	≥32 chars in production or the server refuses to start
<code>SESSION_COOKIE_SAME_SITE</code>	<code>none</code> (prod) / <code>lax</code> (dev)	<code>none</code> is more permissive than needed — see Part VIII
<code>SESSION_COOKIE_SECURE</code>	<code>true</code> (prod) / <code>false</code> (dev)	
<code>SESSION_COOKIE_DOMAIN</code>	—	For cross-origin cookie setups
<code>SESSION_STORE</code>	<code>auto</code>	<code>auto</code> <code>pg</code> <code>memory</code> . <code>pg</code> without <code>DATABASE_URL</code> throws .

Database

Variable	Default	Notes
<code>DB_MODE</code>	<code>local</code>	<code>local</code> (sql.js file) <code>hosted</code> (Postgres)
<code>DB_PROVIDER</code>	from <code>DB_MODE</code>	<code>sqljs</code> <code>postgres</code> (explicit override)
<code>DATABASE_URL</code>	—	Required when provider is <code>postgres</code> ; the app throws without it
<code>SQLJS_PATH</code> / <code>SQLITE_PATH</code>	<code>./data/dev.sqlite</code>	Local file

Resolution: `DB_PROVIDER || (DB_MODE === 'hosted' ? 'postgres' : 'sqljs')`.

Identity

Variable	Default	Notes
AUTH_MODE	local	local sso
SUBMIT_REQUIRES_AUTH	true	Filing needs a signed-in person. <code>false</code> re-opens the anonymous path — a report request still refuses it.
ADMIN_LOGINS	admin	Comma-separated usernames to seed
SEED_ADMIN_USERNAME / SEED_ADMIN_PASSWORD	admin / admin123	Seeding only. Never real credentials.
DEV_IMPERSONATION	false	Dev only; one of three required gates

File storage

Variable	Default	Notes
SUPABASE_URL	—	All three present → Supabase Storage; otherwise local disk
SUPABASE_SERVICE_ROLE_KEY	—	"
SUPABASE_STORAGE_BUCKET	attachments	The bucket is public

Service Desk / EasyVista

Variable	Default	Notes
EASYVISTA_ENABLED	off	Master switch for real transmission
EASYVISTA_DEMO_MODE	on	Present a stub send as real
EASYVISTA_BASE_URL / EASYVISTA_API_KEY	—	Not sufficient to enable
EASYVISTA_REQUESTS_PATH	/requests	Override the unconfirmed endpoint path without a code change
EASYVISTA_ADMIN_MAILS	—	<code>username:mail,...</code> stopgap until <code>users.email</code> is populated
EASYVISTA_CATALOG_GUIDS / _CODES	—	<code>Name:value,Name:value</code> — a catalog per application
EASYVISTA_DEFAULT_APPLICATION	—	Which application the single-catalog form applies to

AI search — all optional

Variable	Default	Notes
AI_PROVIDER	—	Master switch: <code>openai</code> <code>anthropic</code>
ANTHROPIC_API_KEY / OPENAI_API_KEY / VOYAGE_API_KEY	—	Whichever the provider needs
AI_SUMMARY_PROVIDER / EMBEDDINGS_PROVIDER	from <code>AI_PROVIDER</code>	Granular overrides; win if set
AI_MODEL	<code>claude-haiku-4-5</code>	Anthropic summary model
OPENAI_SUMMARY_MODEL	<code>gpt-4o-mini</code>	
EMBEDDINGS_MODEL	per provider	
AI_SEARCH_ENABLED / AI_SEARCH_PUBLIC_ENABLED	<code>true</code>	
AI_SEARCH_TOP_K	<code>20</code>	Candidates sent to the summary model
AI_SEARCH_MIN_SIMILARITY	<code>0.25</code>	Raw-cosine floor; <code>0</code> disables
AI_SEARCH_RECENCY_WEIGHT	<code>0.15</code>	Display blend only
AI_SEARCH_RECENCY_HALFLIFE_DAYS	<code>180</code>	
AI_SEARCH_SAME_TYPE_WEIGHT	<code>0.05</code>	Display blend only, smaller than recency
AI_SEARCH_MAX_QUERY_LENGTH	<code>500</code>	
AI_SEARCH_MAX_INLINE_EMBED	<code>25</code>	Inline self-heal cap per search
AI_SEARCH_PUBLIC_RATE_LIMIT / _WINDOW_MS	<code>20 / 60000</code>	Per-IP bound on the anonymous surface

Client (build time)

Variable	Default	Notes
VITE_API_BASE	<code>''</code>	REST base; empty means same-origin
VITE_SOCKET_URL	hardcoded API host in prod	Socket target

Part VII — Running and verifying

Running the prototype

```
# Server
cd server && npm install
cp .env.example .env           # then edit – see §19
npm run migrate                 # creates tables + seeds lookups
npm run seed:admin              # creates the admin account(s)
npm run seed:team-accounts -- --apply  # the eight working accounts + grants
npm run seed:other-application -- --apply
npm run backfill:embeddings # optional, only if AI search is configured
npm run dev                    # :4000

# Client
cd client && npm install
npm run dev                    # :5173, proxies /api /uploads /socket.io to :4000
```

 Check `server/.env` before you run anything

The database is selected by `DB_PROVIDER` / `DB_MODE`. With `DB_PROVIDER=postgres` you are connected to the **live shared database**, and every maintenance script targets whatever the environment points at.

To force a sandboxed local run without editing the file — `dotenv` does not override real environment variables, so these win:

DB_MODE=`local` DB_PROVIDER=sqljs DATABASE_URL= npm run dev

`[keepAlive]` Supabase heartbeat OK in the log does **not** mean you are on hosted data. That ping runs regardless of provider, and it has misled people.

Test accounts

All seeded with `SEED_ADMIN_PASSWORD` (see `server/.env`; the prototype's value is `admin123`). **Rotate before anything real runs.** Created by `npm run seed:team-accounts -- --apply`, which is their single source of truth — including their display names, which are deliberately descriptive rather than personal.

Username	<code>users.role</code>	Grants	What it demonstrates
<code>admin</code>	admin	super user	Everything, every application, plus Access and the metadata writes
<code>lead_admin</code>	admin	Billing Center (every type), Other (report)	An admin who is not a super user
<code>ops_admin</code>	admin	Policy Center (every type), Other (report)	The same, on the other application
<code>pc_app_admin</code>	admin	Policy Center: defect, enhancement	An application admin with no report access

Username	<code>users.role</code>	Grants	What it demonstrates
<code>bc_app_admin</code>	admin	Billing Center: defect, enhancement	"
<code>pc_report_analyst</code>	admin	Policy Center: report	An analyst — the type-scoped grant, and no defects on screen
<code>bc_report_analyst</code>	admin	Billing Center: report, Other: report	"
<code>pc_owner_analyst</code>	admin	Policy Center: every type; BC + Other: report	Somebody who does both
<code>bc_owner_analyst</code>	admin	Billing Center + Policy Center: every type; Other: report	"
<code>pc_rep</code>	rep	none	A requester. Refused by <code>ensureAdmin</code> , no Admin link, sees only what they filed.
<code>bc_rep</code>	rep	none	" — and the owner of most of the seeded report requests

`is_super_user` is 0 on all ten non-admin accounts. **Nobody holds manager**, so every one of them sees the *personal* throughput view; `admin` sees the team view.

The demonstration data

`server/scripts/seedRealisticSubmissions.js` writes **39 requests** designed to make every surface draw something real: all four kinds of work, three applications, 15 of the 16 statuses as a live value, both report-request branches, 24 hours entries by four analysts across three months, nine approvals, a duplicate pointing at its original, a retired ticket, and one report request routed **Other** → **Billing Center** with its ledger row.

```
node scripts/purgeSubmissions.js           # dry run – the whole plan
node scripts/purgeSubmissions.js --apply --confirm=<n> # n must match the live
count
node scripts/seedRealisticSubmissions.js    # dry run
node scripts/seedRealisticSubmissions.js --apply # refuses a non-empty table
```

Two guards worth copying: `--apply` on the purge **also** requires `--confirm=<count>` matching the live table, so *the plan you read is the plan that executes*; and the seed **refuses a non-empty table** unless `--allow-existing`, because seeding twice would double every ticket indistinguishably.

No attachments are seeded, deliberately: a file needs bytes, and neither a local box nor the deployed instance can be given a file the other can read. A row pointing at bytes that are not there is worse than an empty Files tab.

Verification gates


```

cd server && npm test           # node:test – 378 tests
cd client && npm run lint        # ESLint incl. react-compiler rules – must stay green
cd client && npm run build       # production build

```

The browser harness

`client/scripts/` holds **eight verification scripts** and two shared probes. They are the record of what has been checked by eye and by measurement, and they run against the real app. All except `verify-session-store.mjs` (which brings its own servers) and `verify-docs-pdf.mjs` (which reads built files, not the app) need the server on :4000 and Vite on :5173 already running, and all take an optional `--shots <dir>`.

Script	What it proves	Writes?
<code>verify-admin-data-entry.mjs</code>	The three data-entry dialogs, the redirect dialog, and the report-request Delivery pane	One report request, removed
<code>verify-metadata-page.mjs</code>	The metadata page. Makes ONE reversible write and proves it undid it	Yes, restored
<code>verify-submit-form.mjs</code>	The form's field/height/control counts, the duplicate check's type narrowing, and the session-less wall	One report request, removed
<code>verify-throughput-page.mjs</code>	Both views, in two real sessions (<code>admin</code> is a manager, <code>lead_admin</code> is not)	Two delivered requests + hours, removed
<code>verify-public-board.mjs</code>	The per-type track, the parked state, the type chip, the stage tiles, and that a report request is invisible to a stranger	One public report request, removed
<code>verify-report-import-export.mjs</code>	The spreadsheet round trip: export, re-import the portal's own file, compare every column, and all three refusal paths	Four report requests, removed
<code>verify-session-store.mjs</code>	Sessions outlive a restart. Brings its own servers on :4100 — spawn, sign in, kill, spawn again, present the same cookie	Session rows only
<code>verify-docs-pdf.mjs</code>	Every link in the three PDFs is still a link, resolves to a real page, and lands on the right section of the right file. Needs no server	No
<code>lib/overflow-probe.mjs</code>	Per-container overflow. Read its header before changing it — each exclusion is there because a false positive buried a real finding.	
<code>lib/chart-scale-probe.mjs</code>	Does every bar sit where its own axis says? It exists because a chart drew 27 above the line marked 30 while passing every other check.	

The control is the half that makes `verify-session-store` mean anything. The same sequence runs again with `SESSION_STORE=memory`, where the session **MUST** be lost. Without it, a pass only shows the cookie was accepted, not that Postgres is what accepted it.

A script that writes ends by proving it put the data back, printing the submission count from `server/scripts/removeVerificationSubmissions.js`. That script refuses any id whose summary does not begin with `VERIFY` — **there is no submission DELETE endpoint, on purpose**, so this is the only way a fixture leaves. Ticket ids advance permanently; **the COUNT returning to where it started is the invariant, not the ids.**

The screenshot harness

`client/scripts/capture-screenshots.mjs` — **62 shots**, four sessions, both themes, two viewports, one command.

```
node scripts/capture-screenshots.mjs          # all of them + the manifest
node scripts/capture-screenshots.mjs --list    # names only, no browser
node scripts/capture-screenshots.mjs --only submit # substring filter
node scripts/capture-screenshots.mjs --no-import # skip the one shot that writes
```

The manifest is an OUTPUT, not an input. `handoff/screenshot-manifest.json` is written from the script's `SHOTS` registry at the end of a full run, so it cannot list a shot the script could not take. A manifest read as *input* can, and then the two disagree silently in the direction that makes the documentation wrong.

Design points a rebuild's own harness should keep:

- **Sign in once per account and reuse `storageState`** across every (theme, viewport) context. `/api/auth/login` allows 10 attempts per 15 minutes per IP; the first version logged in six times, so two runs back to back started answering 429. A 429 mid-run looks exactly like a broken check, so the script says so explicitly.
- **`document.fonts.ready`, all images complete, two animation frames** before every shutter. Without it a shot can catch the fallback font, and every width in the picture is a width the app never renders.
- **Every shot is size-checked.** A screenshot call that "worked" and wrote 0 bytes is what produces a manifest describing pictures nobody can open.
- **A failed shot photographs its own failure state** to `_failed-<name>.png`. A timed-out selector tells you what was *not* there; the picture tells you what **was**, which is the half that says whether the probe or the product is wrong.
- **Stale files are pruned** on a fully-green run, and each deletion is named. The 41 orphans left by hand-shooting are what made this harness necessary.
- **It writes exactly once** — two rows through the Excel import, because the client's step 3 is a real import — and removes them through the same `removeVerificationSubmissions.js`, printing the count.

The PDF build

The three documents also ship as PDFs in `docs/pdf/`, **with every link still a link** — the contents list, every cross-reference, and every jump between documents work in a PDF reader exactly as they do in the

markdown. That is the whole point of the build: a 123-page reference nobody can navigate is a worse artefact than the markdown it came from.

```
cd client
node scripts/build-docs-pdf.mjs           # docs/pdf/*.pdf
node scripts/build-docs-pdf.mjs --keep-html # also leave the intermediate HTML
node scripts/verify-docs-pdf.mjs          # assert the produced bytes
```

Markdown → HTML (**marked**) → PDF (Playwright Chromium print). No separate PDF library: Chromium already turns `<a href="#slug">` into a real PDF named destination and `<a href="OTHER.pdf">` into a file link, and it renders the screenshots the manual is mostly made of. `--keep-html` exists only so `verify-docs-pdf.mjs` can run its printable-width overflow check; the HTML is gitignored, and the PDFs are the output.

Decisions worth keeping in a rebuild:

- **Heading slugs must match GitHub's algorithm character for character.** The docs were written against GitHub's anchors, so `githubSlug()` reproduces its lowercase-strip-punctuation-hyphenate-and-disambiguate rules. Anything close but not identical silently breaks the contents list.
- **anchors are validated before anything renders.** A link to a heading that does not exist fails the build. Discovering it as a dead link in a PDF a reader already has is the failure mode this replaces — and it caught one on the first run, a numbered paragraph with no heading to target, fixed with an explicit `<a id>`.
- **Cross-document links need `#nameddest=`, not `#anchor`.** A bare fragment opens the sibling PDF at page 1; `nameddest` is what Acrobat and the Chrome viewer actually honour, and it is the difference between "the link works" and "the link lands on the section".
- **`doc.getDestinations()` returns `{}` for Chromium-produced PDFs.** It stores them in a name tree `pdf.js`'s bulk enumerator does not walk, while per-name `getDestination(name)` resolves fine. The first verifier reported all 66 links broken because of it — **a verifier that fails wholesale is suspect before the thing it measures is**, so it now also proves the links land across 43 *distinct* pages, which an all-page-1 fallback could not fake.
- **overflow-wrap: break-word on table cells, not anywhere.** Both wrap a word too long for its line, but `anywhere` also counts mid-word breaks when the browser computes a column's min-content width — so auto table layout shrank the vocabulary table's first column until ordinary words split (`Submit` / `ion`). `break-word` leaves min-content alone. Long identifiers still break, because `code` keeps `anywhere` and needs it.
- **A part divider opens a page; its first section shares that page.** `h1` and `h2` both break before, then `h1 + h2` cancels it. Without the cancel, either the part title is stranded at the foot of the contents list or it gets a near-empty page of its own.

The four traps this harness was built around

Each cost real time. They are properties of *any* browser harness, not of this one.

1. **Do not pipe a verify script through `head`.** SIGPIPE kills the run before its cleanup and leaves fixtures in the shared database. **Redirect to a file and read the file.**

2. **`/api/auth/login` allows 10 attempts per 15 minutes per IP.** A burst of re-runs starts answering 429, and a 429 mid-run looks exactly like a broken check. Restarting the server resets the limiter (it is in-memory per process, which is why `verify-session-store.mjs` — every boot a fresh process — is exempt).
3. **Read the submission count from the removal script, never from a number written in a document.** It moves.
4. **A browser probe is wrong more often than the code is.** Ten checks have now failed against working code, and every one was the same mistake in different clothes:

The probe did	The truth
Slept a fixed number of ms after changing a filter	It read the previous filter's rows
Waited for "a <code>/api/admin/submissions</code> response"	Resolved on the request already in flight for the previous filter
Used a <code>.dm-pane</code> class	It does not exist
Read <code>.submission</code> on a detail response	The response is flat
Used <code>\s</code> as a word boundary against a sortable header	The caret is glued to the label (<code>▼Status</code>)
Matched <code>button:has-text("Save")</code>	The real button is "Save Changes", and it renders in the header and the footer
Pressed Escape and slept to close a modal	<code>.bs-modal-backdrop</code> stayed and swallowed every later click, surfacing hundreds of lines away as an unrelated failure
Read <code>process.env.DB_PROVIDER</code> from its own process	A script in <code>client/</code> sees none of <code>server/.env</code>
Waited for <code>.xl-maps</code> to be visible	It is inside a collapsed <code><details></code> , and <code>waitForSelector</code> defaults to <code>visible</code>
Matched <code>#(\d+)</code> on a table row's <code>textContent</code>	Cells run together, so <code>#222 + 8/6/2026</code> captured 2228

Click and wait for the RESPONSE whose URL carries the expected query — or better, for the rendered DOM. Prove a modal closed. Read the real DOM before writing the selector.

And one lesson worth carrying beyond the harness: the SIGNED-IN branch is not the branch a requester sees. Two of the owner's five submit-form complaints were only ever visible to a viewer with no session — which was, at the time, everybody who filed a request. `verify-submit-form.mjs` opens a second, session-less context for exactly that reason, and still does now that the branch it checks is a wall.

Scripts

Every destructive maintenance script is **dry-run by default** and reports what it would do.

Command	Purpose
<code>npm run migrate</code>	Sync schema + seed lookups
<code>npm run seed:admin</code>	Create admin users
<code>npm run seed:team-accounts</code>	The eight working accounts and their scoped grants
<code>npm run seed:other-application</code>	The Other application, granted from the grants that exist
<code>npm run seed:realistic</code>	The 39-request demonstration set
<code>npm run seed:unwired-work</code>	The three things that set could not show: a defect and an enhancement in Other (one with a hand-typed incident number), an analyst-created reports-only application with two report requests, and a soft association. Refuses to seed twice. Nothing it adds is Delivered and no hours are logged — that is the modelling rule <code>verify-throughput-page.mjs</code> depends on.
<code>npm run purge:submissions</code>	Every submission and its children. Needs <code>--confirm=<count></code> .
<code>npm run remove:verification-tickets</code>	Remove VERIFY -prefixed fixtures by id
<code>npm run remove:verification-applications</code>	Remove VERIFY -prefixed applications by name, with their grants. Refuses any application a submission still points at — a row with tickets in it is not a fixture, and destroying it would orphan them. Exists because <code>POST /api/admin/applications</code> has no DELETE to match, and the browser check that proves that control works has to use it.
<code>npm run backfill:embeddings</code>	Index existing tickets for AI search; idempotent
<code>npm run backfill:public-visibility</code>	Make existing non-cleanup tickets public; persists the exact id set so it can be reverted precisely
<code>npm run backfill:tracker-history</code>	Relabel historical status events
<code>npm run migrate:money-columns</code>	<code>REAL</code> → <code>DECIMAL(14,2)</code> . No-ops on SQLite.
<code>npm run migrate:report-requests</code>	The report-request schema

Command	Purpose
<code>npm run migrate:report-statuses</code>	The three report-only statuses
<code>npm run migrate:session-store</code>	The <code>user_sessions</code> table
<code>npm run migrate:delivery-notes</code>	<code>submissions.delivery_notes</code>
<code>npm run migrate:easyvista-catalog-columns</code>	The per-application catalog columns
<code>npm run migrate:reports-only-applications</code>	<code>applications.reports_only</code> . The script existed with no npm entry for a pass , while three code comments told you to run it by that exact name.
<code>npm run migrate:working-application</code>	<code>submissions.working_application_id</code> + its index — the soft association
<code>npm run set:demo-catalogs</code>	Give the walkthrough applications a DEMO- catalog so they keep pretending, leaving Other with none. <code>--clear</code> undoes it. Never overwrites a real value.
<code>npm run grant:super-user</code>	Grant/revoke super user

dotenv resolves `server/.env` relative to the CWD, so run these **from `server/`**. Running one from the repo root silently targets the local `sql.js` file and then reports a confident wrong count. Every script prints its dialect first for exactly this reason — **read that line before believing the rest.**

`node -e "..."` does **not** load `.env` at all. Use `node -r dotenv/config -e` for any throwaway check that touches the database.

Part VIII — Deployment and the rebuild decisions

The internal deployment will be on company servers and a company database — not Vercel, Render or Supabase. Read this part as *"here is what the prototype's hosting forced, and here is what you can therefore delete"*, not as a topology to reproduce.

Vite stays — it is the build tool, not a host. `npm run build` still produces `client/dist/`; you serve that from your own web server.

Two things are **not yet decided** and are called out as explicit decisions below: the [database engine](#) and the [hosting shape](#).

How the prototype is deployed today — context, not the target

Component	Host	Notes
Client (React SPA)	Vercel	Static build, SPA fallback rewrite
Server (Express API)	Render	Node web service, no build step
Database	Supabase PostgreSQL	<code>DATABASE_URL</code> , SSL required
File storage	Supabase Storage (public bucket)	Falls back to local disk when unset
EasyVista	External	Off unless explicitly enabled

`client/vercel.json` rewrites `/api/(.*)` to the API host and `/(.*)` to `/index.html`. **There is no `/socket.io` rewrite, and its absence is load-bearing** — see §3. **The API host is hardcoded in two places:** `client/vercel.json` and `client/src/lib/socket.js`'s production fallback. In a rebuild, drive both from configuration.

`NODE_ENV=production` turns on four behaviours at once, which is worth knowing because production and local differ in more than logging:

1. `trust proxy 1` — TLS terminates upstream.
2. **Schema self-sync on boot** — `sync({ alter: true })` plus `findOrCreate` lookup seeds, so a deploy that adds a column needs no manual migrate step. Non-fatal: the server starts anyway and logs the failure. Deliberately gated to production so a local run — which may be pointed at the live DB — never auto-alters it.
3. Generic 5xx bodies.
4. `secure: true`, `sameSite: none` session cookies.

Pushing to `main` deploys, and the deploy migrates the shared database. Production boots with `sync({ alter: true })` against the same database a local `npm run dev` uses, so **a model change lands on live data on push**. Write the explicit migration script anyway (see `scripts/migrateEasyVistaCatalogColumns.js`) so the change is reviewable and re-runnable elsewhere.

The server **refuses to start** in production if `SESSION_SECRET` is the development default or shorter than 32 characters. That fail-closed check is worth keeping.

What you can delete on your own infrastructure

Four things exist **only** because of the prototype's hosting. None is a product decision.

1. The direct WebSocket connection and the whole realtime-token mechanism. Vercel rewrites cannot carry WebSocket upgrades, so a same-origin socket degraded to perpetual HTTP long-polling, "*billing a flood of requests*." The client therefore connects **straight to the API host**, which broke cookie auth, which required a **short-lived HMAC-signed token** from a same-origin endpoint.

Delete `server/src/helpers/realtimeToken.js`, `GET /api/realtime/token`, and the token fetch in `client/src/lib/socket.js`. On your own proxy, carry WebSocket upgrades, keep the socket same-origin, and authenticate it with the session cookie the way everything else is authenticated. **This removes an entire auth path and its signing secret.**

2. **sameSite: 'none'** → **lax**. Same-origin serving makes **none** unnecessary.

3. **keepAlive.js** — a daily **SELECT 1** so a free-tier project is not paused after ~7 days idle. Meaningless on your own database. (Its log line appears **regardless of database provider** and does *not* mean you are connected to Supabase.)

4. **The Supabase Storage backend, and the public-bucket problem with it.** On your own infrastructure this becomes a file share or object store **behind an authorising endpoint**, which is what it should have been.

Also delete once SSO lands: the dev impersonation route, and the browser-remembered "my reports" ids in **useViewer** (the server's **is_mine** supersedes them).

What the internal deployment must provide

Platform-neutral requirements — what the app actually needs, independent of hosting.

#	Requirement	Why	If you skip it
1	Persistent, backed-up attachment storage , served through an authorising endpoint or signed expiring URLs	Attachments are screenshots that may contain customer policy and account data	On ephemeral compute they vanish on restart and leave attachments rows pointing at nothing. On a public URL scheme they are readable by anyone with the link.
2	A shared session store (database or Redis), or SSO with stateless tokens	Partly done — connect-pg-simple against Postgres. Keep a real store on whatever engine you pick.	Every restart signs every admin out, and you cannot run more than one instance
3	A reverse proxy that carries WebSocket upgrades	Lets the socket stay same-origin and use the session cookie	You are stuck reproducing the cross-origin token workaround for no reason
4	TLS terminating in front , with trust proxy configured to match	secure: true cookies require HTTPS; req.ip for rate limiting must be the real client IP	Cookies silently fail to set, or every client shares one rate-limit bucket
5	A secret store for SESSION_SECRET , the DB connection string, and AI provider keys	They are currently in a gitignored .env	Credentials spread through deploy configs and developer machines
6	A reviewable migration step in the deploy pipeline	Production currently self-syncs on boot	Schema changes are unreviewable and irreversible
7	Outbound HTTPS egress to the AI vendor and to EasyVista	See AI egress	AI search self-disables (gracefully); EasyVista sends fail
8	A scheduled job runner if you keep the embedding backfill	Currently manual	New tickets self-heal inline (bounded), so this is a nice-to-have

Requirements 1, 3 and 5 are the ones that most change the shape of the rebuild.

Decision 1: the database engine

Not yet decided. The good news is that the dialect-sensitive surface is **small and inventoried**. This is all of it.

Location	What it does	Portability
<code>db/sequelize.js</code>	Provider selection; Postgres SSL options; <code>normalizeDatabaseUrl</code>	Rewrite for your engine
<code>db/index.js</code> → <code>withReturningIdForInsert</code>	Appends <code>RETURNING id</code> on Postgres only , to recover the new id	SQL Server uses <code>OUTPUT INSERTED.id</code> / <code>SCOPE_IDENTITY()</code> . Or drop the raw-insert path entirely and use the model's <code>create()</code>, which handles this per dialect.
<code>db/models/index.js</code> → <code>RAW_UNIQUE_INDEXES</code> , <code>NO_ALTER_MODEL_NAMES</code> , <code>ensureColumn</code>	Raw <code>CREATE UNIQUE INDEX IF NOT EXISTS</code> for constraints that cannot be declared on the model	Entirely a SQLite workaround. On any real engine, declare composite uniqueness normally and delete all of this. Note <code>IF NOT EXISTS</code> on an index is not valid T-SQL.
<code>services/viewerService.js</code>	<code>SELECT application_id, COUNT(*) ... ORDER BY n DESC ... LIMIT 1</code>	ANSI apart from <code>LIMIT 1</code> → <code>TOP 1 / OFFSET... FETCH</code>
<code>services/accessService.js</code>	<code>SELECT application_id, COUNT(*) ... GROUP BY</code>	Fully portable
<code>services/deliveryService.js</code>	Three raw <code>SELECT</code> s for the throughput numbers (delivered, hours, workers)	ANSI; named replacements. Portable.
<code>keepAlive.js</code>	<code>SELECT 1</code>	Being deleted anyway
<code>submission_embeddings.vector</code>	Vectors as a JSON float array in TEXT , ranked by cosine in application memory	Only to avoid a pgvector dependency on the SQLite path
<code>services/adminViewPreferenceService.js</code>	<code>SELECT *</code> rather than naming <code>pinned_application</code> , so	A consequence of boot- time sync; unnecessary with real migrations

Location	What it does	Portability
	it survives a DB where the boot sync has not added that column	

Everything else goes through Sequelize models, so a dialect change is a configuration change plus the rows above.

Option	Vector search	Schema workarounds	Notes
PostgreSQL	Use pgvector . Replace the JSON-in-TEXT column with a real vector type and rank in the database — removes the in-memory candidate cap.	Delete all of them	Smoothest path; the code already runs on it
SQL Server	No pgvector. Either the 2025 native vector type, or keep vectors in a column and rank in the app, or push AI search to a separate service	Delete the SQLite ones; add TOP/OFFSET-FETCH and identity retrieval	Sequelize supports mssql . Budget time for the raw queries, the insert-id path, and index DDL.
Oracle / DB2 / other	Rank in the app unless the engine has native vector support	As above	Verify Sequelize dialect maturity first; highest-risk option

The money fix (applied)

The money columns **have been fixed**, so you inherit the correct shape. What was wrong, because the same trap is easy to re-introduce:

DataTypes.REAL maps to **single-precision float4** on Postgres — about seven significant digits. The **stored** value was therefore wrong:

```
1234567.89 -> 1234567.875      (displays as $1,234,567.88 – a cent adrift)
99999.99   -> 99999.9921875
0.07       -> 0.07000000029802322
```

SQLite's **REAL** is a double, so **this only ever damaged the hosted data and never reproduced locally** — worth remembering as a class of bug when you keep two dialects around. The Excel export writes these values out raw, so it showed in spreadsheets as well as on screen.

The fix, in three parts:

Change	Where	Note
REAL → DECIMAL(14,	db/models/index.js	—

Change	Where	Note
2)		
Coerce back to a number	<code>helpers/mappers.js</code> → <code>toMoneyNumber</code> , called in <code>mapSubmission</code>	Required. <code>pg</code> returns <code>numeric</code> as a string and Sequelize's Postgres <code>DECIMAL.parse</code> passes it through to preserve precision. Without this the API contract silently changes from <code>1250</code> to <code>"1250.00"</code> . Every submission response and socket payload goes through that one mapper, so it is the only place needed.
Exact integer-cents summation	<code>AdminDashboardPage.jsx</code> → <code>impactTotals</code>	Defensive, not a visible fix. Measured: float64 accumulation stays below half a cent even at 50,000 rows. Done because the totals are read as authoritative and should not depend on rounding to be right.

`toMoneyNumber` keeps null as **null, not 0** — "nobody costed this ticket" and "zero dollars of impact" are different answers, and the totals must not conflate them.

An explicit migration ships with it (`npm run migrate:money-columns`, dry-run by default) rather than letting the production boot-sync perform the `ALTER` unreviewed. **It cannot recover precision `float4` already destroyed**; the dry run reports how many rows carry the damage signature.

For a Postgres rebuild: `numeric` throughout, or integer minor units if you want arithmetic in the application. Either is fine; `REAL` is not.

Timestamps: what a conversion has to handle

Not fixed — deliberately left, because it is not a type change, it is a cross-cutting one. Every model is `timestamps: false`, and `created_at`, `updated_at`, `changed_at`, `routed_at`, `uploaded_at`, `worked_on`, `completed_at`, `approved_at` and `date_time_of_error` are all `TEXT` holding ISO strings.

Use native `timestampz` / `datetime2`. **Four things will break if you only change the column type:**

1. `updated_at` is the optimistic-concurrency token, and it is compared as a string. This is the one that will bite hardest.

```
// read-time check
if (body.base_updated_at && String(rawExisting.updated_at || '') !==
String(body.base_updated_at)) { ... }
// the same value again, inside the UPDATE's WHERE
const updateWhere = { id: Number(id), updated_at: rawExisting.updated_at };
```

The client loads a row, keeps `updated_at` verbatim, and echoes it back on save. With a native timestamp column that round-trip has to survive **JSON serialisation and back with identical precision** — and it will not: Postgres `timestampz` carries microseconds, a JS `Date` carries milliseconds. The string compare then fails

on every save and **every edit returns a spurious 409 conflict**. Fix it by versioning explicitly — an integer `version` column, or an `xmin`/rowversion equivalent — rather than by comparing timestamps.

2. Sorting depends on ISO strings comparing lexicographically. Correct for well-formed ISO-8601, and it is what the server's `compareText` comparators rely on for the date fields. Native types sort properly, so this gets *better* — but verify the comparator mapping rather than assuming.

3. The AI search time-window filter runs in JavaScript on purpose. The comment says it must be "*robust to legacy non-ISO date strings*" — **direct evidence that malformed values already exist in the data**. A native-type migration will reject them. Plan a parse-and-report pass before the conversion, and decide per row: repair, null out, or quarantine.

4. Timezone is unstated. Server-written values come from `new Date().toISOString()` (UTC, with Z). `date_time_of_error` comes from a browser form and may carry local time with no offset. Decide on `timestamp_tz` with an explicit inbound normalisation, and be aware that historical values may not be recoverable to a true instant.

`worked_on` is a **date, not a moment**, on purpose — the day the work happened. Keep it a `date` type, not a timestamp, or the throughput window starts depending on the hour it was typed.

The derived board timestamps (`approved_status_at`, `in_progress_status_at`, ...) are computed from `submission_status_events`, so they follow whatever you do to `changed_at`.

Decision 2: hosting shape, and the reverse proxy

Not yet decided. Rather than prescribing a platform, here is what any of them has to accomplish.

The two processes. A **static client** (`npm run build → client/dist/`; plain HTML/JS/CSS any web server can serve) and a **Node API** (`node src/index.js` on `PORT`, no build step, supervised and restarted on failure).

The proxy must	Notes
Serve <code>client/dist</code> as the document root	—
SPA fallback — any unmatched path returns <code>/index.html</code>	Without it, deep links like <code>/admin/metadata</code> 404 on refresh. IIS: URL Rewrite. nginx: <code>try_files \$uri /index.html</code> .
Reverse-proxy <code>/api/*</code> to the Node process	Keeps the API same-origin, so the session cookie just works
Reverse-proxy <code>/socket.io/*</code> with WebSocket upgrade support	The piece Vercel could not do. IIS: enable the WebSocket Protocol feature + ARR. nginx: <code>proxy_set_header Upgrade \$http_upgrade; proxy_set_header Connection "upgrade";</code> . Getting this right is what lets you delete the realtime-token mechanism.
Reverse-proxy <code>/uploads/*</code>, or replace it	Only if you keep local-disk attachments. Preferably replace with an authorising endpoint.
Terminate TLS, forward the real client IP	<code>X-Forwarded-For</code> / <code>X-Forwarded-Proto</code> , with <code>trust proxy</code> set to match the hop count

The proxy must	Notes
Send X-Content-Type-Options: nosniff on any file-serving path	The app sets it on <code>/uploads</code> ; do not lose it at the proxy

If you containerise, requirements 1 and 3 and the in-memory state are not optional — they are the blockers. Ticket presence and rate-limit counters are plain in-memory maps, so as written the app cannot run more than one replica. Presence degrading is cosmetic; rate limiting is not. (Sessions are now in Postgres, so that one is handled.)

Windows note. Developed on Windows Server, and attachment paths are normalised accordingly (`path.relative(...).replaceAll('\\', '/')`). On Linux that is harmless, but audit any path handling you add.

AI egress

Third-party AI calls are permitted, so the current design carries over unchanged.

Item	Detail
Outbound HTTPS to the chosen vendor	<code>api.anthropic.com</code> or <code>api.openai.com</code> . <code>embeddings.js</code> and the OpenAI path use native fetch , which honours <code>HTTPS_PROXY</code> only if you configure an agent or run Node with proxy support — verify this early, it is a common first-deploy failure .
Keys in the secret store	Never in a dotfile. Rotate the key currently sitting in <code>server/.env</code>.
A cost bound	<code>AI_SEARCH_TOP_K</code> (20) caps candidates per summary call; <code>AI_SEARCH_MAX_INLINE_EMBED</code> (25) caps inline embedding work per search; <code>AI_SEARCH_PUBLIC_RATE_LIMIT</code> (20/min per IP) bounds the anonymous surface. Set these deliberately.
Where embeddings run	<code>AI_PROVIDER=anthropic</code> gives Claude summaries with local, in-process embeddings — no embeddings vendor, no per-call embedding cost, and ticket text never leaves your servers. Given internal hosting, worth considering: it reduces what leaves the network to just the summary call.
First-run behaviour of local embeddings	Downloads ~90MB of weights to the <code>transformers.js</code> cache. In a locked-down or offline environment, pre-seed that cache into the image or a mounted volume.
The feature self-disables	With no summary key, <code>/api/ai-search/status</code> reports <code>enabled: false</code> and every AI surface renders nothing. A blocked or delayed egress approval does not block the deployment.

Deployment gotchas — the ones that bite silently

#	Problem	Why it happens	Carries over?
1	Attachments vanish on every deploy	Ephemeral filesystem plus local-disk storage. attachments rows survive, pointing at nothing.	Yes, if containerised or on any non-persistent disk
2	Every admin is signed out on every deploy	express-session had no store	Fixed — connect-pg-simple . Keep a real store.
3	Presence and rate limiting are single-instance	Plain in-memory maps	Yes, with more than one replica . Presence is cosmetic; rate limiting is not.
4	Attachment URLs are publicly reachable	Uploads go to a public bucket — unguessable but unauthenticated	Only if you copy the scheme . Don't.
5	Cold starts	Free-tier compute spins down; GET /health exists for external ping services	No — but keep /health for load-balancer probes
6	Boot-time alter: true is not a migration strategy	Convenient for a prototype; makes schema changes unreviewable	Yes . Requirement 6.
7	Dev impersonation must stay unreachable	Triple-gated and unregistered in production	Yes until it is deleted

Go-live checklist — internal deployment

Configuration

- ☐ **NODE_ENV=production**
- ☐ **SESSION_SECRET** ≥32 chars, from the secret store (the server refuses to start otherwise)
- ☐ Database connection configured for the chosen engine; **DB_PROVIDER** set explicitly
- ☐ **SESSION_STORE** set deliberately (**pg** on Postgres, so it cannot silently fall back)
- ☐ **CLIENT_ORIGIN** lists exactly the real portal origin(s)
- ☐ AI provider key from the secret store; the key currently in **server/.env** **rotated**
- ☐ **EASYVISTA_ENABLED** set deliberately — off until the contract is confirmed
- ☐ **EASYVISTA_CATALOG_GUIDS** / **_CODES** populated per application by the team that runs EasyVista
- ☐ **SUBMIT_REQUIRES_AUTH** left ON (it is the default) unless there is a named reason
- ☐ **DEV_IMPERSONATION** unset

Infrastructure

- ☐ Persistent, backed-up attachment storage, behind authorisation
- ☐ Reverse proxy: static root, SPA fallback, **/api** proxy, **/socket.io** with **WebSocket upgrade**
- ☐ TLS terminating in front, **trust proxy** matching the hop count
- ☐ Outbound HTTPS egress to the AI vendor verified **through the corporate proxy**
- ☐ Process supervision and restart-on-failure
- ☐ Backup and restore tested for both the database **and** attachments

Data

- ☐ Migrations run as a reviewable deploy step, not on boot
- ☐ Admin account(s) seeded and **every seeded password rotated**
- ☐ At least one super user granted
- ☐ `npm run backfill:embeddings` run if AI search is configured
- ☐ `npm run migrate:money-columns -- --apply` run against the target database, **before** the boot sync reaches it
- ☐ Timestamp columns converted to native types, existing ISO strings parsed and validated
- ☐ The demonstration data removed or replaced with real intake

Deleted before go-live

- ☐ `realtimeToken.js`, `GET /api/realtime/token`, and the socket token fetch
- ☐ `keepAlive.js`
- ☐ `sameSite: 'none' → lax`
- ☐ The Supabase Storage backend (replaced, not merely unconfigured)
- ☐ SQLite portability workarounds (`RAW_UNIQUE_INDEXES`, `NO_ALTER_MODEL_NAMES`, the `sqljs` provider path)
- ☐ Dev impersonation route
- ☐ The browser-remembered "my reports" ids in `useViewer`, once SSO lands

Part IX — Decision record

Why this part exists. The sections above describe the system as it stands. This one records **how it got there, in order**, because several of these decisions were made, then corrected, and the reasoning is the only thing that stops them being un-corrected by a later tidy-up. Where a decision is described in full elsewhere, this part says so rather than repeating it.

Two conventions: **an owner correction is marked as such**, because "the first answer was wrong and here is why" is the most useful kind of entry; and every "~~struck-through~~" line is a decision that was reversed, kept visible so nobody re-derives it.

The shape of the build

Phase	What landed
2026-07 → 08-01	The defect/enhancement portal: intake, triage, the public board, EasyVista, bulk actions, per-admin views, AI search
2026-08-03	Per-application access control, the redirect ledger, reporter binding, the pinned queue, the per-application EasyVista catalog
2026-08-05	The portal rename and <code>TRACKER_LABEL</code> ; three approved redesigns (metadata page, submit form, data-entry dialogs); the money-column fix; the legacy text columns dropped
2026-08-06 → 08-	Report requests, in fifteen passes. Schema, authorisation sweep, submit-form branch, detail-modal fields, throughput page, spreadsheet round trip, the private-report rule, session

Phase	What landed
07	persistence, and eleven items the owner found while testing the deployed site
2026-08-07	Filing requires sign-in for every type; the documentation set and the screenshot harness

The rename, and why the name is not narrower

The portal is "**Service Requests Portal**" / "**Submit · Track · Resolve**". It was briefly "Defects & Enhancements Portal" earlier the same day; that was replaced once the scope widened, because **a portal that also handles report requests is not a defects portal**.

Named for the destination deliberately. The alternative was re-shooting 43+ desktop screenshots a second time when the other request types landed. **Do not narrow it again.**

The downstream ticketing system is `TRACKER_LABEL = 'Service Desk'` — a **display name only**. `easyvista_ticket_id`, the `easyvista-preview` route, `EASYVISTA_*` env vars and `server/src/easyvista.js` keep the vendor name, because renaming them is a migration with no user-facing gain. The Excel import matches on `aliases`, never on `label`, so relabelling an export header is safe — and `service_desk_number` / `service_desk_ticket` / `service_desk_submitted_by` aliases were added so a sheet exported with the new header re-imports.

Report requests: the design decisions, in the order they were taken

It is not one request type — it is a service catalogue

The owner's original list was nine kinds of request, of which report/dashboard requests were one. **Phase 1 was scoped to report requests alone**, and types 3–9 are recorded future scope rather than current work. The extension points Phase 1 needed were built to make a fourth type cheap: the Add-a-ticket dialog's type is **one segmented control plus a computed data-branch**, and a new metadata panel is one `ADMIN_META_CATEGORIES` entry plus its `LOOKUP_TABLES` row.

The schema is additive and all-nullable

14 nullable columns on `submissions`; `attachments.purpose`; `user_application_roles.request_type`; three new tables (`levels_of_effort`, `request_time_entries`, `request_assignments`); the `report` submission type; four seeded effort values. **Every one nullable and null on every existing row, so the migration cannot change what any current ticket means.**

Applied with `npm run migrate:report-requests` (dry-run by default) and verified non-destructive afterwards: the same submission count before and after, both existing grants preserved, no existing row with a report field set, money columns untouched. **Idempotent** — a re-run reports everything already present.

Both child tables had to arrive WITH the feature: neither hours nor the assignment trail can be reconstructed afterwards.

`request_type` is `' '`, not `NULL`

See §10. **Probed with real inserts, not reasoned about.**

ONE status table, scoped per type

See [§7](#) for the full argument. Two alternatives were framed and **neither was taken**, deliberately.

The public board draws a report request on its own track

Owner's call: report requests get their own track rather than staying off the board. Two real defects were found while doing it, both fixed:

1. The type chip read `type === 'enhancement' ? 'Enhancement' : 'Defect'`, so a report request was **called a Defect on the one surface its requester reads**.
2. Every **enhancement** row overflowed its own type column by 2px at desktop width, invisibly, on every load — an 86px chip in an 84px track. **The overflow probe caught it while the third type was being added**; the track is 88px now.

Two questions the owner answered

1. What "Report/Dashboard Approval" is approving. Owner's words: *"basically a manager or supervisor giving them okay to use resources to create the report/dashboard"*. So it is a gate **before** work starts, which is how it is built. **No code change needed**. Two consequences carried forward:

- The approver is often now a portal user, because `manager` exists. The field stays a typed **name** anyway: a supervisor in another department may approve and have no grant here, and the accountability is `approval_recorded_by`, an id the server fills in.
- It is genuinely a **SECOND gate**, not a restatement of the `Approved` status. Triage accepting a request as valid and a manager authorising the spend are different people saying yes to different questions. **Keep them separate, and keep the group labelled "Go-ahead" so the two never read as one.**

2. Report requests get their own status list. Owner: *"we probably aren't submitting to easyvista, but I imagine most statuses can transfer."*

Report request status	From the defect list?
New	yes
Approved	yes — triage accepted it
In progress	new — there was no working state at all
Delivered	new — replaces <code>Deployed</code> , which is deploy language
On hold	new — collapses Backlog / Future Consideration / Deferred, which read as one state to a requester
Rejected	yes
Duplicate	yes
Redirected	yes — moving between application queues is still meaningful
Retired	yes — protected, exists everywhere

Dropped: **Submitted** and **Deployed**, both of which are the Service Desk hand-off a report request never makes.

What that implied, and how each part landed:

- ~~A new lookup, so an 11th Metadata panel.~~ **No new panel.** One table serves every type, so the existing Statuses panel gained three values, each marked **Report requests only**. A second panel over the same table would have let a rename in one change the other silently.
- ~~submissions.status_id → a second column or a resolver.~~ **Neither.**
- ~~The public board's four-stop track.~~ **Built, per type**, plus the two defects hiding behind it.

The spreadsheet round trip

See §15. The headline: **a sheet of report requests could not be imported at all** before `importMode` gained a fourth entry, and the round-trip check (export → re-import the portal's own file → compare every column) is the only kind of test that catches a header which does not survive the trip. Two never did.

Owner correction: `delivery_notes` in Excel

~~"A delivery note is written after the work, and an import loads history from elsewhere, so leave it out."~~ **Exactly backwards, and the owner said so:** a backdated migration is precisely the case where delivery notes **already exist**, and dropping the column that says what was handed over loses the point of the record. It exports and imports now, with aliases for the hand-titled spellings a real spreadsheet will have.

Eleven things the owner found on the deployed site

These are worth reading as a set, because **three of them were not the small changes they looked like**.

1. THE ACCESS PAGE COULD NOT MAKE AN ANALYST — AND UNMADE THE ONES THAT EXISTED. Full detail in §8.7. The lesson is general: **a page that writes a subset of a row's key must write the whole key, or it silently widens what it saves.**

2. A REPORT REQUEST IS PRIVATE. Full detail in §11. The lesson: **one rule, one function, when more than one surface enforces it** — and the leak found while building it proves why.

3. FILING ONE REQUIRES SIGNING IN — the owner's call, given the above. An anonymous report request belongs to nobody. The form says so when the type is picked, keeps what was typed, and disables **both** submit buttons. **There are two** — the readiness rail's and the sticky bar's — and the first version of that check read only the first match and missed that the one a desktop user clicks was still live. *(This was later generalised to every type; see §9.)*

4–7. The detail modal, told apart by type. Release # and Release Notes gone from a report request (nothing ships), Workaround gone from its Triage tab (nothing is broken), **Delivery notes** added to answer the question that does apply, and the Service Desk number made editable **behind an unlock**. **Reviewer fills itself in on SAVE**, not on open: prefilling would mark every ticket edited the moment somebody looked at one.

8. Two queues, two column sets. See §8.4. A row still holding a bare `columns_json` array reads as the default set with no report set, which is right for anyone who saved a view before this.

9. The file picker no longer renders the OS button — **only the button is redrawn**, because the native control carries the accept filter, multi-select and drag-and-drop, and all three would have to be rebuilt to replace it.

10. The semantic search describes, it does not rule. The prompt already banned "Yes, this has been reported" and the model still produced rulings, because **"do not say X" leaves every other way of saying X available**. The SHAPE is prescribed now and the ban is stated as a class.

11. Non-admins see no Admin link, and a rep can sign out from the header — which is the only place they can, because they never reach the admin pages where the account menu lives.

12. Only New wears the queue's left stripe. See §8.2.

13. The EasyVista catalogs card is off the Access page. The owner asked what it was and then made the argument that settles it: **a catalog GUID is an identifier INSIDE EasyVista**, so the team that runs EasyVista owns the value and nobody using this portal has it. The card asked super users for an answer they were never going to have. **Removing it was not enough on its own** — see §14. Also removed: the hook action, the api helper, the PUT behind `ensureSuperUser` and the service function. *An unused write endpoint on a super-user route is a surface with nothing behind it.*

14. Managing metadata is super-user only. See §8.6 — including why the READ stays open, and why a test pins that asymmetry from the router's own stack.

Three earlier corrections worth keeping

The application picker looked like a system dialog from another decade — grey bevel, its own font, nothing to do with the field above it. It was the submit form's **first** select, so `.rs-field select` had never been written: `input` and `textarea` were styled and nothing else. It now shares that rule, with the browser chrome switched off and the caret drawn as a background image — plus **explicit option colours**, or dark mode gives black text on a black sheet.

"Other" is a real application, not a flag. Sometimes the honest answer to "whose data is this?" is "both" or "I do not know", and the request still has to land where somebody will pick it up. Making that somewhere an **application row** buys the existing machinery whole: a queue with grants, an entry in the application filter, and **Redirect** already moves it to its real home and records the hand-off.

The alternative was a null `application_id`, which the portal already has a name for ("No application set"). **That is the wrong home: only a SUPER USER sees those rows**, so a request nobody had claimed would be visible to nobody who works the queue.

`npm run seed:other-application` creates it **last** in the list — it is the answer you reach for when none of the real ones fit — and grants it to **everybody who works report requests anywhere, derived from the grants that exist** rather than a list typed into the script. Their grant on Other is narrowed to `report`, because Other exists for requests whose application is not yet known and that only happens to a report request. **Re-run it after granting somebody new.**

A hardcode this exposed: the import route accepted a default application only from `['Billing Center', 'Policy Center']` — a literal pair — so it would have refused Other outright, and any future application with it. Both that check and the list the dialog offers now come from the applications table.

Two bugs behind owner reports, both one type over from where they were found

Worth recording as a pattern: **the same fault existed in a sibling branch nobody had tested.**

1. **A report request's application was derived and defaulted.** So a report over billing data asked for by somebody in Claims went to whichever queue the fallback named. Fixed: required on the form AND refused at the endpoint. **The bug behind it:** the public submit route pinned every **enhancement** to **'Billing Center'** outright, ignoring the payload — *same fault, one type over*, and invisible because the client had always sent the right value.
2. **A report request's eight requester fields were stored, exported and imported and drawn NOWHERE** in the modal. The Report tab asked a defect's questions. *An analyst could open a report request and not read what had been asked for.*

The assignee's grant is checked on the save, not only offered in the dropdown

The detail modal only ever offered the grant list, but **PUT /api/admin/submissions/:id** **accepted any user id** — so work could be put on somebody who cannot open the ticket. Both the save and the create path ask **isAssignableTo** now, and **an assignee set at creation opens the handover trail** rather than waiting for the first reassignment.

Sessions that survive a deploy

See [§12](#). The seventh pass made the failure **honest**; the eighth removed its **cause**. Both halves were kept on purpose: the honest path should stop firing on deploys, and that is all it should stop doing.

Not done, on purpose

- **The duplicate check no longer warns about somebody else's report request.** It follows from "only the person who filed it may see it". A count without content ("2 similar requests exist") would close the gap without breaking the rule; nobody has asked for it.
- **An admin-created report request has no reporter**, so it is on the admin queue and on nobody's board. Coherent — an admin recording somebody else's request cannot claim it for them — but it means the Add-a-ticket dialog cannot put a report request in front of the person who asked for it.
- **approval_recorded_by is not importable.** That column is the id of whoever entered an approval **in this portal**. Nobody did for an imported row.
- **An imported Duration cannot be split across people or days.** One number in one cell becomes one entry. A per-person hours sheet would be a second import shape, not a column.
- **The throughput page is not live-updating**, and it says when it was built.
- **No true snapshots.** `reported_snapshot` / `easyvista_snapshot` do not exist, so the Report tab shows current saved values **and says so**.
- **Five 401 Unauthorized fetches on the public routes** are still untraced. Pre-existing, harmless-looking, never chased.
- **feat/admin-detail-modal-redesign** is kept on purpose. Its *content* is on `main` (it merged as PR #3, which rewrote the SHAs), but the two commits themselves are not reachable from `main`, so deleting it needs `git branch -D` and would drop the only copy of them.

Session notes worth keeping

These are process lessons, not product ones, and every one cost time.

- **A verification script that writes must prove it put things back.** The metadata script's first version committed a rename and switched a status off in the shared database and undid neither; both had to be repaired by hand. It now takes a before/after fingerprint of **every lookup value** and fails if anything drifted, and its rename checks deliberately never commit.
- **A failing check is a claim about the product until proven otherwise — but a probe can be wrong too.** Four of the overflow probe's exclusions exist because a false positive was burying real findings: `<input>`s scroll their own value; a `clip: rect(0 0 0 0)` screen-reader label holds content far wider than its 1px box *and so do its children*; a deliberate `margin: 0 -24px` full-bleed makes every ancestor up to the clipping element measure wide, so **the allowance has to propagate upward** or the finding just walks one level up the tree each time it is excused. Each was diagnosed by measuring the specific element, **never by loosening the check until it passed.**
- **Restart Vite if a page suddenly fails to mount.** After a long run of edits — especially files rewritten by an external script rather than an editor — the dev server's module graph can go stale and throw `does not provide an export named 'X'` for an export that is plainly there. `npm run build` passing while the dev server does not is the tell. Restart it; do not debug the source.
- **Watch for leaked processes.** Playwright runs that time out leave a browser tree behind, and stacked `npm run dev` invocations add up: with three duplicate dev servers and a leaked Chromium, ESLint died with `Zone Allocation failed – process out of memory` on an 8 GB box. **Not a code problem.**
- **Never slice a stylesheet by line number** to inline it. Doing so cut a rule in half, left an unclosed brace, and the CSS parser silently swallowed everything after it — two panes rendered unstyled while still reporting plausible heights. Inline the whole file.
- **Watch specificity when overriding by class.** `.rs-drop > span` (two classes + a type) outranks `.rs-drop-icon` (two classes) and stole its `order`. And a rule written for `.rs-refbox` did nothing because the real class is `.rs-sub` — **a silent no-op that looked like a design decision that hadn't landed.**
- **The three checks the harness kept earning back:** per-container overflow found two real defects on the admin queue; the field/control-count assertion caught the submit form's counts changing for a reason that turned out to be legitimate (a known reporter is *stated*, not *asked*) and **forced that to be written down rather than guessed**; and checking the export dialog against the server's own field list is what makes the server-side grouping safe.

The eighteenth pass — the analyst's UI, the hand-off affordance, and Other

Three features that turned out to be one story: **what happens to work the Service Desk is not wired up for.**

1. POST /api/admin/applications got a UI. The endpoint had shipped tested in the previous pass and *nothing called it*. It is now one shared control (`AddApplicationControl.jsx`) in both places an analyst hits the wall — the Add-a-ticket dialog's report branch and the Redirect dialog — because the two differ only in what they do with the answer, one picking by name and the other by id. `onCreated` is **awaited** before the picker is pointed at the new value: the caller re-reads the application list there, and selecting an `<option>` that does not exist yet leaves the picker blank, which reads as "it didn't work".

2. Redirect was the fifth write path, and it was not guarded. `helpers/applicationScope.js` was written for four paths that set `application_id` — the public submit, the admin create, the admin update and the import. A **redirect** sets it too, and had no check. A defect could be moved into a reports-only queue, granted only to report workers, where fail-closed scoping makes it **invisible rather than merely unassigned** — and

the sending team has already lost write access by then, so nobody could move it back. The picker now hides it and `redirectService` refuses it; the endpoint is the control, the picker the courtesy.

3. The hand-off button now says what it cannot do, and what to do instead. See §14. The catalog check stopped being live-only, `DEMO-` catalogs keep the walkthrough applications sending, and `Other` demonstrates the manual path.

4. The soft association. The owner's ask — "once they change it from new status to something else, then it soft assigns it to their queue (they can select which of their queues)" — needed a decision first, and it is recorded here because the obvious implementation undoes itself:

A redirect **resets the status to New**, and a soft assign is triggered **by moving the status off New**. Routing it through `redirectService` would immediately undo the change that triggered it.

Two shapes were put to the owner. **(a)** a real move that keeps the status — reuses the ledger, one source of truth, but the ticket leaves `Other` for good. **(b)** a softer association that does not move the ticket. **The owner chose (b)**, and `submissions.working_application_id` is it. What makes a second "whose queue" column safe is that it answers strictly less than the first — see §5 distinction 6.

What `Other` actually is, in the owner's words, and it corrects an earlier framing in this document that had it as *only* "the system is unknown":

"The 'other' option is used for when there isn't an application configured yet to directly submit to EasyVista but we still want to be able to track or create the reports, almost like a task list in a way. A way to still track issues even if for a defect or enhancement the admin has to manually submit to the service desk and then manually enter the ticket number." — and "obviously other can be used too when it is still an unknown application."

Both cases, one queue, for the same reason: the work still has to be tracked and there is nowhere else to put it.

Four things this pass got wrong first

- **A model column added without its migration takes the server down.** Sequelize names every column the MODEL declares, so adding `working_application_id` to `db/models/index.js` while the database lacked it broke queries immediately — the same trap `viewerService.listActiveApplications` and `helpers/lookups.js` already carry defences for. Run the migration in the same breath as the model edit.
- **Editing server source during a browser verification restarts the server under it.** Two runs died on `ECONNREFUSED` and a 500 that looked like product faults and were nodemon doing its job. Finish the code, then verify.
- **`isUnknownQueue` is passed to `resolveSoftAssignment`, not looked up inside it.** The first version required `helpers/lookups` and the test had to monkey-patch a binding that had already been destructured at require time. A rule that decides whose list a ticket lands on should be pinnable without a database.
- **The detail modal sends its WHOLE edit object on every save, so** `hasOwnProperty('working_application_id')` is always true. Treating presence as a deliberate act would have cleared the association on the next unrelated save and made the status trigger unreachable. What identifies a deliberate act is that the value **differs from what is stored**.

And three the harness got wrong, all of them the probe

Added to [the trap table](#):

- **An element handle does not survive a re-render.** `page.$()` then `.click()` across a debounced search gave *"Element is not attached to the DOM"* — which reads exactly like a broken selector. A **locator** re-resolves on every action.
- **The queue's kind switch and application scope are saved per admin**, in `localStorage` and in the server-side pinned application. A ticket in `Other` was simply not in a queue pinned to Billing Center — the queue working correctly. The probe now widens both explicitly, and **puts them back**: leaving the search box filled made a later section time out looking for a ticket it had filtered away.
- **A delta is only safe if nothing else is writing.** `verify-session-store`'s "no session was left behind" check failed because `verify-metadata-page` was signing in **concurrently**, into the same shared `user_sessions` table. Run it alone.

Part X — Known gaps, traps and open questions

Unfinished

Gap	Detail
EasyVista attachment transmission	The only genuinely unimplemented feature. Contract unknown — see §14.
Description is overwritten by EasyVista	EV-side bug. Needs raising with their team before rebuilding.
No true snapshots	<code>reported_snapshot</code> / <code>easyvista_snapshot</code> do not exist. The Report tab shows <i>current saved values</i> and says so. Design these in from the start — an audit surface that shows current values is a footgun.
SSO not wired	<code>AUTH_MODE=local</code> . Every seam is in place; nothing is connected.
users.email is not populated	EasyVista requestor mail comes from a <code>username:mail</code> env map as a stopgap. The code already prefers a real <code>user.email</code> the moment one exists.
AD group names unknown	<code>application_ad_groups</code> is empty. The app works without it.
EasyVista catalogs unset	<code>EASYVISTA_CATALOG_GUIDS</code> is empty, so no application has a catalog and every send is refused. Correct — the values belong to the EasyVista team.
Types 3–9 of the service catalogue	Recorded future scope, not current work. The extension points exist.

Data-type defects

Issue	Status
Currency was single-precision float	FIXED. Now <code>DECIMAL(14,2)</code> , coerced back to numbers at the mapper boundary, with an explicit migration. See the money fix .
Timestamps stored as ISO strings in TEXT	NOT fixed — specified instead. Deliberately: <code>updated_at</code> doubles as the optimistic-concurrency token <i>compared as a string</i> , so a naive type change makes every save return a spurious 409. Malformed legacy values also already exist. Read the conversion notes before starting.

Security shortcuts to close

Issue	Detail
Attachment URLs are unauthenticated	Uploads go to a public bucket. Unguessable, but anyone with the URL can read it — and attachments are screenshots that may contain customer policy and account data. This is the one route around an otherwise carefully allow-listed public boundary. Fix with an authorising endpoint or signed expiring URLs.
<code>sameSite: 'none'</code> in production	More permissive than the proxy setup requires; <code>lax</code> would do.
Sessions are not invalidated server-side on demotion	The <i>rights</i> re-read on every request, which is the important half, but the session row itself persists.
Credentials in a gitignored <code>.env</code>	<code>server/.env</code> is untracked, so no credential is in git history — but each developer's default is whatever they happen to have locally, and it currently points at the shared hosted database. Default to a local database, make production access an explicit opt-in, and put the secrets in a manager.
One shared test password	All eleven seeded accounts use <code>SEED_ADMIN_PASSWORD</code> . Fine on a prototype whose data is test data. Rotate before anybody real signs in.

Won't survive scale — deliberate prototype shortcuts

- **Rate limiting is in-memory** (`server/src/middleware/rateLimit.js`) — single instance only. Needs a shared store behind more than one process. This is the one that matters.
- **Ticket presence is in-memory** — same constraint, but degrading is cosmetic.
- **Cosine ranking happens in application memory** with a cap on candidate rows. Use `pgvector`.
- **The public board sorts in memory** — the list endpoint returns the whole board and the client sorts it. Fine at prototype volume; paginate and sort server-side.
- Sessions are in-memory — **fixed**, see [§12](#).

Portability traps you can probably delete

These exist **only** because the prototype supports SQLite alongside Postgres. On a Postgres-only rebuild, delete them all — but **read them first**, because if you *do* keep a dual-provider setup you will hit every one.

1. **Composite unique indexes cannot be declared on the model.** On SQLite, `sync({ alter: true })` rebuilds the table, reads the composite index back through `describeTable`, and **mis-derives it into standalone per-column UNIQUE constraints**. On `user_application_roles` that means `UNIQUE(user_id)` — silently capping every admin at **one application forever**. Affected tables also skip `alter` entirely and get a plain `CREATE TABLE IF NOT EXISTS`; their uniqueness is created as raw `CREATE UNIQUE INDEX IF NOT EXISTS`.
2. **A table synced without `alter` never gets a new column from the boot sync.** `ensureColumn` adds it explicitly. **Any future column on `user_application_roles`, `submission_embeddings` or `application_ad_groups` needs the same treatment.**
3. **SQLite rejects `ALTER TABLE ... ADD COLUMN ... UNIQUE` outright.** So `users.external_id` is declared plain and gets its unique index raw — declaring it on the model breaks migration locally while succeeding on Postgres.
4. **Vectors as JSON in a TEXT column** instead of `pgvector`.
5. **Raw SQL for one aggregate** in `resolveHomeApplicationId`, so ordering is identical on both dialects.
6. **SELECT * in `adminViewPreferenceService`** rather than naming `pinned_application`, so it keeps working against a DB where the boot-time sync has not added that column yet.
7. **A class of bug you inherit from keeping two dialects:** anything where the two engines' native types differ silently. The money columns are the worked example — SQLite's `REAL` is a double, Postgres's is `float4`, so **the bug only ever damaged hosted data and never reproduced locally**.

All catalogued with reasons in `server/db/models/index.js` (`RAW_UNIQUE_INDEXES`, `NO_ALTER_MODEL_NAMES`, `ensureColumn`).

Operational notes

- **Production self-migrates on boot** (`sync({ alter: true })` + `findOrCreate` seeds), guarded to `NODE_ENV=production` so local runs never auto-alter a DB they might be pointed at. Non-fatal: the server starts anyway and logs the failure. **Reconsider this.** Convenient for a prototype; for production, prefer explicit versioned migrations run as a deploy step.
- **keepAlive.js** pings the DB daily so a free-tier project is not paused. Delete it on paid infrastructure. **Its log line appears regardless of database provider and does not mean you are on hosted data** — this has misled people, including during this handoff.
- **main is the only working branch.** `dev` was retired on 2026-08-05 (24 commits behind, **0 ahead** — zero dev-only commits, so nothing was lost; its last tip was `e0b32ae`). Feature branches are cut from `main` and merged back; the repo auto-deletes a merged head branch.
- **There is no submission DELETE endpoint, on purpose.** A ticket is a record and the portal never destroys one. Two scripts exist for the two operations that genuinely need it — `removeVerificationSubmissions.js` (fixtures, by explicit id, `VERIFY` prefix only) and `purgeSubmissions.js` (everything, with a count confirmation) — and both are dry-run by default.

Open questions for whoever picks this up

1. **Which database engine?** [Decision 1](#).
2. **What hosting shape?** [Decision 2](#).
3. **What does EasyVista actually accept?** The payload shape, endpoint path, response parsing and attachment contract are all unconfirmed, and there is a known EV-side defect. **This is a conversation, not a code task.**

4. **Should the duplicate check tell two people they asked for the same dashboard?** A count without content would do it without breaking the privacy rule.
5. **Who gets *manager*?** Nobody holds it. It gates seeing other people's throughput numbers, and it is a privacy decision.
6. **Do the report-request field lists hold?** The confirmed list is a **sample**. The columns are plain and nullable specifically so adding one stays a one-line migration plus a form control.

Part XI — Rebuild acceptance checklist

Behaviours that are **load-bearing**. Each one either encodes a domain rule or fixes a bug that was actually hit. **If the rebuild breaks one, it is a regression, not a design difference.**

Access and identity

- ☐ An admin with **no grants** sees **no tickets** — not all of them.
- ☐ Scoping runs **before** any query-string filtering; **no parameter can widen** visibility.
- ☐ **Read scope is narrowed by request type, not only by application.** A grant of *report* on an application must not read that application's defects — in the queue, the detail endpoint **or** the export. A filter that happens to hide them is not a permission.
- ☐ The type narrowing applies to **all three** read paths: the application, the soft association, and the hand-off ledger.
- ☐ **Being told is the same permission as being able to see.** A live notification reaches only the admins whose grants cover that row's application *and* type — a broadcast to every admin is the same leak on a different surface.
- ☐ A notification payload that identifies no submission goes to everybody (it cannot be scoped); a notification whose scoping **throws** goes to nobody.
- ☐ A ticket outside read scope returns **404, not 403**.
- ☐ Past owners **keep read** and **lose write** the instant a ticket is redirected.
- ☐ A grant is (*user, application, role, request_type*), and **every write path checks the type**, not just the application.
- ☐ An **analyst** is an admin grant narrowed to one type — not a fourth role.
- ☐ A grant combination the UI cannot express reads as **Mixed** rather than being rounded and rewritten on save.
- ☐ Saving one person's grants **writes the whole key**, so a partial write cannot widen the scope.
- ☐ A **viewer** can read and export, and change nothing.
- ☐ Removing the **last super user** is refused.
- ☐ AD group membership grants **no triage rights**.
- ☐ Super-user and role changes take effect on the **next request**, not the next login.
- ☐ Exactly **one** identity envelope endpoint; no page reads the session directly.
- ☐ A **signed-in** reporter cannot file under someone else's name.
- ☐ **Filing requires a signed-in person**, and a report request requires one even if that switch is off.
- ☐ **Reading the board does not require a session.** The gate is on filing.
- ☐ A lapsed session is reported as **expired**, not as a missing field, and the form keeps what was typed.
- ☐ Metadata **writes** are super-user only; the **read** stays open to every admin, and a test pins the asymmetry.

- ☐ Dev impersonation, if kept, is gated on **three** independent conditions and its route is **not registered** otherwise.

Applications, **Other**, and the soft association

- ☐ **Every path that sets `application_id` refuses a non-report type against a reports-only application** — the public submit, the admin create, the admin update, the import **and the redirect**. The redirect was missed once; a defect landing there is invisible, not merely unassigned.
- ☐ Creating an application **grants it in the same transaction**. A queue nobody can see is worse than no queue.
- ☐ A duplicate name is refused **case-insensitively, including against a switched-off application**, and the message says a super user must switch it back on — because the caller cannot see that row.
- ☐ Creating an application is **not** on the metadata router. Creating touches no existing ticket; renaming touches every ticket that holds the value.
- ☐ `working_application_id` **never** decides who may edit, and **never** widens read access beyond a queue the caller already holds.
- ☐ It is only ever set on a ticket in **Other**, and is **cleared** when the ticket is redirected into a real application.
- ☐ The soft assign is refused for a queue the acting admin does not work in, **type-scoped** — otherwise an admin can put work on another team's list, and that team cannot edit the ticket to get rid of it.
- ☐ The status trigger fires only when leaving **New**, only when nothing is chosen, and only when there is **exactly one** candidate. Two candidates wait to be told.

The Service Desk hand-off affordance

- ☐ An application with no catalog shows the send **disabled with the reason**, on both the demo and the live path — not enabled and failing on click.
- ☐ The button's reason and the endpoint's 400 come from **one** call to `easyVistaCatalogStatus`, so they cannot disagree.
- ☐ The reason states the **whole manual procedure** (raise it by hand → unlock the number → enter it → set `Submitted`), because every step already exists in the UI.
- ☐ A **DEMO-** catalog counts as configured **only while nothing is transmitted**, and reverts to unconfigured the moment the integration goes live. A placeholder must never post a real ticket into a catalog that does not exist.
- ☐ A report request carries **no** hand-off verdict and **no** button.

Data boundary

- ☐ No internal field reaches a public REST response, **a socket broadcast, an AI summary, a public embedding, or a public keyword doc**.
- ☐ A **report request is visible only to the person who filed it**, on all four surfaces, through **one** function.
- ☐ The socket has **three** audiences, not two — everyone, one person, nobody.
- ☐ The routing **note** never reaches a reporter.
- ☐ `is_mine` is computed server-side against `reporter_user_id`.

Concurrency

- ☐ Optimistic concurrency is enforced **both** at save time **and** inside the **UPDATE ... WHERE**.
- ☐ A conflict offers a **field-by-field three-way diff**, not just an error.
- ☐ A pure viewer's open modal **silently re-bases** on a remote change; someone with unsaved edits is **warned**.
- ☐ Authorisation is checked **before** the conflict check.

Bulk

- ☐ The selection scope (**all pages of the filtered set**) is stated in words.
- ☐ Changing filters clears the selection; benign live refreshes do not.
- ☐ The confirmed id set is **snapshotted** at click and **re-intersected** with current rows at apply.
- ☐ Bulk reuses the per-row path so history, sockets and embeddings match exactly.
- ☐ One failing id does not abort the batch.

Lifecycle

- ☐ Nothing hard-deletes a submission.
- ☐ Statuses are **data**; nothing hardcodes the full list.
- ☐ **Which statuses a type may hold is scoped in one place**, enforced on create, update and backdated history, and honoured by the import's analyze step too.
- ☐ A **retired status does not hide a live ticket** (drop the whitelist when all are selected).
- ☐ Board position derives from **current status**, never furthest timestamp.
- ☐ Closed outcomes and parked statuses do **not** draw a pipeline track.
- ☐ Each type draws **its own** track, and the stage tiles name both vocabularies where the words differ.
- ☐ Redirect **moves**; resubmission **forks**; a moved ticket lands as **New** with **status_at_handoff** preserved.
- ☐ **needs_workaround** and **workaround_provided** stay **two** columns; the filter has **three** states.
- ☐ Real defect/enhancement tickets are **public by default**; cleanup-only is private.
- ☐ A cleanup is a **flag** on a defect or enhancement, never a request type.
- ☐ A **cleanup_only** task must have a type **chosen** before it can go to the Service Desk.
- ☐ **A report request is refused by the Service Desk send outright**.
- ☐ Marking a report request Delivered **fills the completion timestamp** the throughput page counts by.

Report requests and delivery

- ☐ The **application is the data's**, asked and never defaulted, and refused at the endpoint if blank.
- ☐ Only the **chosen branch's** fields are stored; the other branch's answers are nulled.
- ☐ **is_new_dashboard** stays **tri-state** — null means "not a report request".
- ☐ **Hours are a child table**, per person per day; **Duration** is **SUM()** computed on read.
- ☐ The **assignment trail** is its own table and ships with the feature.
- ☐ An **approval needs a name AND a date**; the accountability id is server-filled and never accepted from the client.
- ☐ An assignee is **checked against their grant on save**, not merely offered in a dropdown.
- ☐ **is_complete** and **is_approved** are **derived, never stored**.
- ☐ The throughput page's shape is **decided by the server**, and a non-manager's numbers are narrowed **in the query**.

AI search

- ☐ Ticket data in results always comes from the **DB row**, never model text.
- ☐ Top-K is selected by **raw** similarity; recency and same-type only tiebreak display.
- ☐ `has_relevant_match === false` forces an empty match list.
- ☐ Identifiers are matched **literally** and are **not embedded**.
- ☐ Semantic and literal matches stay in **separate labelled sections**.
- ☐ The duplicate check **narrows by type in the query**, and **says what it searched**.
- ☐ Excluding one type does **not** also exclude rows with a null type.
- ☐ A provider failure degrades to literal matches; it never fails the search.
- ☐ The summary **describes and does not rule**.
- ☐ The whole feature **self-disables** with no key configured.
- ☐ The public endpoint is rate-limited.

Service Desk

- ☐ Preview and send are built by the **same** code path.
- ☐ Transmission requires an **explicit** enable — credentials alone are not enough.
- ☐ The repurposed-field mapping lives in **one** place and is **visible in the UI**.
- ☐ The catalog is **per application**, and nothing inherits another application's.
- ☐ `easyvista_application_id` is a **snapshot**.
- ☐ Attachment send failures never fail an already-created ticket.
- ☐ Which fields block a send follows the **chosen** type.

Excel

- ☐ Analyze **writes nothing**; every run is recorded.
- ☐ The mode **forces the type of every row**, and statuses are scoped by it on analyze as well as import.
- ☐ `Assigned To` is resolved to a **user id or nothing** — never stored as text, and refused when ambiguous or ungranted.
- ☐ Rows that landed **minus a field** are reported separately from rows that were skipped.
- ☐ Export reads through the **same access scope** as the queue.
- ☐ The export field list is the **server's**, and a field with no group fails a test.
- ☐ **A round-trip test exists**: export, re-import the file the app itself wrote, compare every column.
- ☐ The shared insert column list is **append-only**, and a test pins the whole list in order.

UI behaviours

- ☐ Both list surfaces implement **all four** data states; the skeleton matches real row height and never leaves stale rows presenting as current.
- ☐ "Whole queue" and "filtered view" counts are **visibly distinguished** and each says whether filters affect it; totals equal the sum of their cards.
- ☐ Applied filters render as **individually removable** chips, derived **once** and shared by badge, chips, summary line and empty state.
- ☐ Hidden filters have their **values reset** so they cannot silently constrain.
- ☐ Sorting is reachable independently of which columns are visible, both paths writing one value.
- ☐ Per-admin view preferences persist **server-side**, allow-listed, sanitised against the **full** registry, and hold **one layout per queue**.

- ☐ A **pin** is distinct from "the last thing I looked at".
- ☐ The kind-of-work switch and the filter panel write the **same** value.
- ☐ Validation appears only **after** a submit attempt, and focus moves to the first problem.
- ☐ The duplicate check runs **before** submit, over **all time**, and offers a re-check when the summary changes.
- ☐ Screenshots can be **pasted** from the clipboard anywhere on the page.
- ☐ The detail modal's identity, alerts and action bar sit **outside** the tab strip.
- ☐ A read-only record tab reads from the **saved** record, not the edit draft.
- ☐ Custom dialogs and notices throughout — **no native** `alert()` / `confirm()`.
- ☐ Every surface works at **390px**; light and dark both fully styled.

Data types

- ☐ Currency is **DECIMAL** or integer minor units — **never** **REAL**/`float4`.
- ☐ If money comes back from the driver as a string, it is coerced at **one** boundary.
- ☐ A money column with no value reads as **null**, **not 0**.
- ☐ Hours are **DECIMAL**, not float.
- ☐ Timestamps are native columns, normalised to one timezone convention; `worked_on` stays a **date**.
- ☐ Optimistic concurrency uses an **explicit version column**, not a timestamp string compare.
- ☐ Malformed legacy date values are found and resolved **before** the type conversion.

Storage, sessions and deployment

- ☐ Attachments survive a restart/redeploy — object storage or a persistent disk.
- ☐ Attachment reads are **authorised**, not merely unguessable — and an `approval` file is never on the unauthenticated path.
- ☐ Uploads validate **extension and MIME type**, and are served with `nosniff`.
- ☐ Sessions survive a restart and are shareable across instances; the fallback path is **tested**, not assumed.
- ☐ Schema changes go through **reviewable versioned migrations**, not boot-time `alter: true`.
- ☐ The API host is **configuration**, not hardcoded in two files.
- ☐ The server **refuses to start** with a weak or default session secret in production.
- ☐ Rate limiting and presence work with more than one instance, or single-instance is an **accepted, documented** constraint.
- ☐ Dev impersonation is deleted, or provably unreachable in every deployed environment.
- ☐ Every destructive maintenance script is **dry-run by default** and prints its dialect before its numbers.

Where to look in the prototype

When this document is not specific enough, these files carry the most decision-density. **The inline comments are unusually dense with *why*** — read the comment above the code rather than inferring intent from the code.

Concern	File
Identity, scoping, the SSO seam	<code>server/src/services/viewerService.js</code>
Who filed a ticket	<code>server/src/services/reporterService.js</code>

Concern	File
Access grants, with scopes	server/src/services/accessService.js
Redirect semantics	server/src/services/redirectService.js
The bulk of triage logic	server/src/services/submissionService.js
Hours, assignments, throughput	server/src/services/deliveryService.js
The public allow-list	server/src/helpers/mappers.js
The private-report rule	server/src/helpers/reportVisibility.js
The positional insert contract	server/src/helpers/submissionInsert.js
Attachment storage (both backends)	server/src/helpers/storage.js
Status history synthesis	server/src/helpers/timeline.js
Session store selection + its fallback	server/src/middleware/session.js
Everything env-driven, with reasons	server/src/config.js
Role ladder, status registries, filingRequiresSignIn	server/src/constants.js
EasyVista payload + repurposed fields	server/src/helpers/easyVistaPayload.js
AI pipeline	server/src/services/aiSearchService.js
Embedding scopes + keyword doc	server/src/services/embeddingIndexService.js
Schema, and every portability trap	server/db/models/index.js
Detail modal orchestration	client/src/hooks/useDetailModal.js
Column/filter/sort registries	client/src/constants/adminConstants.js
Per-type status scope, client side	client/src/constants/statusConstants.js
Board track semantics	client/src/components/public/StatusBoardRow.jsx
Ownership resolution	client/src/hooks/useViewer.js
Throughput window + composition	client/src/hooks/useThroughput.js
What has actually been verified	client/scripts/verify-*.mjs

Also worth reading

Document	Contents
USER_MANUAL.md	Every feature and how to use it, with the current screenshots
NEXT_STEPS.md	The programme decision, and what is being asked of the Customer Interactions team

Document	Contents
server/docs/ai-search.md	AI search presets, cost, tuning, full variable list
server/docs/easyvista-description-format.md	The EasyVista description payload format
CLAUDE.md	Conventions and skills for AI-assisted work in this repo
plan.md	The running project plan and its dated record

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